Advanced Void Automation

Survival and Factory Guide

Follow the survival path from your first tree through powered automation, advanced materials and the Star Forge. Use the reference section when you need a recipe or drop chance.

Open the contents p. 2

Find the next build for your stage of the island.

Start on a new island p. 3

Install the pack and make your first renewable supplies.

Find a recipe or a drop p. 75

Jump to the crafting atlas, processor recipes or sieve tables.

Diagrams use pack textures. HUD illustrations are schematics. Vanilla item art is from Mojang/Microsoft.

Contents

First island	3
Sieves and first metals	12
How to make obsidian	17
Power, tanks and item pipes	21
Smeltery, alloys and tools	32
MV Storage and factory automation	38
Polymer and advanced machines	50
Essence gear and travel	56
Armor and tool comparison	66
Void mining and late-game power	70
Wither Chamber and Star Forge	74
Reference and recipes	80

First island

Turn one tree into a renewable supply of wood, dirt, water and cobblestone.

1. Install both packs p. 4

Enable the required experiment before loading the world.

2. Make a crook and compost saplings p. 6

Save a sapling to replant before composting extras.

3. Collect water and make a bucket p. 8

Use water and dust to get clay for your early containers.

4. Make lava and renew your cobblestone p. 11

Set up a generator before you start crushing blocks in bulk.

Continue to Sieves and first metals p. 12

Open the next chapter when your current setup is ready.

Look up ingredients and recipes p. 75

Find the illustrated crafting grids and machine input amounts.

Start here: installation and using this handbook

 *MV System* Crafting recipe	 *Sieve* Crafting recipe		

This survival handbook follows Advanced Void Automation 0.2.56 Beta, from the first tree through the Star Forge. Use the contents to find the next build or jump to the recipe appendix.

Import the mcaddon and activate BOTH Behavior and Resource Packs at the same version. Keep the Script API/Beta APIs required by the current pack enabled. Restart Minecraft if the library still displays an older cached version.

Existing worlds do not need to be deleted. Back up and close the world, then update both embedded packs or select both matching imported versions. An old embedded pack and a newer shared-library pack can appear separately in the version list.

For native Supremium flight, enable Education features and Cheats. The other factory systems do not require Education. A normal addon cannot switch on smooth native Creative flight through the read-only Player API alone; this build does not silently restore the old jittery simulation.

Use the clickable contents to jump to a system, the machine directory to identify a block, and the recipe index for the added-block crafting grids. Shaped grids show exact positions; shapeless grids show ingredients. Construction images are labeled schematics. Item and ingredient pictures use current pack textures.

The handheld Wireless MV Remote binds directly to MV Storage. Use the wired MV Link Unit to connect item automation to storage.

JUMP TO: *Machine directory* | *Block recipes* | *Drop tables* | *Essence equipment recipes* | *Tree to Supremium costs*

Quick start for your first island

Import the .mcaddon into Minecraft Bedrock. Activate the matching Behavior and Resource Packs in your skyblock world and enable Beta APIs. Back up an existing world before updating it.

This walkthrough starts on an island with a tree and a patch of dirt. “Use” means right-click, tap Interact, or press your controller’s interact button. Save one sapling for the next tree and keep the original dirt to plant it on.

 *Wood Crook*	 *Silkworm*	 *Sieve* Crafting recipe	 *Pebble*

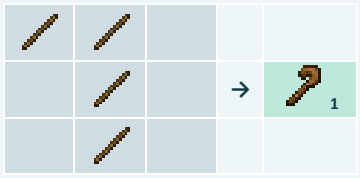
1. Make tools from the first tree. Turn Oak Logs into Oak Planks. Make a Crafting Table from four planks, then craft sticks from two planks stacked vertically. Make a Wood Crook before breaking the leaves.

2. Grow more trees and get string. Break normal oak leaves with the crook for saplings and a chance at Silkworms. Use a Silkworm on an intact oak leaf block, wait for nearby oak leaves to become infested, then crook the infested leaves for String.

3. Build a Sieve and Composter. Craft the Sieve with four Oak Planks and four String in any positions. It comes with a basic mesh. Make a regular Composter from seven wooden slabs in a U shape.

4. Make your first Cobblestone. Use eight spare saplings on the empty Composter. When the green filling turns to dirt, use it with an empty hand to collect one Dirt. Use the Dirt on the Sieve, then keep interacting until the batch finishes. The loading interaction counts as the first of six. Collect the pebbles and craft four ordinary Pebbles together for one Cobblestone. Repeat to build a stockpile.

Wood Crook



Wood Hammer



Composter (wooden slabs)



GO TO *Sieve* | *Dirt build* | *Sieve drop rates* | *Next: make water*

First island: wood, saplings, string and cobblestone

 *Wood Crook*	 *Silkworm*	 *Sieve* Crafting recipe	 *Pebble*

Protect the initial tree and saplings. Expand before harvesting everything so drops do not fall into the void. Make a crafting table, sticks and a Wood Crook. Replant before using the last sapling.

Use a crook on normal leaves for its species-matching bonus sapling roll and Silkworm roll. Shared bonus rolls are 15% sapling and 3% Silkworm per leaf, in addition to native leaf loot. Better crooks mainly improve speed/durability; they do not change those bonus odds.

Use a Silkworm on oak leaves to infest them. Infestation spreads to neighboring oak leaves. Harvest infested leaves for string and establish the sieve/mesh progression. Sneak vein mining applies the same rolls separately to every leaf.

Use eight surplus saplings in an empty Composter for one Dirt. Sieve Dirt for ordinary Pebbles, then craft four together into one Cobblestone. Specialty pebbles make their own stone types.

JUMP TO: *Machine directory* | *Block recipes* | *Drop tables* | *Essence equipment recipes* | *Tree to Supremium costs*

Eight saplings into dirt



INPUT: *8 saplings*

OUTPUT: *1 Dirt*

1. Place an empty composter. Keep water out of this batch.

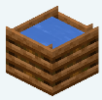
2. Use eight saplings on the empty Composter. All eight fill steps are green.

3. Wait for the full batch to compost into dirt, then collect it before refilling.

READ MORE: *Composting, water, clay and mycelium* | *All build examples*

CRAFT THE BLOCKS: *Moisture Collector* | *Sieve*

Make water, clay and your first bucket

			
Rainwater Composter	*Dust*	*Clay Ball*	*Clay Bucket*

1. Collect rain for your first water. Place an empty Composter under open sky in the Overworld. Rain fills it in four stages. Leave it on ordinary ground so the water stays fresh. A full Composter holds 1,000 mB, or one bucket.

2. Make water any time with leaves. Use shears to collect four ordinary leaf blocks and add them to an empty Composter, one at a time. Each adds 250 mB. To make shears, hammer Cobblestone into Gravel and sieve it for Iron Fragments. Craft four fragments into one Raw Iron and smelt it in a Furnace made from eight Cobblestone around an empty center. Fuel it with wood. Repeat to get the two ingots for shears.

3. Make Clay. Use a Wood Hammer on Cobblestone, then Gravel, then Sand to get Dust. Use one Dust on the filled Composter to turn the entire water fill into one Clay Block. Break the Clay Block to get four Clay Balls.

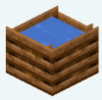
4. Make a bucket and refill the Composter. Shape three Clay Balls into an Unfired Clay Bucket and fire it in a regular Furnace. Refill the Composter with rain or leaf blocks, then use the empty Clay Bucket on it to draw the water. An Iron Bucket crafted with three Iron Ingots also works.

Unfired Clay Bucket	**Make Mud**	**Make Grout**
	Use one Dirt on a full ordinary-water Composter. This uses all 1,000 mB.	Craft one Clay Ball, one Sand and one Gravel together to get two Grout.

For a repeatable water supply later, place a Moisture Collector on solid ground and put an empty Composter directly above it. Feed the collector a Lava Bucket or connect a lava Tank. It uses 10 mB of lava to produce 50 mB of water per second while there is room for the water.

GO TO *Water and clay illustration* / *Pumped water build* / *Next: make lava*

Composting, water, clay and mycelium

			
Rainwater Composter	*Moisture Collector* Crafting recipe	*Ancient Spores*	*Mycelium*

For Dirt, use eight saplings in an empty Composter. Wait for the green filling to finish, then use an empty hand to collect the Dirt.

For your first water, collect rain in an empty Composter under open sky or add four ordinary leaf blocks. Use an empty bucket on the full Composter, or use one Dust to make a Clay Block or one Dirt to make Mud.

For renewable water later, place a Moisture Collector on solid ground with a Composter directly above it. Supply lava to the collector. Use a Fluid Pump to move ordinary water into a dedicated water Tank.

For Witch Water, sieve Sand or Red Sand for Ancient Spores, use the spores on Dirt, then place a water-filled Composter directly on the Mycelium. Use Sand once it becomes Witch Water to make Soul Sand.

The Stone Barrel holds lava for Netherrack, End Stone and Obsidian. Use the regular Composter for water-based clay and mud recipes.

STEP BY STEP *Make water and clay* | *Make lava* | *Barrel and Witch Water recipes*

BUILD EXAMPLES: *Eight saplings into dirt* | *Witch Water into Soul Sand* | *Water conversions: clay and mud*

JUMP TO: *Machine directory* | *Block recipes* | *Drop tables* | *Essence equipment recipes* | *Tree to Supremium costs*

Water conversions: clay and mud



INPUT: *1,000 mB ordinary water + Dust OR Dirt*

OUTPUT: *Clay OR Mud*

1. Fill an empty Composter with four ordinary leaf blocks or one Water Bucket. Set it on ordinary ground for these recipes.

2. For clay, use one Dust on the water. For mud, use one Dirt.

3. Collect the finished result before changing recipes. Refill one bucket of water for each new conversion.

READ MORE: *Composting, water, clay and mycelium* | *All build examples*

CRAFT THE BLOCKS: *Stone Barrel*

Make lava and a renewable cobblestone supply

			
Kiln	*Tank*	*Fluid Pump*	*Stone Barrel*
Crafting recipe	Crafting recipe	Crafting recipe	Crafting recipe

1. Craft a Kiln. Place eight Cobblestone in the Crafting Table, leaving the bottom-center slot empty. It needs no power or external heat.

2. Melt Cobblestone. Hold extra Cobblestone and use it on the placed Kiln. Each piece yields 1,000 mB of lava, or one bucket, after about 10 seconds at normal tick speed.

3. Take the lava. Use an empty Clay Bucket or Iron Bucket on the Kiln once it has at least 1,000 mB. Keep lava in a nonflammable area away from your tree and wood platform.

4. Make a cobblestone generator. Dig a trough in a fireproof platform. Place a water source at one end and put a one-block-deep catch pit in the next cell. Leave an open meeting cell beyond the pit, then place the lava source at the far end. Water falls into the pit and flowing lava reaches the water from the other side. Mine the Cobblestone as it forms.

If you need	Do this
Trough from above	Water source | Deep catch pit | Open mining cell | Lava source. Build a floor and solid walls around the trough to contain the flow.
A bigger lava supply	Place a Fluid Pump by the Kiln and run purple Transfer Pipes to a compatible Tank.
A way to store water	Use a separate Tank group for water. Connected Tanks share storage, so keep the water and lava groups apart.
A Stone Barrel	Craft seven Cobblestone in a U. Fill it with lava for the recipes on the next page.

GO TO *Kiln* | *Stone Barrel* | *Pumps and tanks* | *Next: lava and Witch Water recipes*

Sieves and first metals

Use the renewable Cobblestone supply to build your first resource workshop.

1. Make a hammer, sieve and mesh p. 13
Crush materials, then sieve the inputs for each drop you need.

2. Build a compact sieve workshop p. 14
Keep the input blocks, meshes and outputs within reach.

3. Turn your drops into ingots p. 15
Choose the fragment, crushed ore or powder route for each metal.

4. Unlock progression blocks p. 16
Make the materials for later builds, including Netherrack and Soul Sand.

Continue to Power, tanks and item pipes p. 20
Open the next chapter when your current setup is ready.

Look up ingredients and recipes p. 75
Find the illustrated crafting grids and machine input amounts.

Hammers, sieving and mesh progression

			
Wood Hammer	*Dust*	*Iron Sieve Mesh*	*Netherite Sieve Mesh*

Use a hammer for supported crushing conversions. The early chain starts Cobblestone -> Gravel -> Sand -> Dust. Other supported rocks have their own outputs. A better hammer does not automatically improve sieve odds.

Use a supported block on a Sieve, then keep interacting until its six work clicks complete. The loading interaction counts as the first click. Collect the physical drops on a safe platform.

Use Iron, Diamond or Netherite Mesh on the sieve to upgrade it. Basic/String, Iron, Diamond and Netherite apply x1, x1.25, x1.6 and x2 to independent base rolls, capped at 100%. Sneak-use empty-handed to remove an upgrade and return the mesh.

Dirt has a 5% base Inferium Essence roll. With Iron, Diamond and Netherite Mesh that becomes 6.25%, 8% and 10%. The drop tables include the weighted fallback used when every independent roll fails.

BUILD EXAMPLES: *A compact sieve workshop*

JUMP TO: *Machine directory* | *Block recipes* | *Drop tables* | *Essence equipment recipes* | *Tree to Supremium costs*

A compact sieve workshop



INPUT: *One sieve input + installed mesh*

OUTPUT: *Resource drops listed by input and mesh*

1. Use the supported hammer chain: Cobblestone to Gravel to Sand to Dust. Choose the stage you need from the drop tables.

2. Install a mesh in the Sieve. Load one supported input and work it through six clicks.

3. Keep room for ejected drops. A nearby Vacuum Hopper is a collection example; add a Transfer Node if you want automatic export to the chest.

4. Upgraded meshes change the independent drop rolls. Follow the linked drop tables for exact chances and quantities.

READ MORE: *Hammers, sieving and mesh progression* | *All build examples*

CRAFT THE BLOCKS: *Sieve* | *Vacuum Hopper*

DROP CHANCES: *All inputs and meshes*

Fragments, crushed ore, powder and ingots

			
Iron Fragment	*Crushed Iron Ore*	*Iron Ore Powder*	*Iron Ingot*

Gravel yields fragments, Sand yields crushed ore, and Dust yields powders. For iron, copper and gold, combine four matching fragments or crushed pieces into one Raw Ore and smelt it in a Furnace. Four matching powders craft one ingot directly. Custom metals can use their own combine recipes.

Fresh raw ore processing can have an ore bonus, but recycled ingots must not multiply forever. Four Iron, Gold or Copper Powders combine into an INGOT, not raw ore. Pulverizing a finished ingot and recombining its powder is neutral recycling.

The Smeltery accounts for powder, ingots and fresh raw ore using different molten quantities. Four powders represent one ingot. Supply machine fuel/power and free output space before assuming a processor has stalled.

JUMP TO: *Machine directory* | *Block recipes* | *Drop tables* | *Essence equipment recipes* | *Tree to Supremium costs*

Make Obsidian, Netherrack, End Stone and Soul Sand

A Stone Barrel holds lava only. Use a Lava Bucket on an empty Stone Barrel. Each recipe uses 100 mB, so a full 1,000 mB bucket can supply ten conversions.

Output	Use this ingredient on the lava-filled Stone Barrel
1 Obsidian	1 Water Bucket or Clay Water Bucket. The water is a recipe ingredient and you get the empty bucket back.
1 Netherrack	1 Redstone Dust. Sieve Sand or Dust to get Redstone. Hammer Netherrack into Crushed Netherrack for further sieving.
1 End Stone	1 Glowstone Dust. Sieve Sand or Dust to get Glowstone Dust.

Make Witch Water and Soul Sand

1. Make Mycelium. Sieve Sand or Red Sand for Ancient Spores. The base roll is 5% per input with the basic mesh. Use the spores on a Dirt block to turn it into Mycelium.

2. Brew Witch Water. Place a regular Composter directly on the Mycelium. Fill it with one bucket of water or four normal leaf blocks. Wait for the status to say Witch Water. Keep this setup separate from your ordinary water supply.

3. Make Soul Sand. Use one Sand on the finished Witch Water. Collect the Soul Sand and refill the Composter to brew another batch. You can sieve the Soul Sand for Quartz and a chance at Nether Wart or a Ghast Tear.

To start the smeltery, smelt Grout in a Furnace to get Seared Brick items, then craft four of them in a square to make one Seared Bricks block.

GO TO *Stone Barrel illustration* | *Witch Water illustration* | *Build the smeltery* | *Contents*

How to make obsidian

Manual method

Put a lava source in a nonflammable basin or channel. Let water flow onto that source. Mine the resulting obsidian with a diamond or netherite pickaxe, or aim a Transfer Node directly at the obsidian and connect the node to storage. The lava source is consumed during conversion. Replace it to repeat.

Automatic pipe junction

1. Put a water source under one Transfer Node and a lava source under a second node. Check that both nodes report DOWN when built this way.

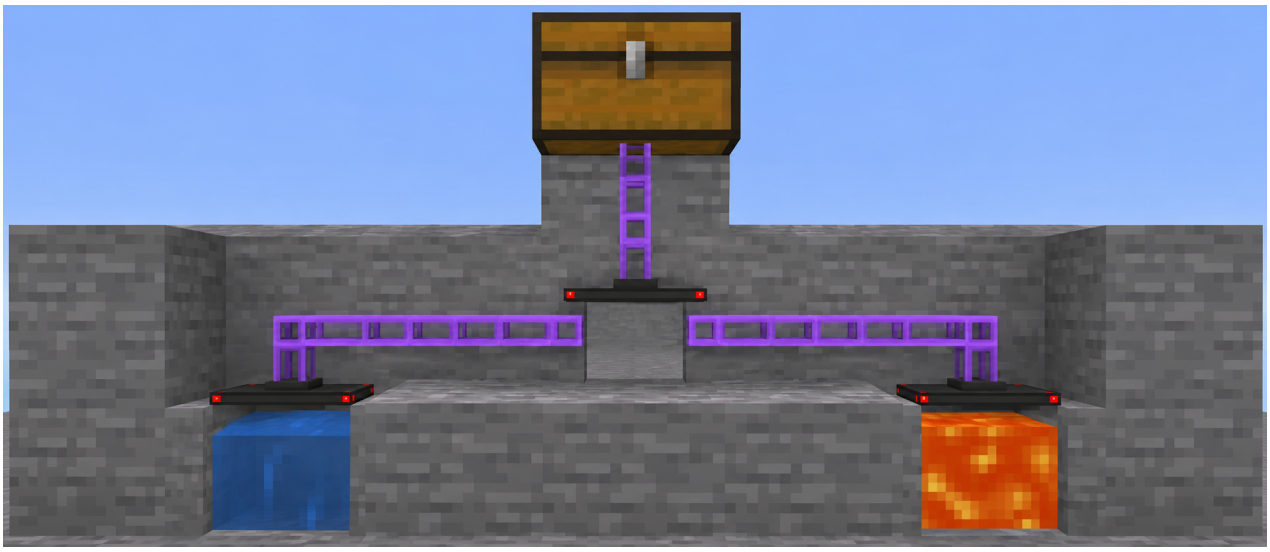
2. Run a purple Transfer Pipe from each node. Leave exactly one air block between two opposing pipe ends. Keep the two pipe branches separate around the gap.

3. Attach a chest to either branch. Leave room for obsidian. The junction creates one block on a 10-second cooldown without consuming the two fluid source blocks.

4. If obsidian stays in the gap, check storage space and Smart Transfer Node filters. A separate collector can face the obsidian directly in 0.2.86.

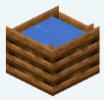
Water branch	Junction	Lava branch
Water source under intake node	One air block	Lava source under intake node
Intake node to purple pipe	Obsidian forms here	Purple pipe to intake node

The pipes beside the gap sit at the same height in this layout. The fluid sources sit below their respective intake nodes.



Place the center Transfer Node one block above the air gap, facing down.
Leave the gap empty. Run the purple pipe from the collector up to the output chest.

Material recipes: End Stone and other progression blocks

 *Stone Barrel* Crafting recipe	 *Fluid Pump* Crafting recipe	 *Moisture Collector* Crafting recipe	 *Rainwater Composter*
 *Ancient Spores*			

END STONE: place a Stone Barrel. Supply at least 100 mB of lava, then insert 1 Glowstone Dust to produce 1 End Stone. A lava bucket holds 1,000 mB and supports ten conversions. For automation, connect a lava tank through the fluid network and insert Glowstone Dust through a Transfer Node or MV Link. Extract End Stone from the barrel output. No Watt power is needed by the barrel itself.

NETHERRACK: use the same Stone Barrel with 100 mB lava plus 1 Redstone for 1 Netherrack. A hammer breaks Netherrack into Crushed Netherrack; sieve the crushed material for Nether resources. Consult the linked sieve tables for exact rolls and mesh multipliers.

OBSIDIAN: insert 1 Water Bucket into a Stone Barrel containing at least 100 mB lava. This pack produces 1 Obsidian and returns the empty bucket. A Clay Water Bucket also works and returns a Clay Bucket. These are pack-specific conversion costs.

MYCELIUM: obtain Ancient Spores by sieving the supported sand inputs, then use Ancient Spores on Dirt. Mycelium is required below the water-filled composter for Witch Water. See the sieve tables for the current chance for your input and mesh.

SOUL SAND. Put a full ordinary-water Composter above Mycelium and wait for Witch Water. Use one Sand to make one Soul Sand, then refill and wait for the next batch. Bring water by bucket or use four ordinary leaf blocks in the empty Composter.

CLAY AND GROUT. Hammer Sand into Dust, then use one Dust in a full ordinary-water Composter for one Clay Block. Break it into Clay Balls. Craft one Clay Ball, one Sand and one Gravel into two Grout. Smelt Grout into Seared Brick in a regular Furnace or Magma Furnace.

STEEL: build a valid smeltery and provide lava fuel. The Steel alloy uses 1 Iron + 1 Coal + 1 Blaze Powder to produce 1 Steel. Select the stored metal to cast it through a Drain and Faucet into a suitable cast. Selecting Steel does not itself supply missing alloy ingredients.

NETHERITE. Make Netherrack from lava and Redstone Dust. Hammer it into Crushed Netherrack and sieve it for Netherite Scrap. The Basic mesh gives 1%, Iron 1.25%, Diamond 1.6% and Netherite 2% per input. Combine four Netherite Scrap and four Gold Ingots into one Netherite Ingot.

ESSENCE EQUIPMENT: start with Prosperity Shards and Inferium from the documented sieve/crop routes. Grow essence crops to maturity before harvesting. Use the infusion progression and tiered ingot recipes in the complete essence recipe section, then follow each armor or tool recipe. The seed is not itself an ingot; repeated mature harvests supply its ingredients.

BUILD EXAMPLES: *End Stone: Glowstone Dust and lava*

JUMP TO: *Machine directory* | *Block recipes* | *Drop tables* | *Essence equipment recipes* | *Tree to Supremium costs*

Witch Water into Soul Sand



INPUT: *Mycelium + composter + 1,000 mB water + 1 Sand*

OUTPUT: *1 Soul Sand per conversion*

1. Sieve Sand or Red Sand for Ancient Spores. Use the spores on Dirt to make Mycelium.

2. Place the composter directly on Mycelium. Fill it with one bucket of ordinary water (1,000 mB).

3. Keep the area loaded for 30 one-second processing ticks. Its status changes to Witch Water. Do not connect an ordinary-water extraction route.

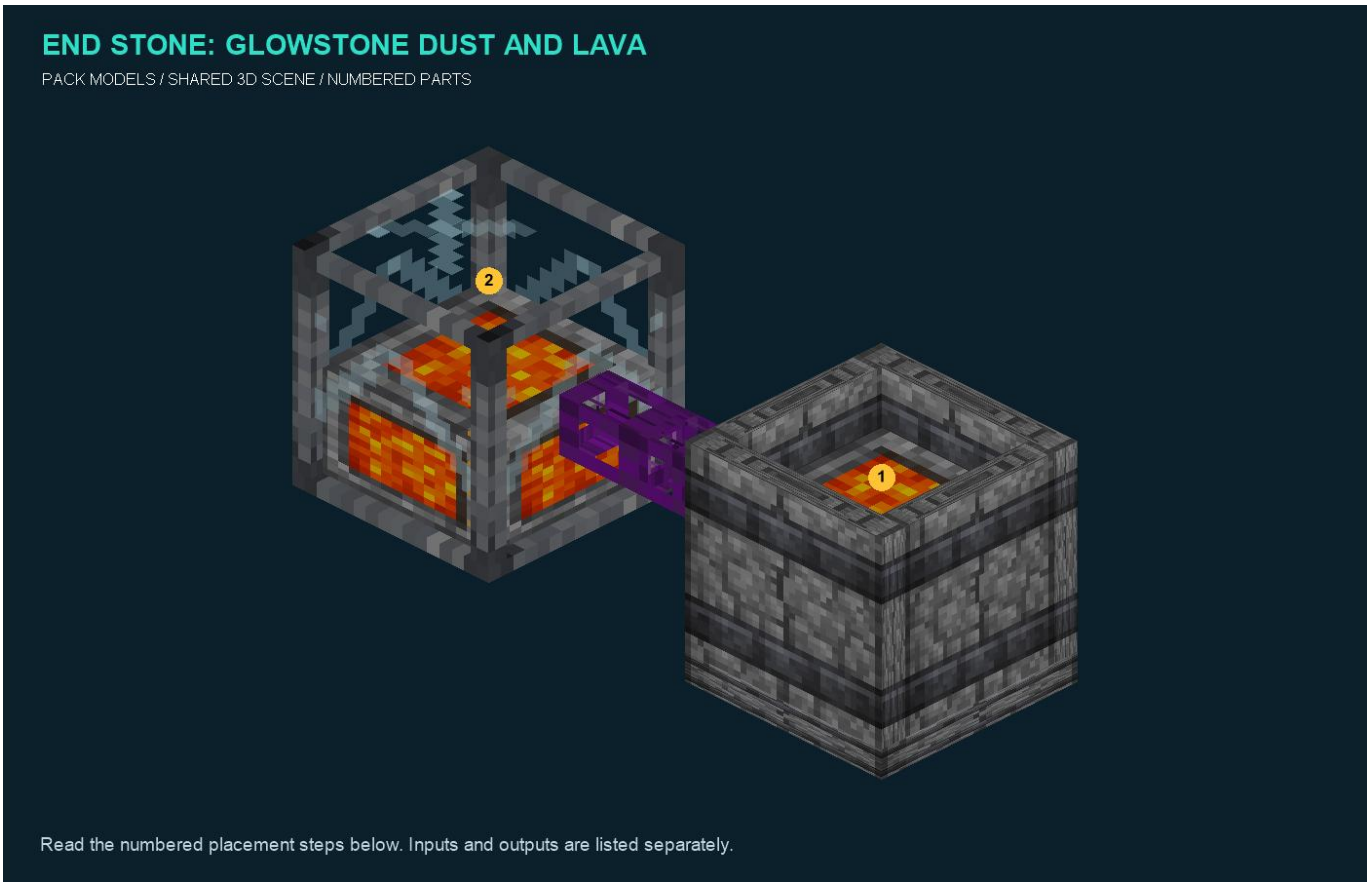
4. Use one Sand on the ready Witch Water. Collect the Soul Sand, refill the water, and repeat.

READ MORE: *Composting, water, clay and mycelium* | *All build examples*

CRAFT THE BLOCKS: *Sieve*

DROP CHANCES: *All inputs and meshes*

End Stone: Glowstone Dust and lava



INPUT: *1 Glowstone Dust + 100 mB lava*

OUTPUT: *1 End Stone*

1. Place Stone Barrel 1. For manual operation, fill it with a Lava Bucket; each End Stone uses only 100 mB of the 1,000 mB bucket.

2. Insert 1 Glowstone Dust for each End Stone. Collect the result from the barrel output before it fills. Redstone in the same barrel makes Netherrack instead.

3. Optional renewable supply: connect lava Tank 2 through a Transfer Pipe to the barrel. The barrel pulls only the lava needed for its next conversion while output has room. Fill the tank from your lava production first.

4. For item automation, use a Transfer Node to supply Glowstone Dust and another correctly facing extraction node to collect End Stone. Keep the input filter on Glowstone Dust. The two-block illustration shows only the barrel and optional lava supply.

READ MORE: *Material recipes: End Stone and other progression blocks* | *All build examples*

CRAFT THE BLOCKS: *Stone Barrel* | *Tank* | *Transfer Pipe*

Power, tanks and item pipes

Power one machine and prove one item route before connecting the whole island.

1. Connect your first Watt line p. 22

Start with a generator. Add a battery when you want backup power.

2. Move fluids into a tank p. 24

Keep water and lava in separate tanks and connect the right pump route.

3. Build a filtered item route p. 26

Use a Transfer Node to extract from the source inventory.

4. Compare all three connections p. 30

Use this plan to distinguish power, fluid and item routes.

Continue to Smeltery, alloys and tools p. 31

Open the next chapter when your current setup is ready.

Look up ingredients and recipes p. 75

Find the illustrated crafting grids and machine input amounts.

From hand tools to your first factory

1. Make enough metal for a power line

Start with the basic Sieve and a renewable Cobblestone supply. Hammer Cobblestone into Gravel for Coal, Iron Fragments, Nickel Fragments and Aluminum Scrap. Hammer Gravel into Sand for Crushed Copper. Sieve Dust or Sand for Redstone Dust.

Craft four Nickel Fragments into a Nickel Ingot and four Aluminum Scraps into an Aluminum Ingot. Craft four Crushed Copper into one Raw Copper, then smelt it in a Furnace. Make two Copper Ingots for your first Combustion Generator.

Craft and place the generator, add logs or charcoal to its fuel slots, and run red Watt Pipes from it to one machine. Start with a short line. The generator can supply its stored energy while the flame is idle.

CRAFT *Combustion Generator* / *Watt Pipe*

2. Get one item route working

Put a test stack in a chest. Attach a Transfer Node with its intake facing that chest, run purple Transfer Pipes, and end at a second chest with free space. Once the stack moves, reuse the same source-to-destination layout to feed inputs or collect output from a machine.

BUILD *Transfer Node* / *Transfer Pipe* / *Item routing and filters*

3. Build a one-layer smeltery and cast your first ingot

Start with 21 Seared Bricks blocks for the floor and nine for the walls. Leave the four outer corners open on both levels. Fill three wall positions with the Controller, Drain and Seared Tank. Leave the 3 × 3 center above the floor empty.

Place the Faucet on the outside of the Drain and the Casting Table directly below the Faucet. The 30 Seared Bricks blocks and five functional pieces take 152 individual Seared Bricks, plus one Furnace for the Controller and one Glass for the Tank. Smelt 152 Grout to make the brick items.

Fill the Seared Tank with lava. Craft a Stone Ingot Pattern from two Cobblestone side by side and place it in the Casting Table. Melt two Gold Ingots, select Gold on the Controller and use the Faucet. Collect the reusable Gold Ingot Cast and put it back in the Table. Then melt and select another metal. Pour one ingot’s worth and collect it once cooled.

BUILD *Smeltery layout* / *Smeltery Controller* / *Smeltery Drain* / *Seared Tank* / *Seared Faucet* / *Casting Table*

4. Unlock MV Storage

Use the smeltery to make Steel. Make Netherite from sieved Crushed Netherrack, save nine Diamonds for a Diamond Block, and collect the other metals on the MV recipe card. Craft the Universal Crafter first, then use it as an ingredient for MV Storage. Power MV with Watt Pipes and test deposit and withdrawal before adding MV Links.

CRAFT *Universal Crafter* / *MV System* / *Steel and Netherite routes* / *Connect MV Storage*

Generators, Watt Pipes and battery backup

			
Combustion Generator	*Solar Panel*	*Pink Slime Generator*	*Battery Bank*
Crafting recipe	Crafting recipe	Crafting recipe	Crafting recipe

A Combustion Generator generates 80 W per working second. Its 10,000 W buffer continues supplying connected machines during standby and while remaining charge lasts after fuel runs out. It resumes burning once there is room for another 80 W. If a powered machine stops while the generator is charged, check its own status and Watt Pipe connections.

Red Watt Pipes form the electrical network. Purple Transfer Pipes carry items/fluids. Choose a generator supported by your supplies: Combustion, Solar/Advanced Solar, lava conversion or the late Pink Slime route. Read the live HUD for power and fuel conditions.

An empty source cannot power machines using stale pipe energy. Solar depends on its active lighting/day conditions. Check source fuel, red pipe connections and workload when a machine reports no power.

Place touching Battery blocks to combine their storage. Connect the bank to sources through Watt Pipes. Multiple sources can contribute spare charging power; full banks stop charging. Charge remains through reloads.

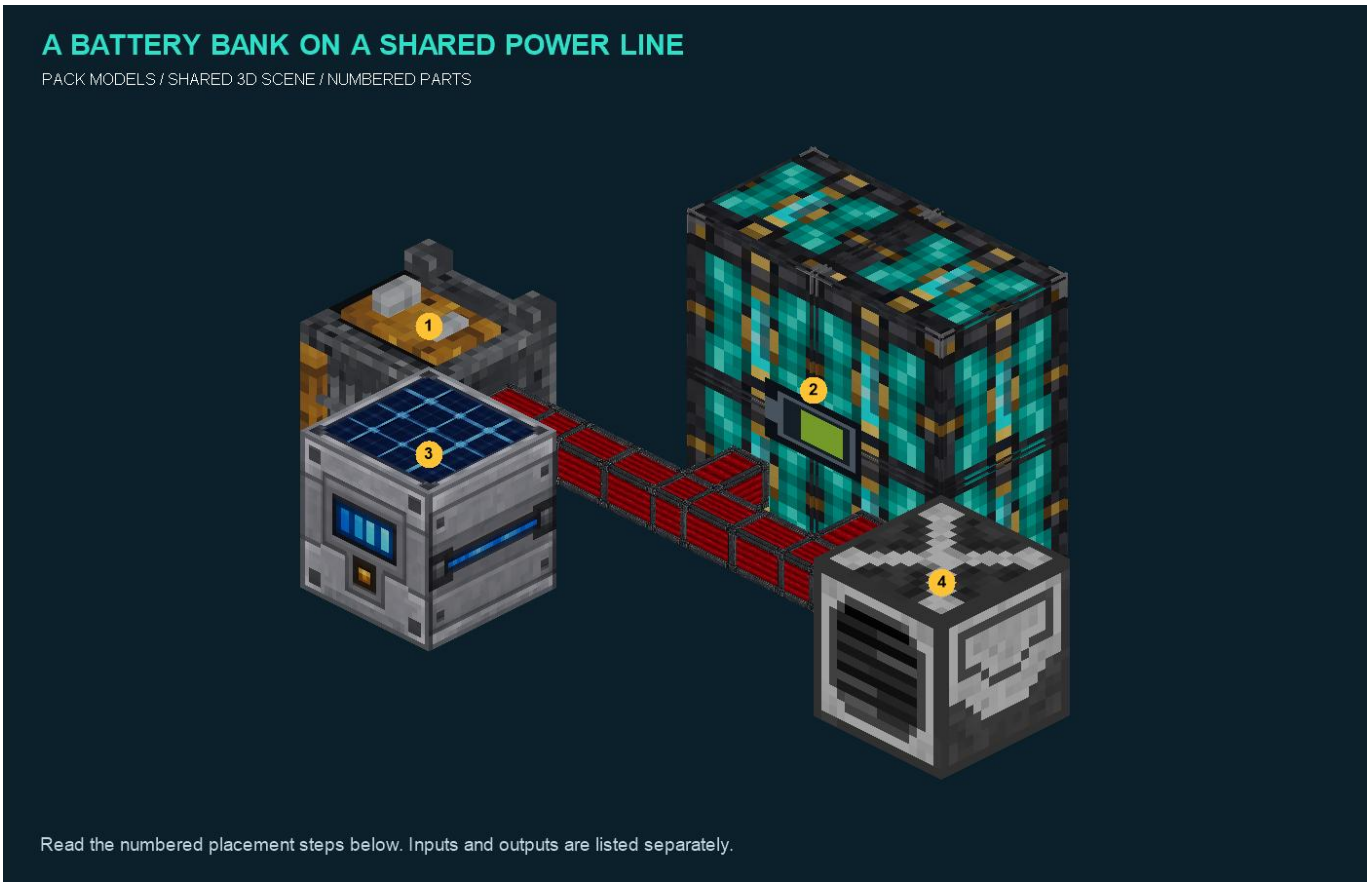
Batteries remain standby while primary generation is active, then discharge when primary sources go down. They do not supplement an undersized but still-active primary source. Remove source fuel for a safe outage test.

The connected bank displays a centered horizontal gauge on one front face, red when low and green when high, plus numeric status. A broken battery drops empty; plan storage changes before dismantling a charged bank.

BUILD EXAMPLES: *A battery bank on a shared power line*

JUMP TO: *Machine directory* | *Block recipes* | *Drop tables* | *Essence equipment recipes* | *Tree to Supremium costs*

A battery bank on a shared power line



INPUT: *Active primary power sources*

OUTPUT: *Stored backup energy when primary generation stops*

1. Connect the generator and machine to the same Watt Pipe network. Batteries are a connected branch, not a mandatory pass-through.

2. Place batteries adjacent to combine storage capacity. Add more active sources, such as solar, to the same network to increase available charging power.

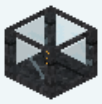
3. Active main sources supply the network while charging the bank. A full bank stops drawing charging power.

4. When no active primary source is available, the bank supplies its stored energy. An active but undersized source does not automatically become a battery-assisted mode.

READ MORE: *Generators, Watt Pipes and battery backup* | *All build examples*

CRAFT THE BLOCKS: *Battery Bank* | *Watt Pipe* | *Solar Panel*

Fluid Pumps, tanks and lava handling

			
Fluid Pump	*Tank*	*Seared Tank*	*Kiln*
Crafting recipe	Crafting recipe	Crafting recipe	Crafting recipe

For water, place an empty Composter on top of a lava-fed Moisture Collector. Use a Fluid Pump and purple Transfer Pipes to connect the source to a compatible water Tank. A full tank or a tank containing another fluid cannot accept more.

For lava, use Cobblestone on the Kiln. Each piece melts into one bucket of lava without power. Use an empty bucket on the Kiln or connect a Fluid Pump and a dedicated lava Tank.

The Kiln is an early lava route. Supply its supported solid input and connected output storage, then wait for processing. A blocked/full output tank can prevent further production.

Seared Tanks supply the smeltery fuel arrangement. Do not mix different fluids into an occupied group. Molten metals can ignite entities like lava; build safe walkways rather than stepping into the molten volume.

BUILD EXAMPLES: *Water collector, pump and tank*

JUMP TO: *Machine directory* | *Block recipes* | *Drop tables* | *Essence equipment recipes* | *Tree to Supremium costs*

Water collector, pump and tank



INPUT: *Lava for the Moisture Collector*

OUTPUT: *Water stored in connected tanks*

1. Place a Moisture Collector on solid ground, with an empty Composter directly above it.

2. Use a Lava Bucket on the collector or connect a lava Tank. The collector adds water to the Composter as long as it has lava and there is space.

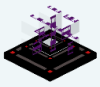
3. Place the Fluid Pump against the water source and run its purple Transfer Pipe route into matching tanks.

4. Watch the source and tank totals. A full or incompatible destination must wait without deleting source water.

READ MORE: *Fluid Pumps, tanks and lava handling* | *All build examples*

CRAFT THE BLOCKS: *Fluid Pump* | *Tank* | *Transfer Pipe*

Transfer Nodes, filters, priorities and bulk storage

			
Transfer Node	*Transfer Pipe*	*Polymer Circuit Board*	*Black Hole Unit*
Crafting recipe	Crafting recipe		Crafting recipe

Place a Transfer Node against the source inventory and route purple Transfer Pipes to a destination. Keep intake direction unambiguous. Sneak-use rotates the intake. Start with one source chest and one destination when troubleshooting.

Use a Polymer Circuit Board on a node to enable smart routing. Open its HUD and place actual example items into persistent filter slots. Remove them manually when you want to change the filter.

ALLOW listed items transfers only matching items; BLOCK listed items excludes them. An empty filter permits all items. The button label names the active mode. Destination priority tries High, then Normal, then Overflow, skipping full or incompatible inventories.

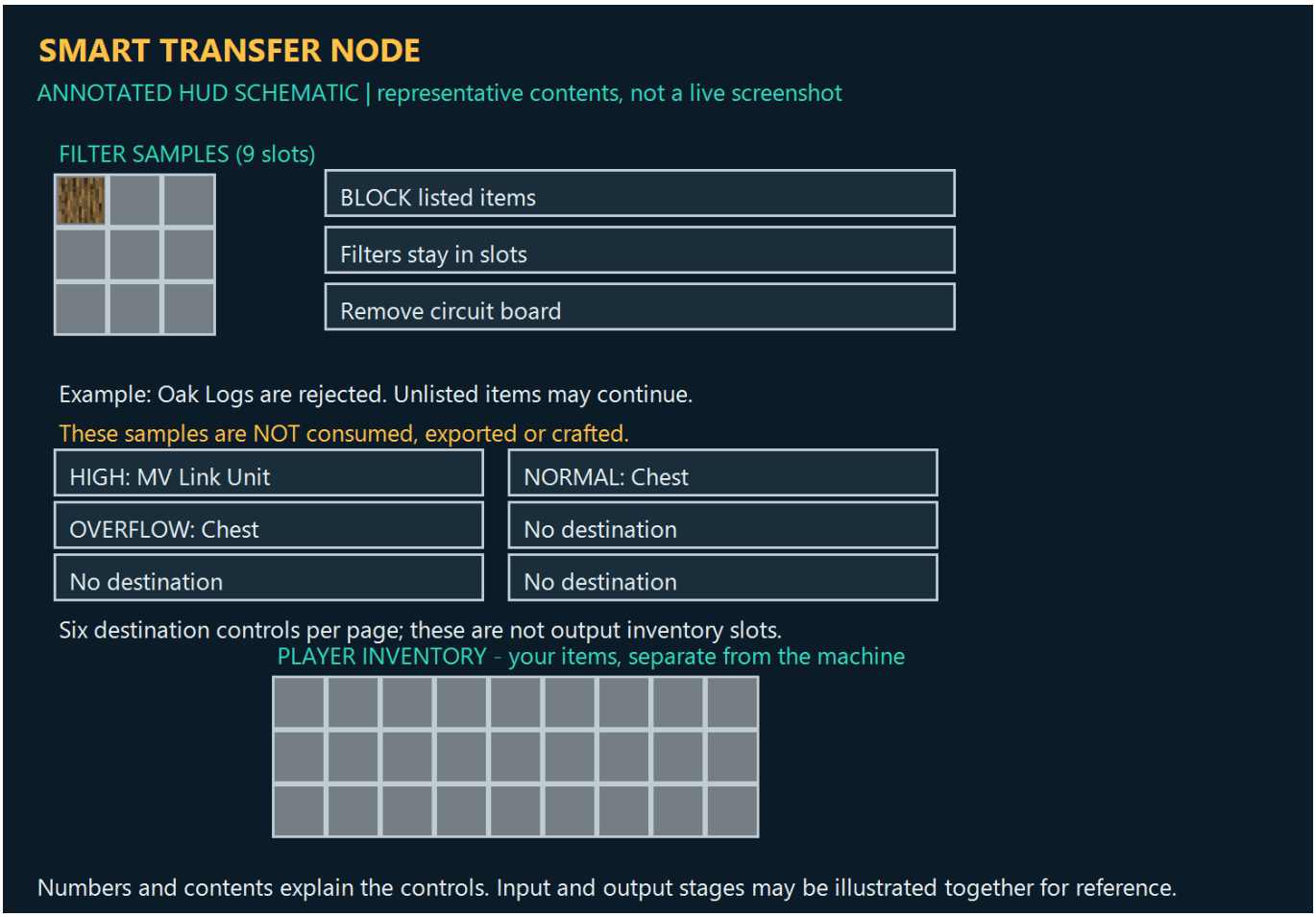
A Black Hole Unit can accept and supply items through the transfer network. Its capacity does not remove the need for a correct source/destination route. A Vacuum Hopper collects loose drops; conveyors and Express Conveyors move physical items. Filter overflow after reserving production inputs.

HUD EXAMPLES: *Smart Transfer Node: filters are not output slots*

BUILD EXAMPLES: *Filtered chest-to-chest transport* | *Bulk storage off a farm line*

JUMP TO: *Machine directory* | *Block recipes* | *Drop tables* | *Essence equipment recipes* | *Tree to Supremium costs*

Smart Transfer Node: filters are not output slots



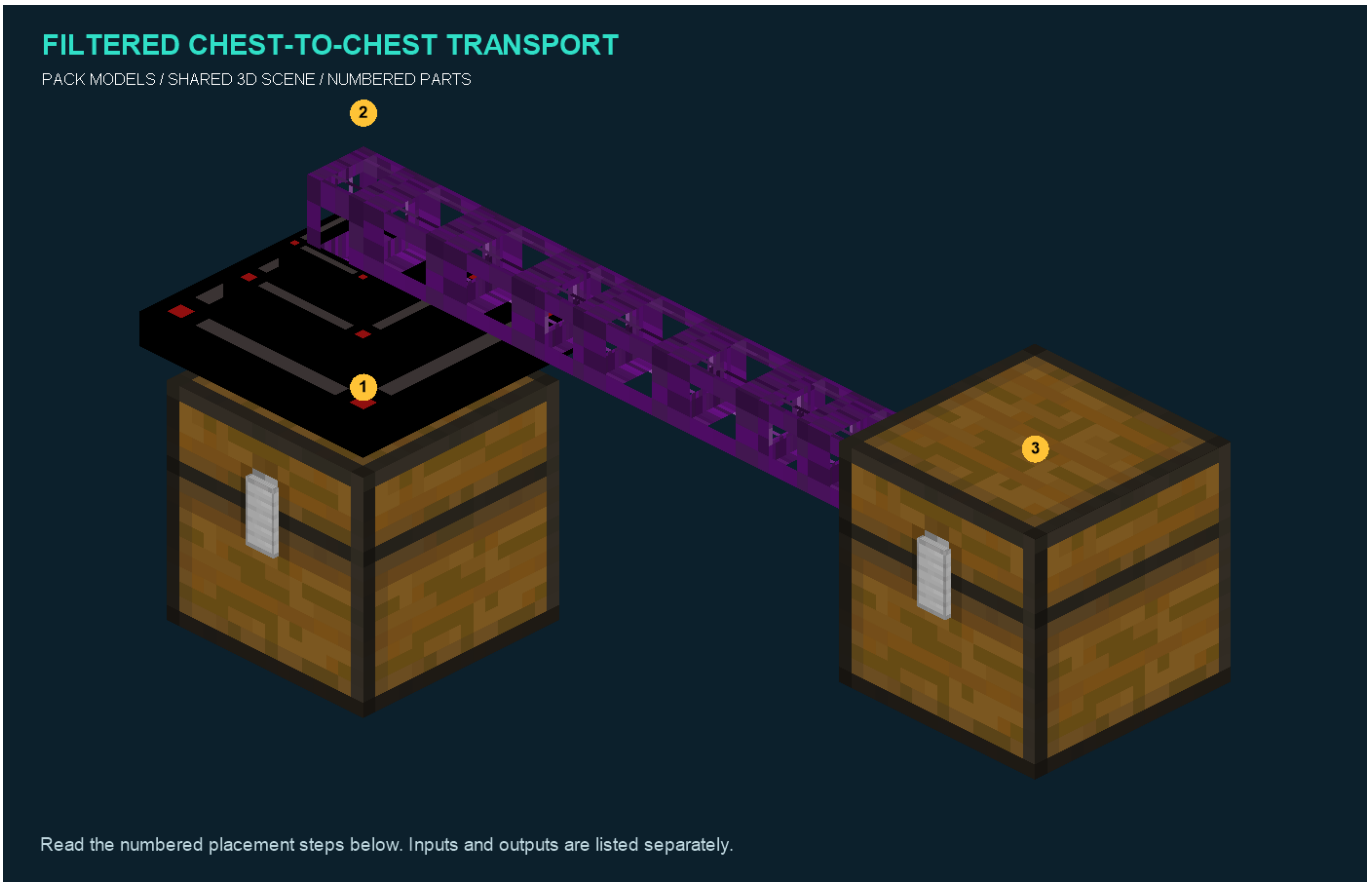
1. The 3 x 3 filter area holds real sample items and keeps them until you remove them. They are not crafting ingredients or exported products.

2. Click the mode control to choose ALLOW or BLOCK. In this example, BLOCK plus an Oak Log sample rejects Oak Logs while unlisted items remain eligible.

3. The six destination controls set routing priorities. HIGH is tried before NORMAL, then OVERFLOW. They are buttons, not inventories. The node still needs a correctly aimed source intake and a connected item route.

BUILD THIS: *Placement example* | *Transfer Nodes, filters, priorities and bulk storage* | *All HUD examples*

Filtered chest-to-chest transport



INPUT: *Source inventory + listed filter items*

OUTPUT: *Only permitted items reach destinations*

1. Place a Transfer Node on top of the source chest; the default intake is down. Confirm the intake side if you rotate or reconfigure it.

2. Connect purple Transfer Pipes to a destination inventory. The second chest is the destination in this example.

3. Install the circuit board to enable smart routing. Put real filter examples in its persistent filter slots.

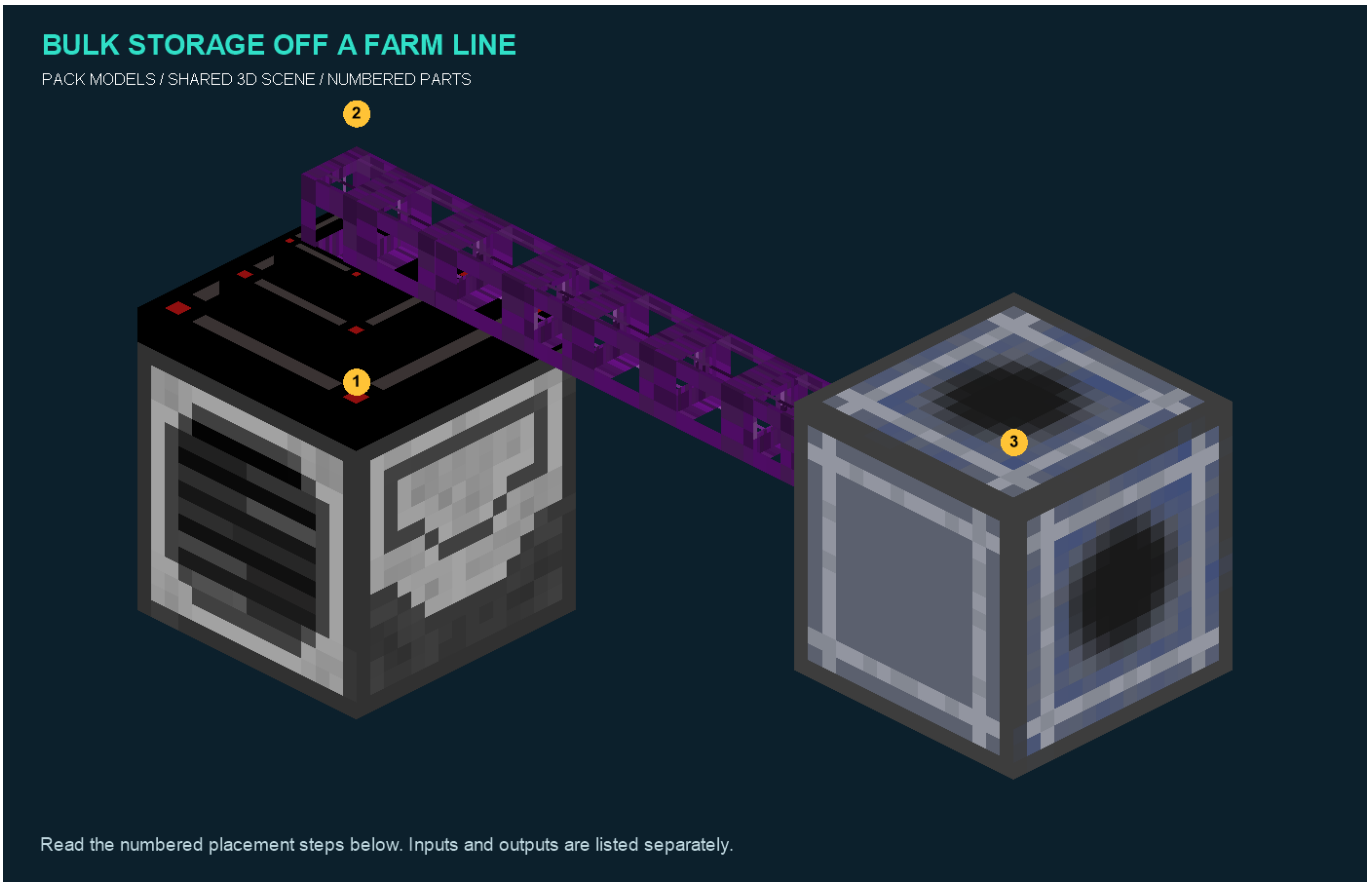
4. ALLOW sends only the listed types; BLOCK rejects the listed types. Test one allowed and one blocked item before turning on the full farm. Set destination priorities if several inventories are connected.

READ MORE: *Transfer Nodes, filters, priorities and bulk storage* | *All build examples*

HUD: *Inputs, outputs and controls*

CRAFT THE BLOCKS: *Transfer Node* | *Transfer Pipe* | *MV Link*

Bulk storage off a farm line



INPUT: *A high-volume single item type*

OUTPUT: *Bulk storage with compatible pipe access*

1. Attach an extraction node to the farm's output inventory. Confirm its intake is aimed at the farm.

2. Run its Transfer Pipe route to the Black Hole Unit. Reserve the unit for the chosen bulk item type.

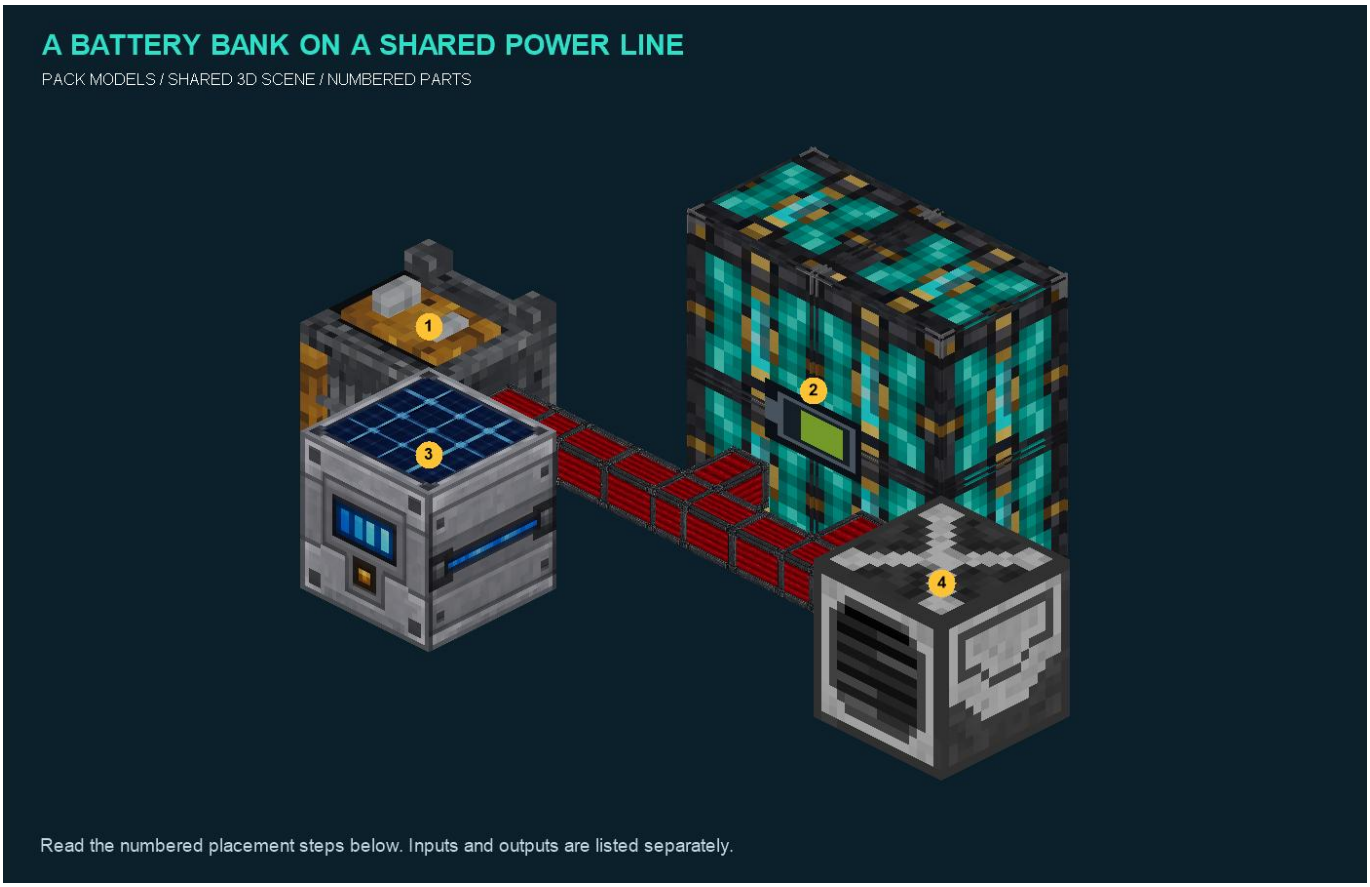
3. Use smart filters when other drops share the source inventory. Route secondary drops to other storage destinations.

4. The Black Hole Unit supports insertion and extraction through compatible transfer routes. Test both directions with a small batch.

READ MORE: *Transfer Nodes, filters, priorities and bulk storage* | *All build examples*

CRAFT THE BLOCKS: *Black Hole Unit* | *Transfer Node*

Power, fluid and item build plans



Power: connect sources, batteries and machines to the same Watt Pipe network. Battery backup is a branch of this shared network, not a required pass-through block.

Water: place the water-producing composter above a Moisture Collector. Place the Fluid Pump against the valid water source and connect its tank route.

Items: attach a Transfer Node to the machine output and connect its Transfer Pipe route to an MV Link connected to an MV system. Test a small batch before extending the network.

Smeltery, alloys and tools

With lava and metals on hand, build the Smeltery to make alloys and cast parts.

1. Build and fuel the Smeltery p. 32

Place the Controller, Drain and Seared Tank in the wall.

2. Pour through the Drain and Faucet p. 34

Place the Casting Table or Basin below the Faucet.

3. Make alloys and reusable casts p. 35

Select the metal to pour and assemble modular tools in the Tool Forge.

4. Compare tool roles and use Vein Miner p. 36

Choose the right tool for the material you want to harvest.

Continue to MV Storage and factory automation p. 37

Open the next chapter when your current setup is ready.

Look up ingredients and recipes p. 75

Find the illustrated crafting grids and machine input amounts.

Build and run a seared smeltery

 *Seared Bricks* Crafting recipe	 *Smeltery Controller* Crafting recipe	 *Smeltery Drain* Crafting recipe	 *Seared Tank* Crafting recipe
 *Seared Faucet* Crafting recipe			

Build a 5 × 5 Seared Brick floor and a perimeter wall above it. The four outside corners of the floor and each wall layer are optional. Leave the center 3 × 3 open. Put the Controller, a Drain and a lava-filled Seared Tank in the wall. Place the Faucet outside the Drain and a Casting Table or Casting Basin immediately below the Faucet.

Put both metals in the Controller input. The smeltery melts one item, then advances to the next eligible slot. For example, copper in slot 1 and tin in slot 9 both get processed while the stacks stay where you placed them. The selected molten metal controls pouring. Choose the alloy recipe separately to mix it.

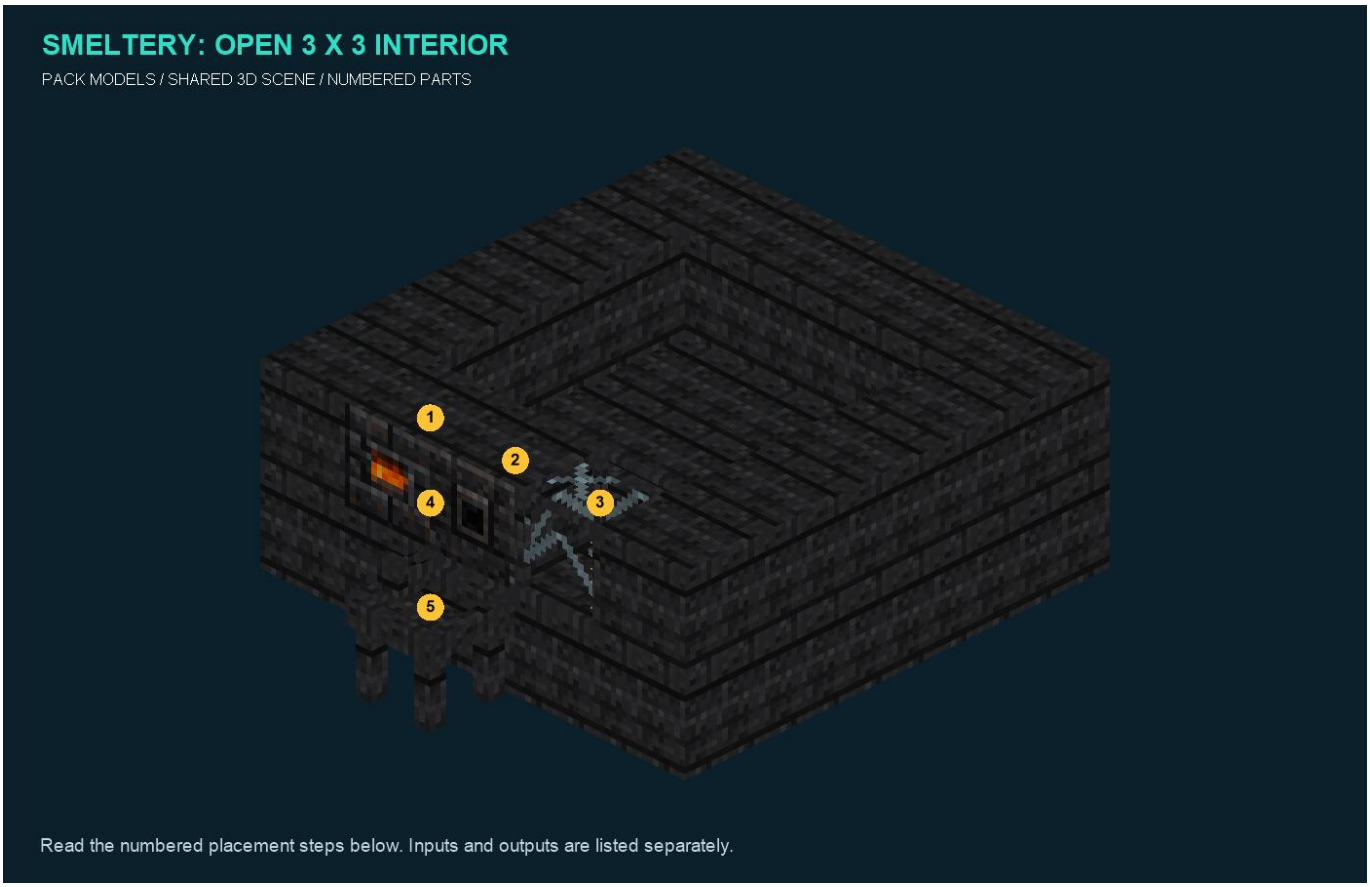
Supply lava through the connected Seared Tank arrangement. Insert supported solids in the controller, wait for melting, and select the metal you intend to inspect or pour. Distinct molten metals remain tracked separately.

Route a Drain/Faucet to the proper casting receiver below it. The Table handles reusable casts and parts; the Basin handles bulk block casting. Let a pour finish and cool before collecting.

Occupied/incompatible receivers stop a pour. Retrieve stored contents before rebuilding the structure. Powder, ingots and raw ore have different molten values; one powder is not one full ingot.

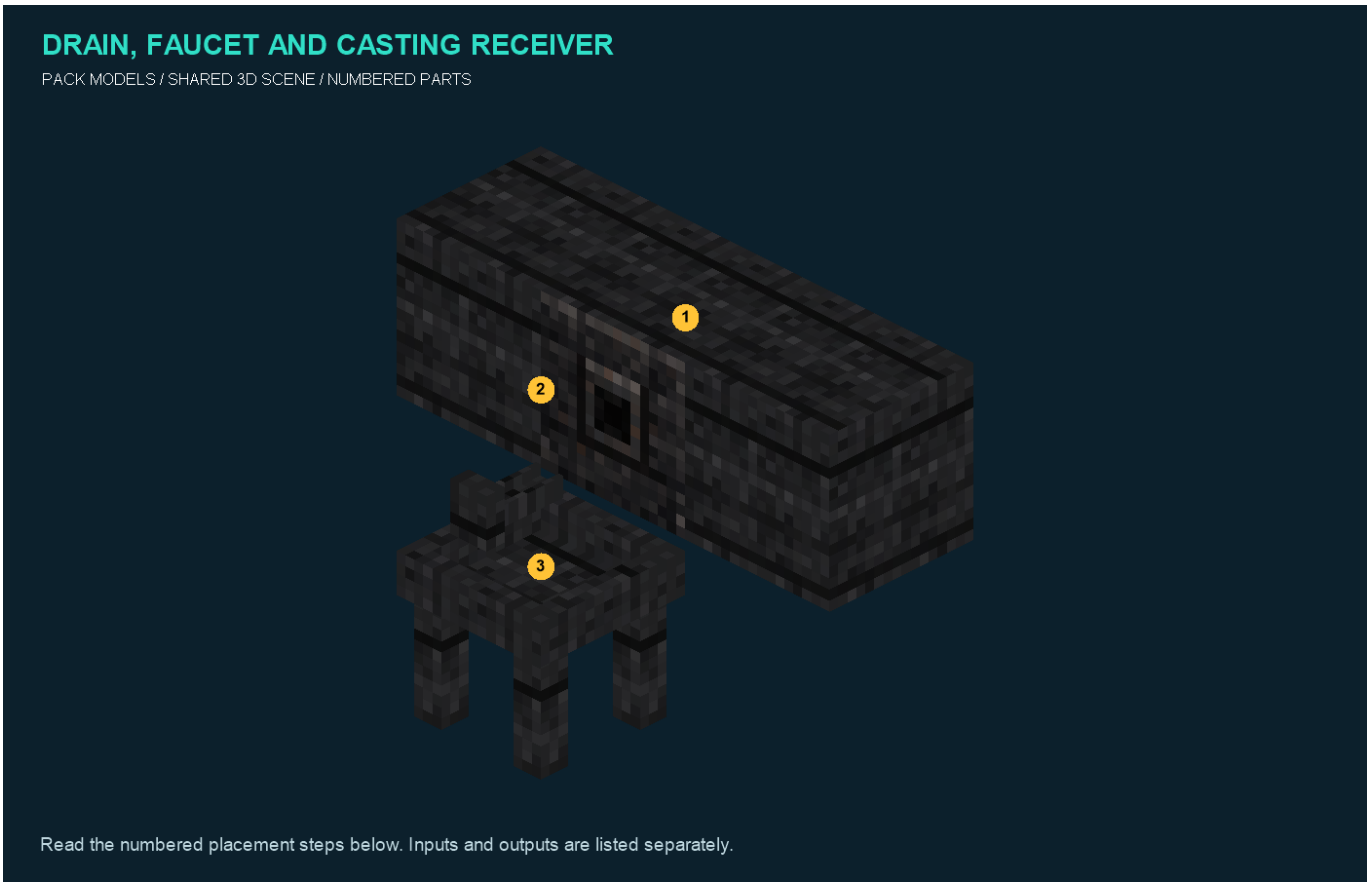
JUMP TO: *Machine directory* | *Block recipes* | *Drop tables* | *Essence equipment recipes* | *Tree to Supremium costs*

Smeltery construction plan



Build a 5 × 5 Seared Brick floor and a perimeter wall above it. The four outside corners of the floor and each wall layer are optional. Leave the center 3 × 3 open. Put the Controller, a Drain and a lava-filled Seared Tank in the wall. Place the Faucet outside the Drain and a Casting Table or Casting Basin immediately below the Faucet.

Drain, faucet and casting receiver



INPUT: *A valid smeltery + selected molten metal + correct cast/receiver*

OUTPUT: *Cooled ingot, part or metal block*

1. Build a valid smeltery first. Put the Drain in its wall and select the intended molten material at the Controller.

2. Place a Faucet outside the Drain, with the Casting Table directly below its pour path. A Casting Basin can replace the Table for bulk recipes; it is not a second outlet.

3. Use the Table for compatible casts and tool parts. Use the Basin for bulk block casting. Supply enough metal for the selected recipe.

4. Wait for the pour and cooling to finish, then collect. Do not replace the receiver while it contains an unfinished pour.

READ MORE: *Alloys, reusable casts and the Tool Forge* | *All build examples*

CRAFT THE BLOCKS: *Smeltery Drain* | *Seared Faucet* | *Casting Table* | *Casting Basin*

Alloys, reusable casts and the Tool Forge

			
Casting Table	*Casting Basin*	*Tool Forge*	*Steel Ingot*
Crafting recipe	Crafting recipe	Crafting recipe	

Make a reusable gold part cast by placing its stone pattern on the Table and pouring the listed Gold amount. Standard part-cast creation uses two Gold Ingots. Pour the desired metal into the cast, cool it and collect the part; the gold cast remains.

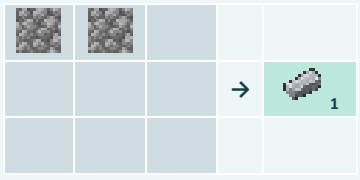
Alloy batches: 3 Copper + 1 Tin -> 4 Bronze; 1 Gold + 1 Silver -> 2 Electrum; 2 Iron + 1 Nickel -> 3 Invar. Steel uses 1 molten Iron plus 1 Coal and 1 Blaze Powder from solid input -> 1 Steel.

Select the desired alloy and enable Auto alloy. When its molten ingredients and any solid catalysts are ready, the controller makes the alloy. Select the finished metal to pour it into a suitable cast.

Cast a Head, Rod and Binding and assemble the supported modular pickaxe/hammer in the Tool Forge. Materials may differ between parts. The atlas includes the supported combinations.

Forge modifiers are not machine boards. Reinforcement Plate adds durability; Efficiency Module applies Efficiency II; Fortune Module applies Fortune I to a compatible pickaxe. Hold a modifier, sneak-use the Forge and select a tool. Restrictions on incompatible/equal-or-higher enchantments still apply.

Stone Ingot Pattern



Use the stone pattern on the Casting Table, then pour two ingots? worth of molten Gold to make a reusable Gold Ingot Cast.

BUILD EXAMPLES: *Drain, faucet and casting receiver*

JUMP TO: *Machine directory* | *Block recipes* | *Drop tables* | *Essence equipment recipes* | *Tree to Supremium costs*

Tool roles, metal armor and vein mining

 *Supremium Pickaxe*	 *Supremium Crook*	 *Supremium Hammer*	 *Supremium Paxel*

Pickaxes mine stone/ores; axes mine wood and strip logs; shovels dig soil and make paths; hoes till suitable soil. Paxels combine pickaxe/axe/shovel mining and applicable stripping/path actions.

Crooks use leaf bonus rolls. Hammers crush supported blocks. Sickles and scythes harvest nearby mature crops, with the scythe covering the larger area. Swords provide combat damage. Tool wear and enchantment compatibility depend on the tool.

Sneak-mine with a supported pickaxe, axe, shovel, paxel, hammer or crook to mine connected matching blocks. Cap: 64 total, or earlier when a finite tool breaks. Release Sneak or switch tools to stop queued work.

Crook leaf rolls and hammer conversions happen per block in a vein. Unbreakable Supremium tools still obey the 64-block cap. Fortune is not a universal multiplier for every scripted crushing or machine output.

Ordinary metal armor uses normal crafting-table armor shapes with its own ingots. Modular tools use the Forge. Essence gear uses a different tier-upgrade route in the next chapter.

JUMP TO: *Machine directory* / *Block recipes* / *Drop tables* / *Essence equipment recipes* / *Tree to Supremium costs*

MV Storage and factory automation

Bring your production lines into one network, then automate a repeatable recipe.

1. Set up MV Storage p. 38
Power the terminal, deposit materials and try a withdrawal.

2. Connect a Universal Crafter p. 41
Save a recipe and return its output to the MV network.

3. Add tree and crop automation p. 42
Pair a Planter and Harvester. Use an Activator or Sprinkler where needed.

4. Add machine upgrades p. 48
Choose upgrades after testing your line at base speed.

Continue to Polymer and advanced machines p. 49
Open the next chapter when your current setup is ready.

Look up ingredients and recipes p. 75
Find the illustrated crafting grids and machine input amounts.

MV Storage, in-screen crafting and wireless access

			
MV System	*MV Link*	*Wireless MV Controller*	
Crafting recipe	Crafting recipe		

Build and power an MV System. Open its custom storage HUD, search stored items with Find, and use its deposit/withdrawal controls. Selected withdrawal amount is separate from the total stored quantity.

The 3x3 manual crafting pattern is inside MV Storage. Place a pattern and use its output to craft from stored ingredients. The removed recipe-search controls and separate MV Pattern block are not required.

MV Links connect item networks to storage. Choose Transfer or Pull as needed. Filters, pauses and reach still apply. A linked automatic crafter may need both input and output directions permitted.

Sneak-use the Wireless MV Remote on the desired MV to bind it, then use the remote normally to open that MV HUD. MV, storage and power must be loaded in the same dimension. The remote does not force-load chunks. The retired Wireless MV Link access point is not required. Keep the wired MV Link Unit for automated item transfers.

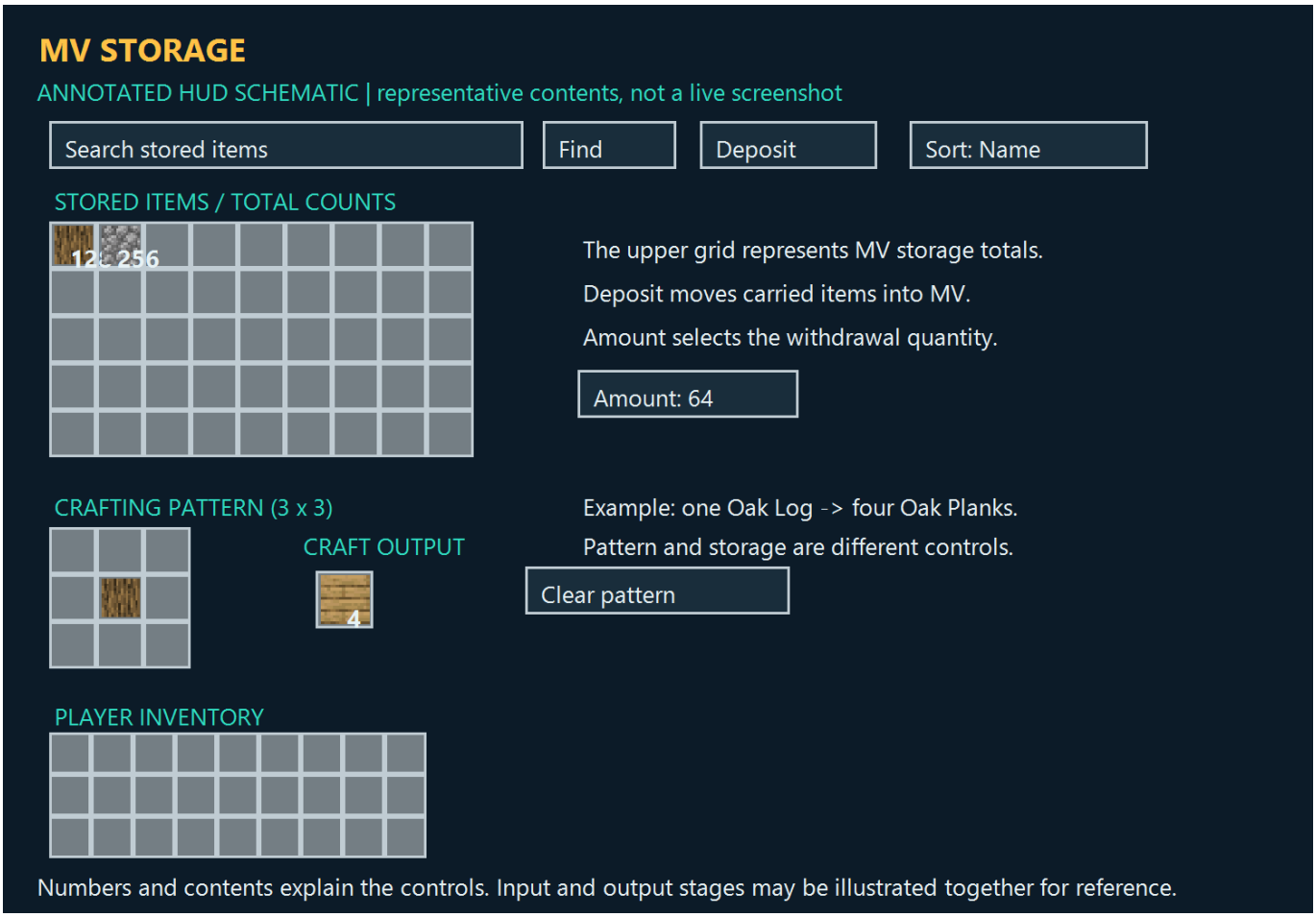
Empty valuable storage and internal inventories before dismantling. Do not delete hidden inventory hosts to force a machine to break; they can hold real items.

HUD EXAMPLES: *MV Storage: storage totals, recipe pattern and player inventory*

BUILD EXAMPLES: *MV Storage with a handheld remote*

JUMP TO: *Machine directory* | *Block recipes* | *Drop tables* | *Essence equipment recipes* | *Tree to Supremium costs*

MV Storage: storage totals, recipe pattern and player inventory



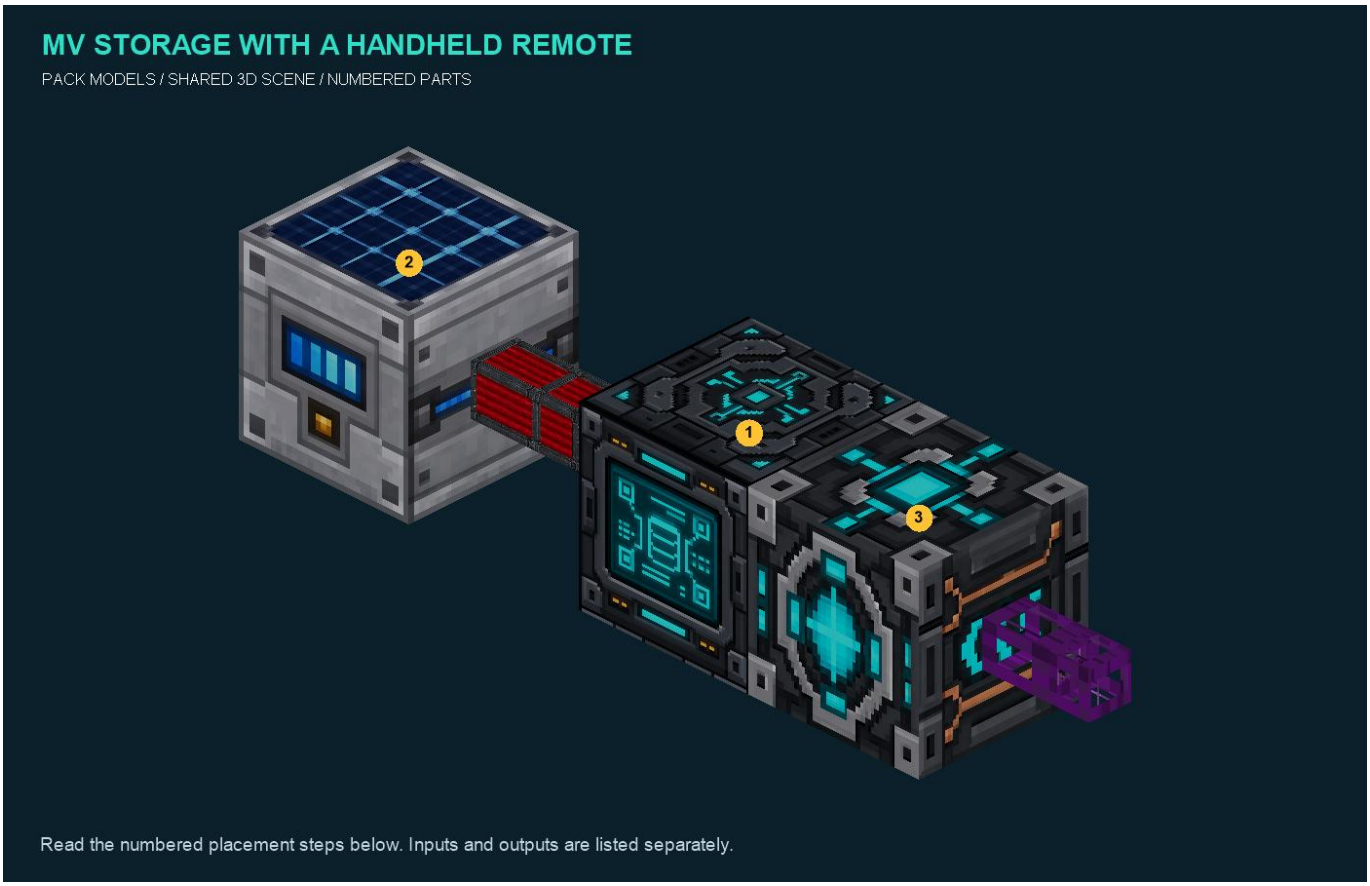
1. Search and Find act on stored items. Deposit sends carried items into MV. The upper grid shows storage contents and totals; the player inventory at the bottom is separate.

2. Amount selects the withdrawal quantity. Use the integrated 3 x 3 crafting pattern to choose a recipe and the output control to craft when the required ingredients are available.

3. The example illustrates Oak Log to Oak Planks. Clear empties the pattern selection. The handheld remote opens this same storage interface after binding to the powered MV system.

BUILD THIS: *Placement example* | *MV Storage, in-screen crafting and wireless access* | *All HUD examples*

MV Storage with a handheld remote



INPUT: *Powered MV Storage + bound handheld remote*

OUTPUT: *Remote storage and integrated crafting access*

1. Build and power MV Storage. The remote opening check requires at least 100 W available at the bound MV system.

2. Sneak-use the handheld Wireless MV Remote directly on MV Storage to bind it. Release Sneak, then use the remote to open storage.

3. Keep the target loaded in the same dimension. The retired Wireless MV Link access point is not required.

4. For automatic pipe insertion/extraction, keep the wired MV Link Unit. Remote access does not replace item-transfer plumbing.

READ MORE: *MV Storage, in-screen crafting and wireless access* | *All build examples*

HUD: *Inputs, outputs and controls*

CRAFT THE BLOCKS: *MV System* | *MV Link*

Universal Crafter and MV automation

 *Universal Crafter* Crafting recipe	 *Crafting Planner*		

The Universal Crafter uses a 3 x 3 sample recipe. Drag ingredients into the grid, then return the remaining cursor stack to your inventory before using its controls. Samples are returned; recipes consume ingredients from the connected MV storage.

Click Add automation to save and start a valid recipe immediately. No preliminary manual craft is needed. The sample grid clears after adding. Up to ten saved recipe boxes appear in Recipe / Queue. Click a box to stop, resume, remove or load it.

Mode: Auto crafts continuously while ingredients, MV room and power are available. Optional stock limits wait when their target is reached and resume after withdrawals. Crafted results return directly to MV.

Connect the Crafter directly through its network to MV Storage, or use an enabled bound MV Link through Transfer Pipes. Supply the Crafter with Watt power; remote MV input and output also require link transfer power. Missing materials or power cause a recoverable wait.

MV Storage keeps ordinary stackable items as counts. Use Find to filter stored names, including custom items. Shift-click deposits go into storage; deliberate grid placement sets the crafting recipe. MV Link has the same name-matching Find logic. Show all items resets its search; clicking an item selects its routing filter.

JUMP TO: *Machine directory* | *Block recipes* | *Drop tables* | *Essence equipment recipes* | *Tree to Supremium costs*

Planters, harvesters, sprinklers and Activators

			
Planter	*Harvester*	*Automated Sprinkler*	*Activator*
Crafting recipe	Crafting recipe	Crafting recipe	Crafting recipe

Provide a Planter with seeds/saplings, suitable soil, free space and power. Place the powered Harvester so its area covers mature crops/trees. Export its internal output with a transfer node. Tree harvesting has built-in crook behavior.

Use the Automated Sprinkler and its supported water/operation arrangement for growth assistance. Watering cans and bone meal provide manual help. Growing faster and harvesting faster are different problems.

Configure an Activator deliberately: face, aim, use/break click, sneaking, redstone and order. Tool durability is spent on the first usable tool one at a time, not randomly spread across tools.

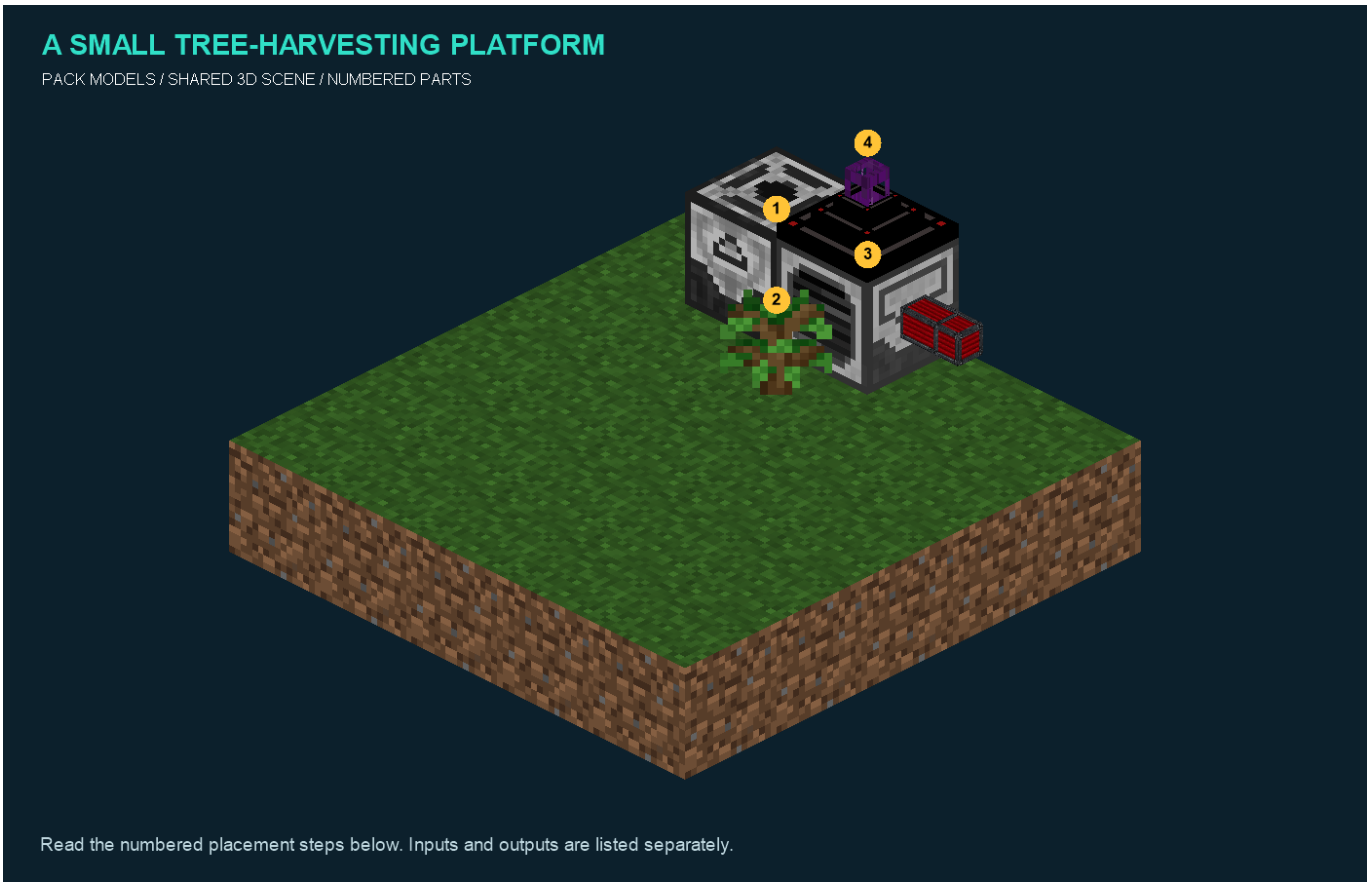
Ore crops have four harvest tiers. Inferium crops have five and always grow Inferium, with larger per-harvest yields. Check power and output capacity before adding speed boards.

HUD EXAMPLES: *Activator: input tools, world action and returned output* | *Harvester: collected output and machine upgrades*

BUILD EXAMPLES: *A small tree-harvesting platform* | *Crop growth with a water-fed sprinkler* | *An Activator aimed at a sapling*

JUMP TO: *Machine directory* | *Block recipes* | *Drop tables* | *Essence equipment recipes* | *Tree to Supremium costs*

A small tree-harvesting platform



INPUT: *Valid soil, supported planting inputs and power*

OUTPUT: *Harvested wood, leaves and tree drops*

1. Build a soil platform with open space above and around the tree. The drawing is a small starter example; leave clearance for the species you grow.

2. Place the Planter next to the Harvester and supply supported seeds or saplings. This example keeps the tree one block in front of the Harvester, inside its unmodified range.

3. Place the powered Harvester within working range. Its default working area and upgrade range must cover the tree.

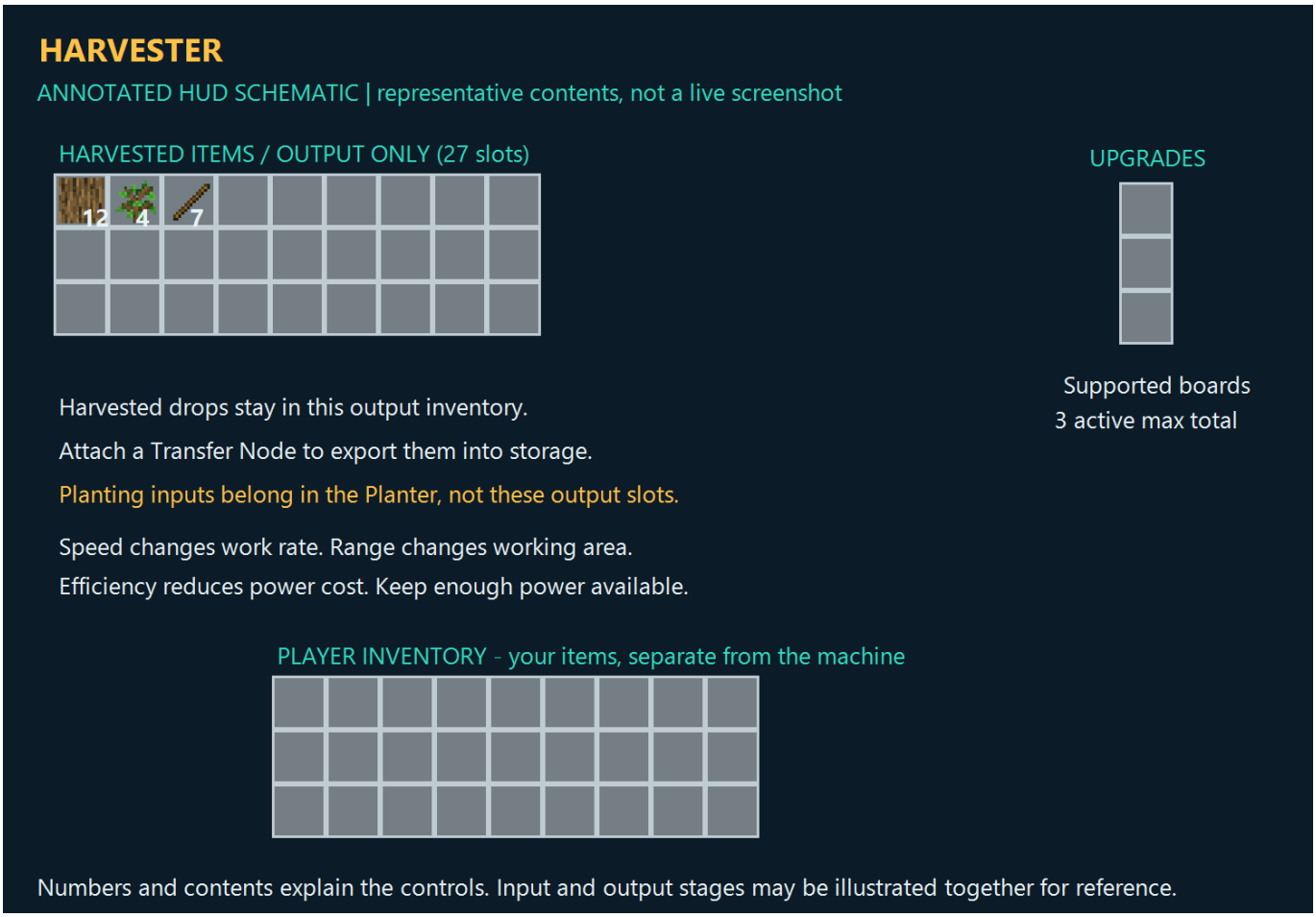
4. Attach an extraction node to the Harvester output and export drops. Keep a separate seed return supply for the Planter.

READ MORE: *Planters, harvesters, sprinklers and Activators* | *All build examples*

HUD: *Inputs, outputs and controls*

CRAFT THE BLOCKS: *Planter* | *Harvester* | *Transfer Node*

Harvester: collected output and machine upgrades



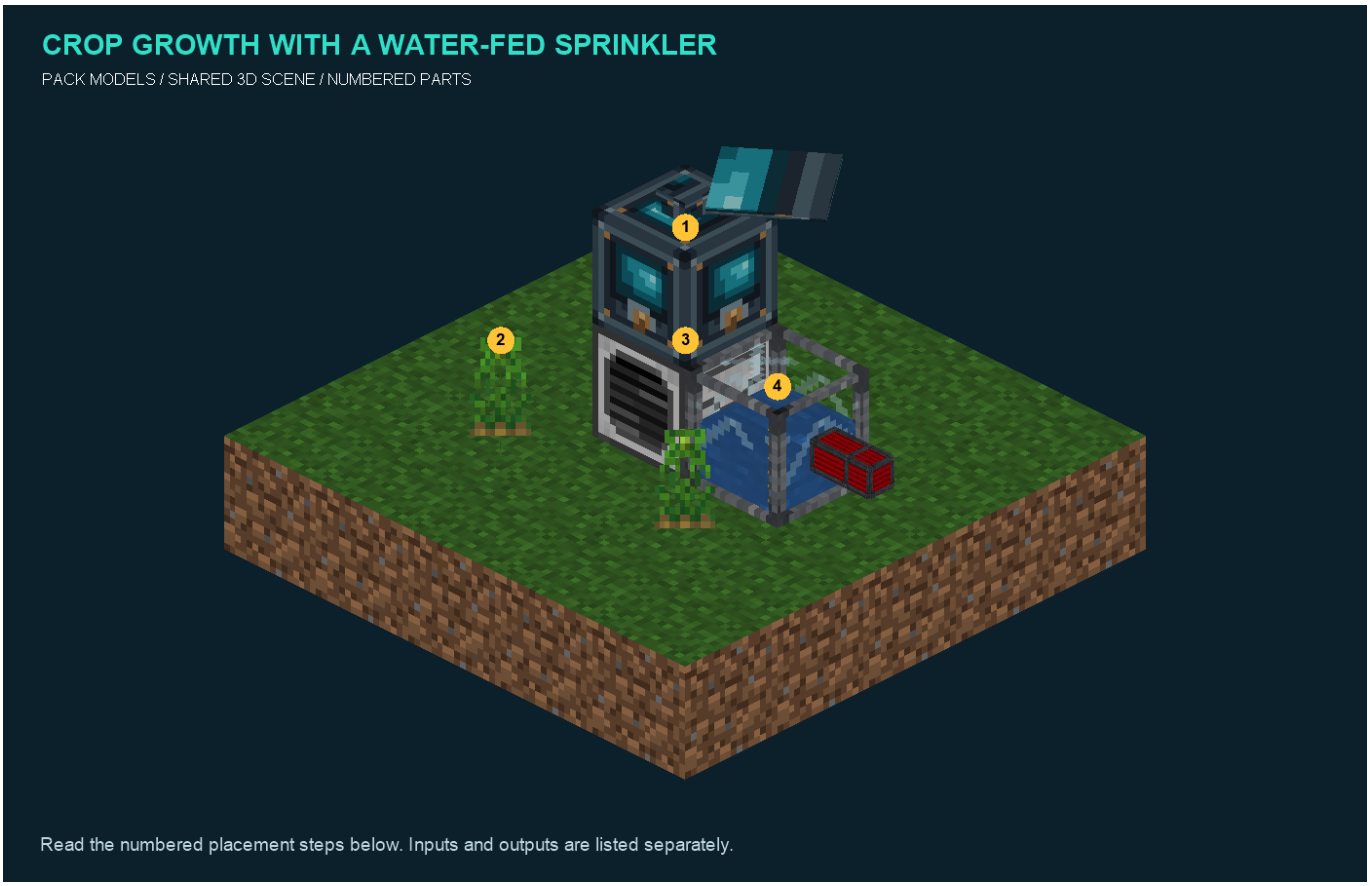
1. The Harvester grid is OUTPUT storage for collected drops. Do not treat it as a planting-input inventory. Seeds and saplings for planting belong in the Planter.

2. Attach a Transfer Node aimed at the Harvester to export stored drops. A full output can stop useful work; check extraction before adding more Speed upgrades.

3. Speed changes work rate, Range changes the working area, and Efficiency reduces the power cost. The supported upgrade slots share a maximum of three active upgrades total.

BUILD THIS: *Placement example* | *Planters, harvesters, sprinklers and Activators* | *All HUD examples*

Crop growth with a water-fed sprinkler



INPUT: *Growing crops + water + power*

OUTPUT: *Faster growth; separate machine harvest*

1. Build the soil platform and plant supported crops. Center the Harvester and place the sprinkler one block above it, so the surrounding plants are within both machines' unmodified horizontal range.

2. Keep growing plants within the sprinkler's 5 x 5 candidate area, including its supported height band.

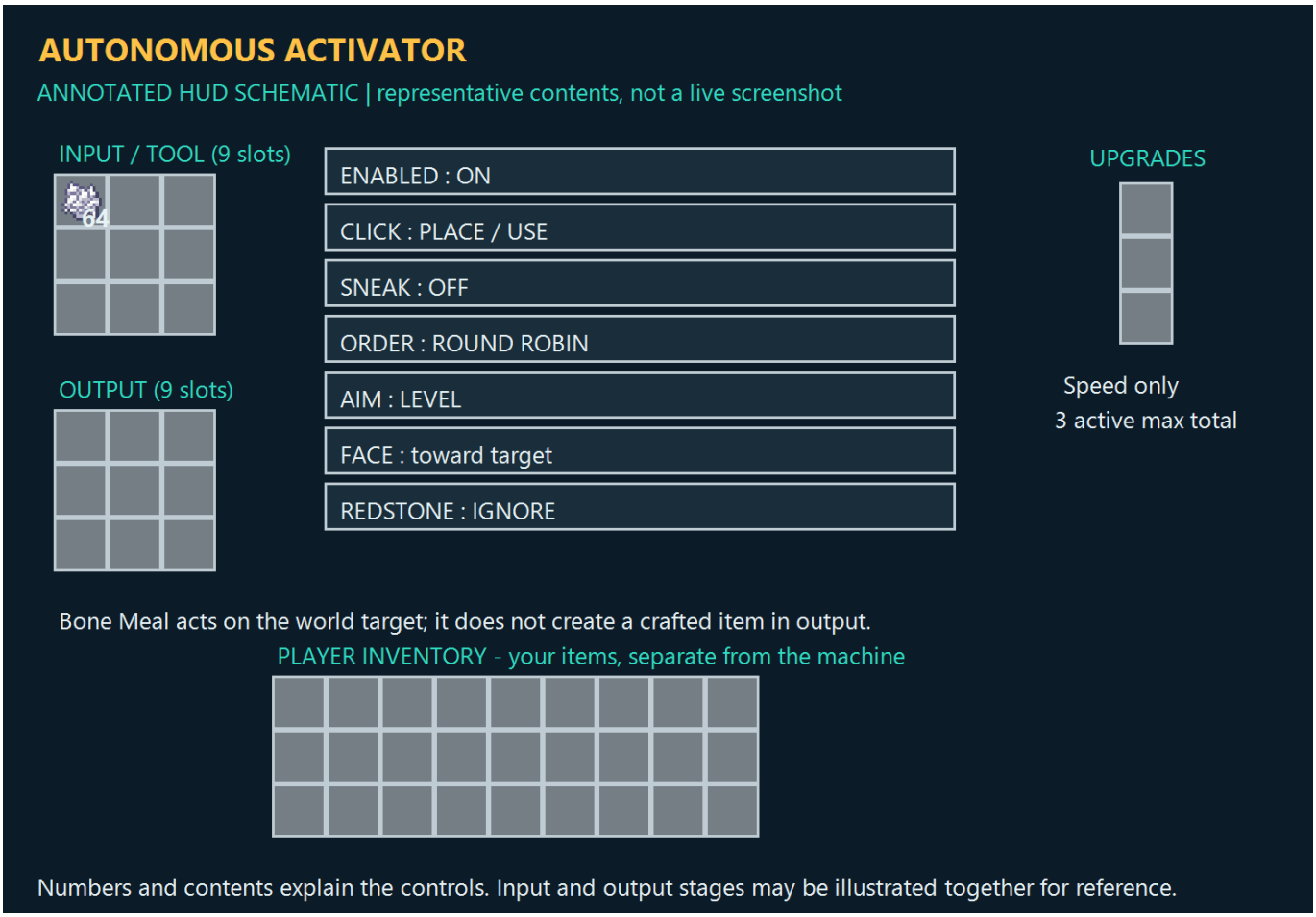
3. Supply water and power. A working sprinkler pulse needs 25 mB water and 10 W before efficiency changes; empty supplies pause it.

4. The sprinkler grows crops; a separate Harvester collects mature crops and a Planter replenishes planting inputs.

READ MORE: *Planters, harvesters, sprinklers and Activators* | *All build examples*

CRAFT THE BLOCKS: *Automated Sprinkler* | *Harvester* | *Planter*

Activator: input tools, world action and returned output



1. Put Bone Meal, tools or other supported usable items in INPUT / TOOL. Enable the machine, choose the click mode, and set its face and aim toward the target.

2. The OUTPUT slots hold any returned or produced items from supported actions. Bone Meal growth changes the target in the world; a crafted item is not expected in OUTPUT.

3. Tool durability is spent one tool at a time in slot order. This machine supports Speed upgrades only. Removing boards restores the normal action interval.

BUILD THIS: *Placement example* | *Planters, harvesters, sprinklers and Activators* | *All HUD examples*

An Activator aimed at a sapling



INPUT: *A valid target + usable input/tool*

OUTPUT: *Repeated configured tool/use actions*

1. Place the target where the Activator is aimed. The drawing only shows relative positions; set its actual working face in the HUD.

2. Load Bone Meal or the supported tool into input/tool slots. Choose Place/Use, enable the machine, and check its face and aim.

3. It spends tools in first-slot order and stops when the current tool wears out. Keep output storage clear.

4. Use only supported upgrade slots. Speed boards increase action rate, with three active upgrades total; removing them restores the base rate.

READ MORE: *Planters, harvesters, sprinklers and Activators* | *All build examples*

HUD: *Inputs, outputs and controls*

CRAFT THE BLOCKS: *Activator*

Upgrade slots: speed, range and efficiency

 *Speed Upgrade*	 *Range Upgrade*	 *Efficiency Upgrade*	

Put real boards into compatible HUD slots. Silhouettes show supported kinds. Activators accept Speed only; Harvesters/Grinders accept Speed, Range and Efficiency; processors accept Speed and Efficiency.

The active cap is THREE boards TOTAL across the machine, not three per type. Stack duplicates in one slot or spread them across slots. Unsupported boards are rejected and excess boards are returned/ejected.

Speed produces x1 base, then x2/x3/x4 with one/two/three boards. Faster processor work increases its working power use. Each Range board adds two blocks to a supported working radius.

Each Efficiency board multiplies applicable working power cost by 0.8: 20%, 36% or about 49% saved before rounding. It does not reduce ingredients. Removing boards restores base behavior. Read live HUD stats and remove boards before breaking a protected machine.

JUMP TO: *Machine directory* | *Block recipes* | *Drop tables* | *Essence equipment recipes* | *Tree to Supremium costs*

Polymer and advanced machines

Turn your farm output into the parts and alloys that feed the endgame.

1. Produce Polymer p. 50

Feed crops or saplings, charcoal and water into the Pressure Reactor.

2. Assemble frames and circuits p. 53

Use the Precision Assembler to build components for later machines.

3. Make Crystalline Alloy p. 105

Feed diamonds, Steel, Nickel, Polymer and water to the Material Infuser.

4. Use the Factory Core p. 106

Check the machine recipes before connecting permanent supply lines.

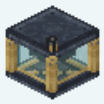
Continue to Essence gear and travel p. 55

Open the next chapter when your current setup is ready.

Look up ingredients and recipes p. 75

Find the illustrated crafting grids and machine input amounts.

Processors and the Polymer factory

			
Pulverizer	*Dehydrator*	*Glacial Chiller*	*Pressure Reactor*
Crafting recipe	Crafting recipe	Crafting recipe	Crafting recipe

Put listed ingredients into INPUT, provide power/fluid as required, unpause and keep OUTPUT clear. Use the machine named on a card; a 3x3-looking inventory is not automatically a crafting bench.

Pulverizer makes its listed powders and crushed products. Dehydrator handles dry processing, Glacial Chiller handles cooling and water, and Magma Furnace provides its smelting route. Exact accepted inputs/outputs appear in the atlas.

Pressure Reactor batches use 16 accepted biomass items, Charcoal, water and power to make Polymer. Supported biomass includes its configured crops and saplings, not removed mulch. Mix accepted crop types when making a batch.

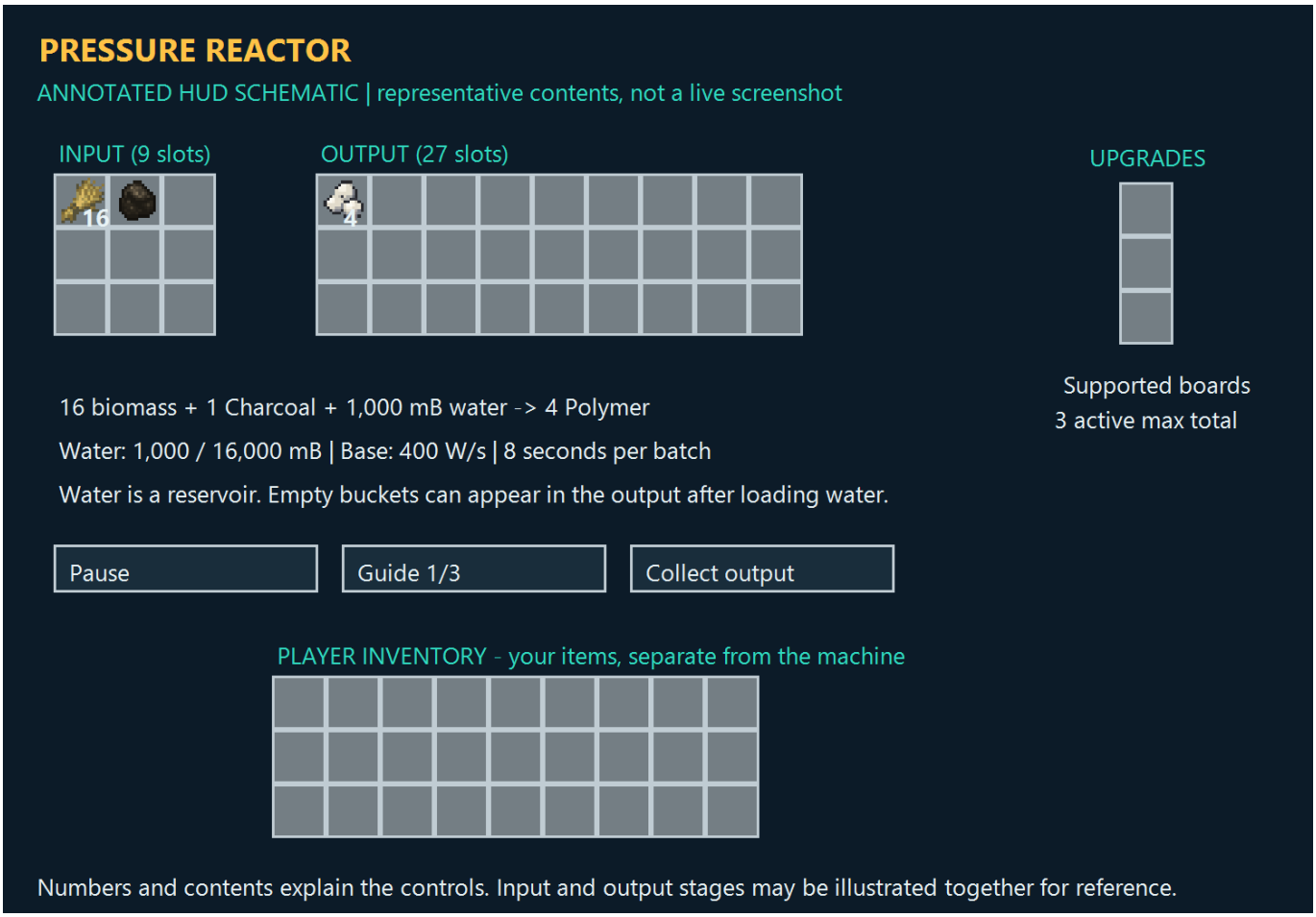
Polymer leads to Circuit Boards, upgrades and advanced factory components. Feed input and export finished output through nodes. Returned buckets need output space too. Verify a basic working line before adding speed.

HUD EXAMPLES: *Pressure Reactor: inputs, water and Polymer output*

BUILD EXAMPLES: *A first Polymer production cell*

JUMP TO: *Machine directory* | *Block recipes* | *Drop tables* | *Essence equipment recipes* | *Tree to Supremium costs*

Pressure Reactor: inputs, water and Polymer output



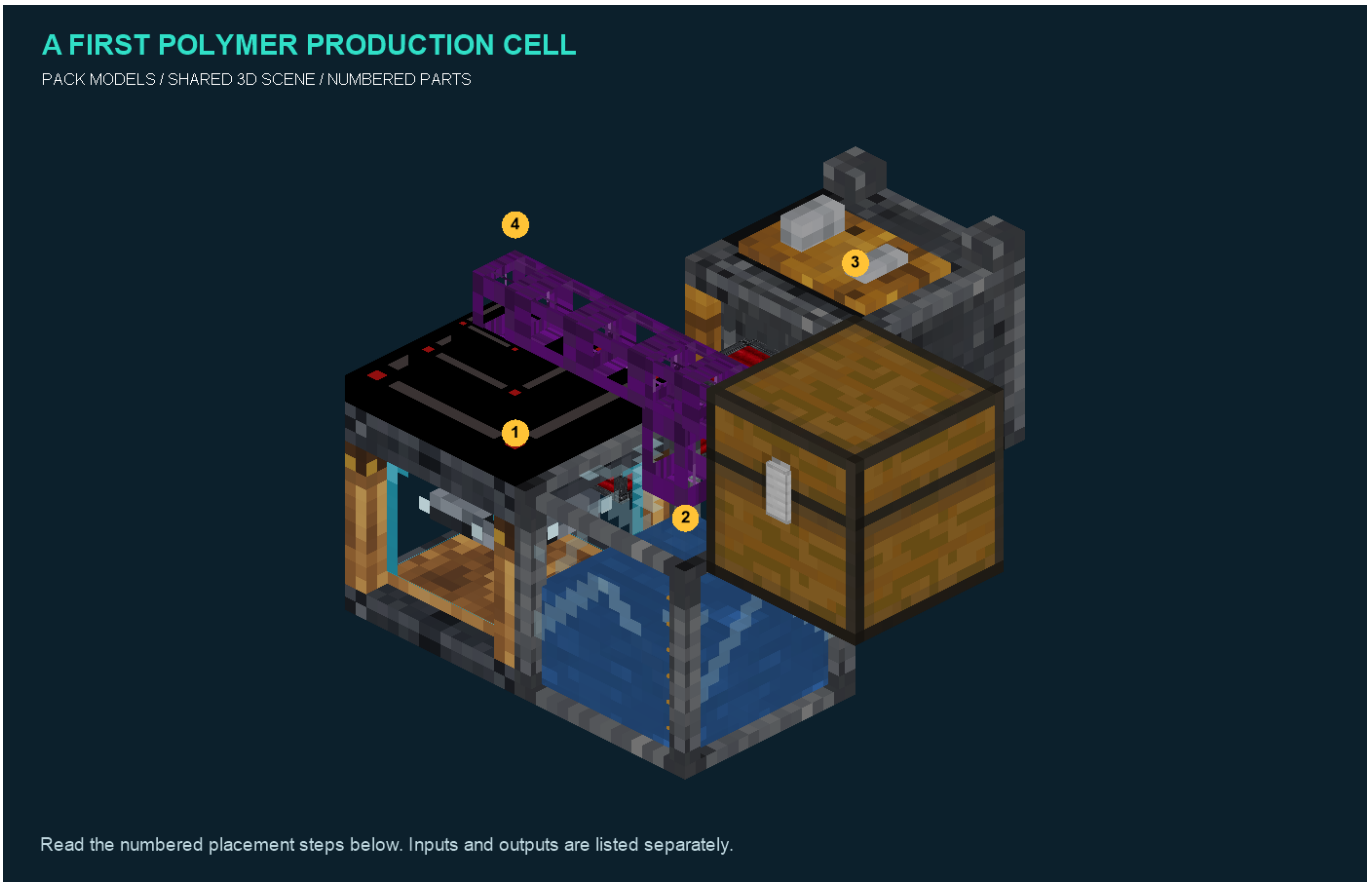
1. Put biomass and Charcoal in the nine INPUT slots. The example uses 16 Wheat and 1 Charcoal. Supply 1,000 mB water in the machine reservoir.

2. The 27 OUTPUT slots hold finished Polymer and returned empty buckets. The player inventory below is separate storage; items left there are not automatically processor inputs.

3. Pause stops the job. Guide cycles recipe help. Collect output moves finished items to your inventory. Use only supported boards in the upgrade slots, with three active upgrades total.

BUILD THIS: *Placement example* / *Processors and the Polymer factory* / *All HUD examples*

A first Polymer production cell



INPUT: *16 biomass + 1 Charcoal + 1,000 mB water + power*

OUTPUT: *4 Polymer per completed batch*

1. Place the Pressure Reactor and connect a sufficient power source. Its unmodified working cost is 400 W per working second.

2. Supply 16 accepted crop/sapling biomass items, one Charcoal and 1,000 mB water. An adjacent connected water-tank route can refill the machine.

3. A base batch needs eight progress steps and makes four Polymer. Speed and efficiency upgrades change work rate and power use.

4. Provide output extraction. Use Polymer for circuit boards and the advanced factory chain instead of stockpiling crops without an outlet.

READ MORE: *Processors and the Polymer factory* | *All build examples*

HUD: *Inputs, outputs and controls*

CRAFT THE BLOCKS: *Pressure Reactor* | *Precision Assembler* | *Material Infuser* | *Factory Core*

Advanced assembly, infusion and the Factory Core

			
Precision Assembler	*Material Infuser*	*Factory Core*	
Crafting recipe	Crafting recipe	Crafting recipe	

Precision Assembler makes the listed Reinforced Frames, Advanced Circuits, Conductive Cores, Growth Essence and modifiers. Supply ingredients and Watts; the first complete matching recipe is processed.

Material Infuser needs solid ingredients plus water and power for Crystalline Alloy or Enriched Essence. Its fluid supply and returned containers are part of your production line.

Factory Core uses late components and substantial power for its listed high-throughput recipes. Build crop/Charcoal production, Polymer, circuits, assembly components and infusion materials in stages before scaling the core.

Keep buffers between stages. Use High-priority input destinations before bulk overflow. A blocked output or missing machine-made component can stop a large factory even when most materials are abundant.

JUMP TO: *Machine directory* | *Block recipes* | *Drop tables* | *Essence equipment recipes* | *Tree to Supremium costs*

Scale the factory and finish progression

			
Factory Core	*Black Hole Unit*	*Battery Bank*	
Crafting recipe	Crafting recipe	Crafting recipe	

Build stable loops in stages. Start with trees, Cobblestone and sieve outputs. Make early metals and power, then build the smeltery. Make Steel and Netherite before attempting MV Storage. Expand with Polymer, automated crops and advanced processing.

Use the optional achievements checklist to track milestones. Keep a working supply of intermediate ingredients for each new production tier.

Buffer intermediate products and prioritize active production before bulk overflow. Monitor whether a bottleneck is farm growth, power, a wrong filter, loaded chunks or full output.

Batteries protect outages but do not replace sufficient primary generation. Increase sources, input and extraction together. Fewer loose item entities and purposeful layouts can perform better than many redundant networks.

JUMP TO: *Machine directory* / *Block recipes* / *Drop tables* / *Essence equipment recipes* / *Tree to Supremium costs*

Essence gear and travel

Use the farm to grow essence. Upgrade gear through all five tiers and choose how to travel.

1. Plant essence and upgrade your gear p. 56
Use the Essence Tinkering Table to move through the tiers.

2. Plan the path to Supremium p. 60
Estimate seed and essence costs before scaling the farm.

3. Install compatible charms p. 62
Choose perks that match your armor pieces or tools.

4. Build a flying ship p. 64
Try a small isolated deck before assembling a larger ship.

Continue to Void mining and late-game power p. 65
Open the next chapter when your current setup is ready.

Look up ingredients and recipes p. 75
Find the illustrated crafting grids and machine input amounts.

Essence armor: the complete five-tier progression

			
Inferium Essence	*Infusion Crystal*	*Essence Tinkering Table*	*Supremium Chestplate*
		Crafting recipe	

Sieve Dirt for Inferium Essence and Sand for Prosperity Shards. Make an Essence Crafting Seed with one Wheat Seed and four Prosperity Shards in a cross. Surround that seed with four Inferium Essence in the same cross. Plant the resulting Inferium Essence Seeds and interact with mature plants to harvest without removing the seed.

Craft the Infusion Crystal at a Crafting Table using the pictured recipe in the Essence appendix. Then put the finished crystal into the Essence Tinkering Table.

Craft the Essence Tinkering Table at a Crafting Table. Follow its pictured recipe. Place and open it. It runs without Watt power.

Insert four lower essence plus the already-crafted crystal into its input slots. Unpause and wait four seconds. It batches up to 64 output and retains the crystal, spending one use per infused output. Chain: Inferium -> Prudentium -> Intermedium -> Superium -> Supremium, always 4:1.

The crystal lasts 1,000 infused outputs. A Master Crystal has unlimited uses: normal crystal in the center, four Prosperity Ingots on the edges and four Supremium Essence in the corners.

Craft ingots at a normal table: Iron Ingot + four Prosperity Shards -> Prosperity Ingot. Prosperity Ingot + four Inferium Essence -> Inferium Ingot. Each later ingot uses one previous-tier ingot + four matching next-tier essence. All are shapeless.

Craft initial Inferium armor in normal armor shapes: helmet 5, chestplate 8, leggings 7, boots 4; 24 Inferium Ingots total. Upgrade one previous armor piece + four NEXT-tier ingots in the Essence Tinkering Table. Every complete set upgrade uses 16 next-tier ingots.

Full-set bonuses: Inferium is base armor; Prudentium gives water breathing; Intermedium gives step assist; Superium prevents fall damage; Supremium prevents fall damage and grants native flight. Wear all four matching pieces. Higher sets do not automatically inherit all earlier bonuses. Supremium gear is unbreakable.

For all armor through Supremium, plan for 24 Inferium Ingots, then 16 each Prudentium/Intermedium/Superium/Supremium Ingots. Make EXTRA lower-tier ingots for those higher ingots: ingots already consumed by armor cannot be reused. Gear upgrades preserve names, enchantments and proportion of durability used.

Upgrade seed tiers at a crafting table with the previous seed plus four of the next essence in a cross. They yield 1/2/3/4/5 INFERIUM per harvest, never higher essence directly. Build the farm first, then infuse its output for gear.

Native flight requires Education features and Cheats. Equip the full Supremium set in Survival/Adventure, wait for ready, then double-tap Jump. Jump rises, Sneak descends, sprint boosts speed; double-tap stops flight. No flight charm is needed. Removing a piece revokes and resets active flight, restoring the original Survival/Adventure mode. Creative and Spectator flight are left alone.

PICTURED RECIPES *Infusion Crystal* | *Essence Tinkering Table*

HUD EXAMPLES: *Essence Table: crystal and essence input*

BUILD EXAMPLES: *Essence infusion without power*

JUMP TO: *Machine directory* | *Block recipes* | *Drop tables* | *Essence equipment recipes* | *Tree to Supremium costs*

Essence infusion without power



INPUT: *A real Infusion Crystal + 4 lower-tier essence*

OUTPUT: *1 next-tier essence per operation*

1. Craft the Infusion Crystal at a normal Crafting Table: four Prosperity Shards, four Gold Ingots and a Diamond. The Essence Table does not craft this raw recipe.

2. In the Essence Tinkering Table, place the crystal plus four lower-tier essence. No generator is required.

3. Process the colored essence tiers in order. Crops produce Inferium Essence; they do not directly replace the table's 4-to-1 tier upgrades.

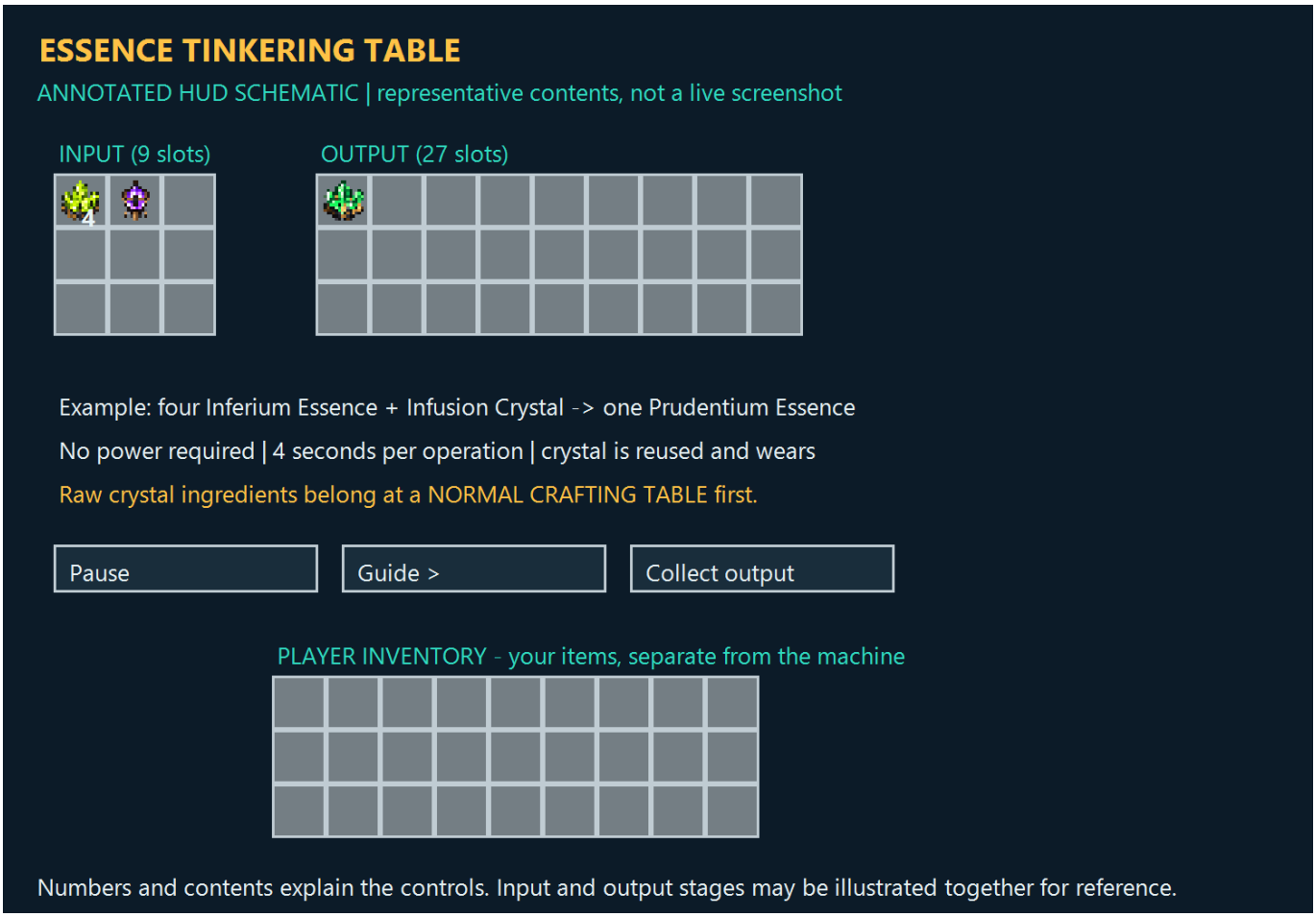
4. Use the linked armor chapter for ingots, full sets and charms. The table's input and output inventories must have room.

READ MORE: *Essence armor: the complete five-tier progression* | *All build examples*

HUD: *Inputs, outputs and controls*

CRAFT THE BLOCKS: *Essence Tinkering Table*

Essence Table: crystal and essence input



1. Put four lower-tier essence and the real Infusion Crystal in INPUT. One operation produces one next-tier essence in OUTPUT. The crystal remains available and spends durability.

2. This machine does not need power. Make the raw Infusion Crystal recipe at a normal Crafting Table first. Raw Gold, Diamond and Prosperity Shards do not perform infusion.

3. The same table also supports eligible equipment upgrades and charms. Use the named recipe requirements; an item picture alone does not make it a valid ingredient.

BUILD THIS: *Placement example* / *Essence armor: the complete five-tier progression* / *All HUD examples*

Essence charms and watering cans

 *Blank Essence Charm*	 *Speed Essence Charm*	 *Inferium Watering Can*	 *Supremium Watering Can*

Blank Essence Charm: one Prosperity Ingot + four Supremium Essence. A finished charm uses the blank, two Supremium Essence and its specialty ingredient. Put it with a compatible Supremium piece in the Essence Table to install one charm. Charmed gear plus Flint removes and returns the charm.

Armor charms include Speed, Night Vision, Water Breathing, Fire Resistance, Resistance and No Fall. Tool charms include compatible Strength, Fortune, Silk Touch, Efficiency and Mining Area effects. One charm per item. See the atlas for every compatible target and ingredient.

Craft an Inferium Watering Can from its ingots and a Bucket. Later can recipes use the previous can. Use on water to fill, then on a crop/soil to water the nearby area. Higher tiers have larger areas.

Remaining water percentage appears while held, with exact water/capacity in the tooltip. An accepted watering action spends one unit immediately, even on mature crops. The first successful plant growth is included; additional plants spend another unit each. Refill when empty.

A short hand stream and target splashes animate watering. Effects are throttled and spend no extra water. Switching the selected slot cancels remaining queued work. Watering a mature plant does not create an additional harvest.

JUMP TO: *Machine directory* | *Block recipes* | *Drop tables* | *Essence equipment recipes* | *Tree to Supremium costs*

Starting tree to Supremium: costs and difficulty

Supremium is a late-game farming project. Getting the first seeds is the early hurdle; producing tens of thousands of Inferium is the long part. There is no reliable real-time completion estimate because farm size, growth assistance, loading and collection determine production. The numbers below are material calculations for this pack, not SkyFactory Java defaults.

Stage	What to build or collect
1. Starting tree	Harvest wood and saplings. Compost eight accepted ingredients per dirt. Establish renewable trees before spending your last saplings.
2. Early resources	Make the sieve and hammer chain. Sieve Dirt for Inferium and Wheat Seeds; crush Cobblestone to Gravel to Sand. Sieve Sand for Prosperity Shards. Basic mesh rolls: Inferium 5% per Dirt, Prosperity 8% per Sand.
3. Start essence farming	Craft an Essence Crafting Seed (Wheat Seed + four Prosperity Shards), then the first Inferium seed (crafting seed + four Inferium in a cross). Plant on valid soil and harvest mature crops; the seed remains planted.
4. Infuse and expand	Craft the normal Infusion Crystal at a Crafting Table (four shards, four gold, one diamond). Build the unpowered Essence Tinkering Table. Infuse four lower essence into one higher essence. Expand crops and collection; upgrade seeds for greater Inferium yield.
5. Armor progression	Make Prosperity Ingots, then Inferium Ingots. Craft the initial four Inferium armor pieces. Upgrade every piece through Prudentium, Intermedium, Superium and Supremium using four next-tier ingots per piece in the Essence Table.
6. Supremium	Wear all four pieces for the set bonus. Native flight still requires Education features and Cheats. Tools and optional charms are separate expenses; they are not prerequisites for the armor set.

Exact material budget for one armor set

Armor reached	Ingots used at this step	Cumulative Inferium equivalent
Inferium	24 Inferium Ingots	96
Prudentium	16 Prudentium Ingots	416
Intermedium	16 Intermedium Ingots	1,760
Superium	16 Superium Ingots	7,200
Supremium	16 Supremium Ingots	29,024

Final armor-only total: 88 Iron Ingots, 352 Prosperity Shards and 29,024 Inferium-equivalent essence. This includes the lower-tier ingots consumed inside higher-tier ingots. It excludes all seeds, extra tools, crystals, the workbench and farm machinery. Do not count the ingots already used in lower armor as reusable ingredients.

Make the grind manageable

Essence values: 1 Prudentium = 4 Inferium; 1 Intermedium = 16; 1 Superium = 64; 1 Supremium = 256. Ingots also consume the previous ingot, so a Supremium Ingot costs 1 Prosperity Ingot plus 4 essence at EACH of the five tiers: 1,364 Inferium equivalent.

Direct basic-mesh sieving would average 580,480 Dirt for the armor-only essence budget. With Netherite Mesh (10% Inferium roll), the average is 290,240 Dirt. These are expected values, not guarantees, and exclude the other resources. Use sieving to start the farm rather than to finish the set.

Farm example	Output per complete harvest	Armor-only harvest rounds
64 tier-1 plants	64 Inferium	454 rounds
64 tier-5 plants	320 Inferium	91 rounds

These examples assume every planted crop matures and is collected once per round. They exclude the essence and shards needed to establish or upgrade the farm, and do not imply a fixed growth time. Higher-tier seeds still produce Inferium, not their matching higher essence.

The armor-only infusion chain uses 9,344 crystal charges: 7,168 Prudentium, 1,728 Intermedium, 384 Superium and 64 Supremium outputs. Budget ten fresh normal crystals at 1,000 uses each, or make a Master Crystal. The Master costs an extra four Supremium Essence (1,024 Inferium equivalent), four Prosperity Ingots and a normal crystal; it then has unlimited uses. Farm upgrades add further infusion work.

Recommended route: get the first seed, expand the renewable farm, add growth assistance and harvesting/collection, then upgrade seeds and infuse stockpiles. A larger reliable farm helps more than manually sieving toward 29,024 essence. Check water, power where required, output space and loaded chunks before investing in speed upgrades.

RECIPES: *All essence recipes* | *Essence workstation setup* | *Crop farm example*

Charm installation and removal

Finished charm crafting recipes appear above. Install one charm by combining it with a compatible Supremium item in the Essence Tinkering Table. Gear plus Flint removes and returns the charm. Armor flight is the full Supremium set bonus; it does not require a charm.

Charm	Compatible Supremium targets
Speed Essence Charm	Supremium Boots, Supremium Leggings
Night Vision Essence Charm	Supremium Helmet
Water Breathing Essence Charm	Supremium Helmet
Fire Resistance Essence Charm	Supremium Chestplate, Supremium Boots
Resistance Essence Charm	Supremium Chestplate
Strength Essence Charm	Supremium Sword
Fortune Essence Charm	Supremium Pickaxe, Supremium Paxel
Silk Touch Essence Charm	Supremium Pickaxe, Supremium Paxel
Efficiency Essence Charm	Supremium Pickaxe, Supremium Axe, Supremium Shovel, Supremium Hoe, Supremium Hammer, Supremium Crook, Supremium Paxel, Supremium Sickle, Supremium Scythe
Mining Aoe Essence Charm	Supremium Pickaxe, Supremium Axe, Supremium Shovel, Supremium Paxel, Supremium Hammer
No Fall Essence Charm	Supremium Boots

BACK TO: *Essence recipe index* | *Tree-to-Supremium budget*

Builder's Wand and early flight equipment

 *Builder's Wand*	 *Jetpack*		

Hold the Builder’s Wand and Sneak + use to open its block selector. Choose a building block available in your inventory or MV stock, and choose Fill or Replace and Box or Dome. Close the HUD, aim at a block within 32 blocks, and left-click to select the start. Aim to resize the preview and left-click again to begin building. Sneak + left-click cancels the selection or active build. Supply the selected blocks; the wand is not a free-block generator.

Jetpacks are chest equipment for earlier held-Jump lift. Equip one and hold Jump to rise; release Jump or Sneak to stop its lift. Flight consumes durability over time, and the pack stops when it wears out. This is a different control scheme from the full Supremium set’s native flight. Test above a safe platform and do not assume a jetpack provides the full armor set’s fall protection.

For Void platform mode, select the first corner under your feet, move or fly to the second corner, then left-click to build a flat platform between them. In other modes, Sneak + left-click cancels the selection or active build. Both use the selected blocks from your inventory or MV stock.

JUMP TO: *Machine directory* | *Block recipes* | *Drop tables* | *Essence equipment recipes* | *Tree to Supremium costs*

Flying ships

Build a 3 x 3 deck of Oak Planks and place a Ship Helm and Ship Balloon on it. Use a Ship Dock to isolate the build from your island. Open the Helm, let it scan, and select Captain (Launch). One Balloon supports up to 32 total ship blocks.

Use normal movement controls. Look up or down while moving forward to change altitude. Jump opens the Helm menu while piloting. Use Dock Ship or dismount in a clear area to restore real blocks. Single chests and barrels retain their cargo. Separate the ship from AVA machines before scanning.

Helm I uses a Compass, Bronze, Redstone and Oak Planks. The Helm II recipe uses Helm I with Reinforced Frames, Steel Ingots and Circuit Boards. The Helm III recipe uses Helm II with Reinforced Frames, Advanced Circuits and Conductive Cores. Use the pictured patterns on the next page.

GO TO *Helm and Balloon crafting grids* | *Contents*

Advanced Void Automation equipment comparison

Pack version 0.2.86 Beta. The tables use the pack's item definitions and scripts. Charged and uncharged Star gear share the same base stats.

Armor sets

Piece order is helmet / chestplate / leggings / boots. Protection is in armor points before effects or enchantments.

Set	Total	Protection by piece	Durability by piece	Perks
Supremium	26	5 / 9 / 8 / 4	Unbreakable	Unbreakable. Full-set flight. Each piece accepts one compatible charm.
Superium	23	4 / 8 / 7 / 4	4000 / 4000 / 4000 / 4000	Standard protection
Platinum	20	3 / 8 / 6 / 3	440 / 640 / 600 / 520	Standard protection
Star	20	3 / 8 / 6 / 3	640 / 960 / 900 / 780	Full charged set grants Resistance I, Speed I and Jump Boost I. Approximately 20 minutes from full charge.
Intermedium	19	3 / 7 / 6 / 3	2000 / 2000 / 2000 / 2000	Full set automatically steps up one-block ledges.
Steel	19	3 / 7 / 6 / 3	275 / 400 / 375 / 325	Standard protection
Bronze	17	3 / 6 / 6 / 2	198 / 288 / 270 / 234	Standard protection
Invar	16	2 / 7 / 5 / 2	231 / 336 / 315 / 273	Standard protection
Prudentium	15	2 / 6 / 5 / 2	1000 / 1000 / 1000 / 1000	Full set grants Water Breathing.
Lead	14	2 / 5 / 4 / 3	165 / 240 / 225 / 195	Standard protection
Nickel	14	2 / 5 / 5 / 2	165 / 240 / 225 / 195	Standard protection
Aluminum	13	2 / 5 / 4 / 2	132 / 192 / 180 / 156	Standard protection
Inferium	13	2 / 5 / 4 / 2	500 / 500 / 500 / 500	Standard protection
Electrum	12	2 / 4 / 4 / 2	88 / 128 / 120 / 104	Standard protection
Silver	11	2 / 4 / 4 / 1	121 / 176 / 165 / 143	Standard protection
Tin	9	1 / 4 / 3 / 1	88 / 128 / 120 / 104	Standard protection
Copper	8	1 / 3 / 3 / 1	66 / 96 / 90 / 78	Legacy AVA set. Vanilla Copper is separate.

Jetpacks and other wearable equipment

Equip a Jetpack in the chest slot and hold Jump for lift. Flight uses one durability per second of lift. Release Jump or sneak to stop.

Piece	Armor points	Durability
Aluminum Armored Jetpack	9	640
Chainmail Armored Jetpack	5	240
Diamond Armored Jetpack	8	528
Golden Armored Jetpack	5	112
Iron Armored Jetpack	6	240
Jetpack	2	240
Leather Armored Jetpack	3	80
Netherite Armored Jetpack	8	592
Upgraded Emerald Jetpack	2	240

Tools and weapons

Each cell shows durability / attack bonus / digger speed. Attack bonus is the item's configured damage component before enchantments or effects. Speed applies to the tool's configured block targets. A dash within a cell means the item does not declare that stat. An empty cell means the pack has no standard item of that type for that material.

Material	Sword	Battleaxe	Pickaxe	Axe	Shovel	Hoe
Copper	175 / 4 / 2		175 / 2 / 4	175 / 3 / 4	175 / 1 / 4	175 / 1 / 4
Aluminum	220 / 4 / 2		220 / 2 / 5	220 / 3 / 5	220 / 1 / 5	220 / 1 / 5
Bronze	500 / 5 / 2		500 / 3 / 6	500 / 4 / 6	500 / 2 / 6	500 / 2 / 6
Electrum	100 / 4 / 2		100 / 2 / 14	100 / 3 / 14	100 / 1 / 14	100 / 1 / 14
Invar	450 / 6 / 2		450 / 4 / 7	450 / 5 / 7	450 / 3 / 7	450 / 3 / 7
Lead	150 / 4 / 2		150 / 2 / 5	150 / 3 / 5	150 / 1 / 5	150 / 1 / 5
Nickel	300 / 6 / 2		300 / 4 / 7	300 / 5 / 7	300 / 3 / 7	300 / 3 / 7
Platinum	1700 / 7 / 2		1700 / 5 / 9	1700 / 6 / 9	1700 / 4 / 9	1700 / 4 / 9
Silver	200 / 5 / 2		200 / 3 / 6	200 / 4 / 6	200 / 2 / 6	200 / 2 / 6
Steel	750 / 6 / 2		750 / 4 / 8	750 / 5 / 8	750 / 3 / 8	750 / 3 / 8
Tin	200 / 4 / 2		200 / 2 / 5	200 / 3 / 5	200 / 1 / 5	200 / 1 / 5
Inferium	500 / 5 / 2		500 / 3 / 5	500 / 3 / 5	500 / 3 / 5	500 / 3 / 5
Prudentium	1000 / 7 / 2		1000 / 5 / 7	1000 / 5 / 7	1000 / 5 / 7	1000 / 5 / 7
Intermedium	2000 / 10 / 2		2000 / 8 / 9	2000 / 8 / 9	2000 / 8 / 9	2000 / 8 / 9
Superium	4000 / 14 / 2		4000 / 12 / 12	4000 / 12 / 12	4000 / 12 / 12	4000 / 12 / 12
Supremium	Unbreakable / 20 / 2		Unbreakable / 18 / 16	Unbreakable / 18 / 16	Unbreakable / 18 / 16	Unbreakable / 18 / 16
Star	2700 / 19 / 2	2700 / 16 / —	2700 / 8 / 14	2700 / 10 / 14	2700 / 7 / 14	2700 / 6 / 14

Material	Hammer	Crook	Paxel	Sickle	Scythe	Watering Can
Wood	59 / — / 2	59 / — / 3				
Stone	131 / — / 4	131 / — / 4				
Copper	175 / 2 / 4	175 / 1 / 4				
Gold	32 / — / 12	32 / — / 4				
Iron	250 / — / 6	250 / — / 4				
Diamond	1561 / — / 8	1561 / — / 4				
Netherite	2031 / — / 9	2031 / — / 4				
Aluminum	220 / 2 / 5	220 / 1 / 4				
Bronze	500 / 3 / 6	500 / 2 / 4				
Electrum	100 / 2 / 14	100 / 1 / 4				
Invar	450 / 4 / 7	450 / 3 / 4				
Lead	150 / 2 / 5	150 / 1 / 4				
Nickel	300 / 4 / 7	300 / 3 / 4				
Platinum	1700 / 5 / 9	1700 / 4 / 4				
Silver	200 / 3 / 6	200 / 2 / 4				
Steel	750 / 4 / 8	750 / 3 / 4				
Tin	200 / 2 / 5	200 / 1 / 4				
Inferium	500 / 3 / 5	500 / 3 / 4	500 / 3 / 5	500 / 3 / 5	500 / 3 / 5	— / — / —

Material	Hammer	Crook	Paxel	Sickle	Scythe	Watering Can
Prudentium	1000 / 5 / 7	1000 / 5 / 4	1000 / 5 / 7	1000 / 5 / 7	1000 / 5 / 7	— / — / —
Intermedium	2000 / 8 / 9	2000 / 8 / 4	2000 / 8 / 9	2000 / 8 / 9	2000 / 8 / 9	— / — / —
Superium	4000 / 12 / 12	4000 / 12 / 4	4000 / 12 / 12	4000 / 12 / 12	4000 / 12 / 12	— / — / —
Supremium	Unbreakable / 18 / 16	Unbreakable / 18 / 4	Unbreakable / 18 / 16	Unbreakable / 18 / 16	Unbreakable / 18 / 16	— / — / —
Star	2700 / 9 / 14	2700 / 4 / 13				

- Hammer crushes supported blocks into progression materials such as gravel, sand and dust.

- Crook harvests leaf drops such as saplings and silkworms.

- Paxel combines pickaxe, axe and shovel roles. Sickle and Scythe harvest groups of crops.

- Watering Can refills from water and speeds growth. From Inferium through Supremium, the five capacity values are 64, 128, 256, 512 and 1,024 water charges.

- Star Sword and Battleaxe grant Strength I while held and charged. Charged Star mining tools grant Haste I while held. A full charge holds up to 1,200 seconds.

- Supremium tools and weapons are unbreakable and support one compatible charm per item.

Supremium charms

Native full-set flight uses the world’s Education features. Use a No Fall charm on the boots for fall protection.

Charm	Compatible equipment	Effect
Speed	Boots, leggings	Speed II
Night Vision	Helmet	Night Vision
Water Breathing	Helmet	Water Breathing
Fire Resistance	Chestplate, boots	Fire Resistance
Resistance	Chestplate	Resistance
No Fall	Boots	Fall damage protection
Strength	Sword	Sharpness V
Fortune	Pickaxe, paxel	Fortune III
Silk Touch	Pickaxe, paxel	Silk Touch I
Efficiency	Pickaxe, axe, shovel, hoe, hammer, crook, paxel, sickle, scythe	Efficiency V
Mining AOE	Pickaxe, axe, shovel, paxel, hammer	Area mining

Modular Tool Forge equipment

The Tool Forge makes a pickaxe or hammer. The head sets the mining speed and base durability. Add the rod bonus and binding bonus to get total durability.

Metal	Head durability	Head speed	Rod or binding bonus	Matching set durability
Aluminum	180	5	+40	260
Bronze	500	7	+100	700
Copper	180	4	+40	260
Electrum	350	12	+90	530
Gold	80	12	+20	120

Metal	Head durability	Head speed	Rod or binding bonus	Matching set durability
Invar	850	8	+180	1210
Iron	300	6	+80	460
Lead	240	4	+60	360
Nickel	400	6	+120	640
Platinum	650	8	+160	970
Silver	220	7	+60	340
Steel	1200	9	+200	1600
Tin	160	4	+30	220

- Invar binding saves durability every fourth use.

- Electrum binding doubles the hammer’s product every fifth use.

- Reinforcement Plate adds 25% of original assembled durability, rounded up. Efficiency Module adds Efficiency II. Fortune Module adds Fortune I to modular pickaxes.

- Special utility items include the Builder’s Wand for building between corners, Safari Net for capturing and releasing mobs, and the Wireless MV Remote for accessing bound MV Storage.

Void mining and late-game power

Add renewable metal production and a larger power network once the factory can supply the parts.

1. Build the Void Ore Miner p. 66
Make a single flat 3 x 3 layer over a clear shaft.

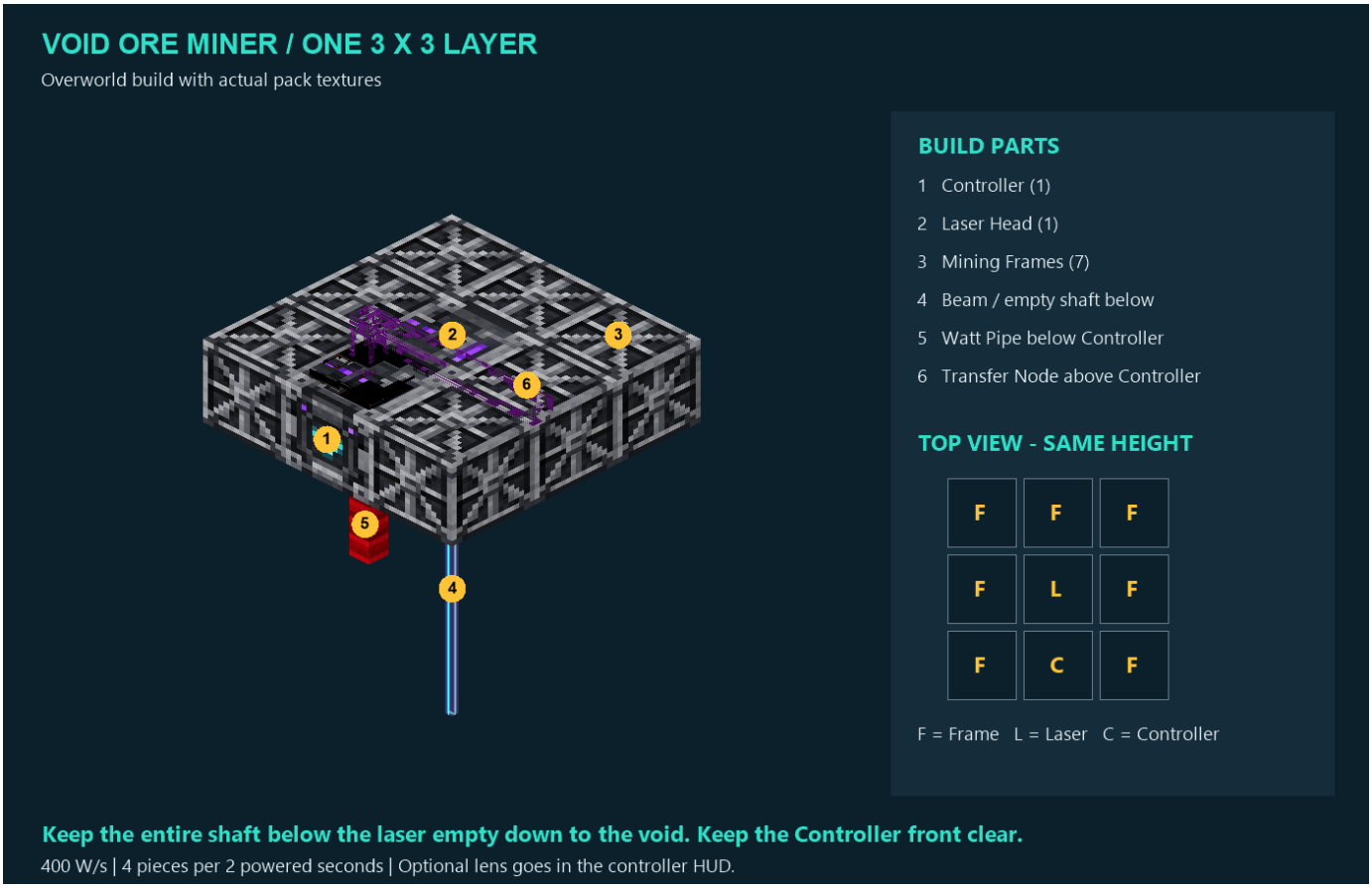
2. Choose a Focus Lens p. 67
Bias ore output toward the metal your factory needs.

3. Build a reactor or distribute power wirelessly p. 68
Use the nuclear and relay section to plan your next power source.

Continue to Wither Chamber and Star Forge p. 69
Open the next chapter when your current setup is ready.

Look up ingredients and recipes p. 75
Find the illustrated crafting grids and machine input amounts.

Build the Void Ore Miner



1. Make a flat 3 x 3 platform. Place seven Mining Frames and one Controller around the rim. The Controller goes in the middle of any edge. Put the Laser Head in the center, level with the ring.

2. Build over open Overworld void. Leave the whole shaft directly below the Laser Head clear. Blocks, water and lava stop the beam.

3. Run power and items separately. The rendering shows a red Watt Pipe below the Controller and an extraction Transfer Node above it. Keep the front open for the HUD. Send the purple item route to a chest or an MV Link in Transfer mode.

4. Start mining. Supply 400 W per working second. The miner produces four pieces of raw or crushed metal per two-second cycle. A reusable Focus Lens in the Controller is optional.

Craft the Mining Frame recipe twice to get eight frames. Craft two Laser Heads: the Controller recipe consumes one and the second goes in the center.

Void miner: power, lenses and output

Setting	Current behavior
Production	4 pieces of the selected raw/crushed metal per completed 2-second cycle. No input blocks are consumed.
Working power	400 W per working second; 800 W per four-piece batch. Supply sustained generation or a charged battery reserve.
Lens slot	The single slot at the left of the Controller HUD is optional. Search Focus Lens to find Iron, Copper, Gold or Alloy. Lenses are reusable; stacking does not increase their effect.
Controls	Pause/Resume enables production. Rescan Structure immediately rechecks the ring and shaft. Collect Output moves results into available player inventory space.
Output storage	27 output slots. Connect extraction to MV or another inventory. Normal right-click opens the HUD; sneak-right-click places a Transfer Node against the machine.

Focus Lens effects

Lens	Favored output	Approximate chance per batch
None	Default mixed metal distribution	See base chances below
Iron	Raw Iron	68.3% instead of 35%
Copper	Raw Copper	57.1% instead of 25%
Gold	Raw Gold	35.3% instead of 12%
Alloy	Six custom crushed metals combined	60.9% instead of 28%

Lenses multiply matching weights by four and then normalize the table. They do not guarantee a metal, increase total output, speed up the miner or change power consumption. All four pieces in one batch are the same selected metal.

Base chances and processing

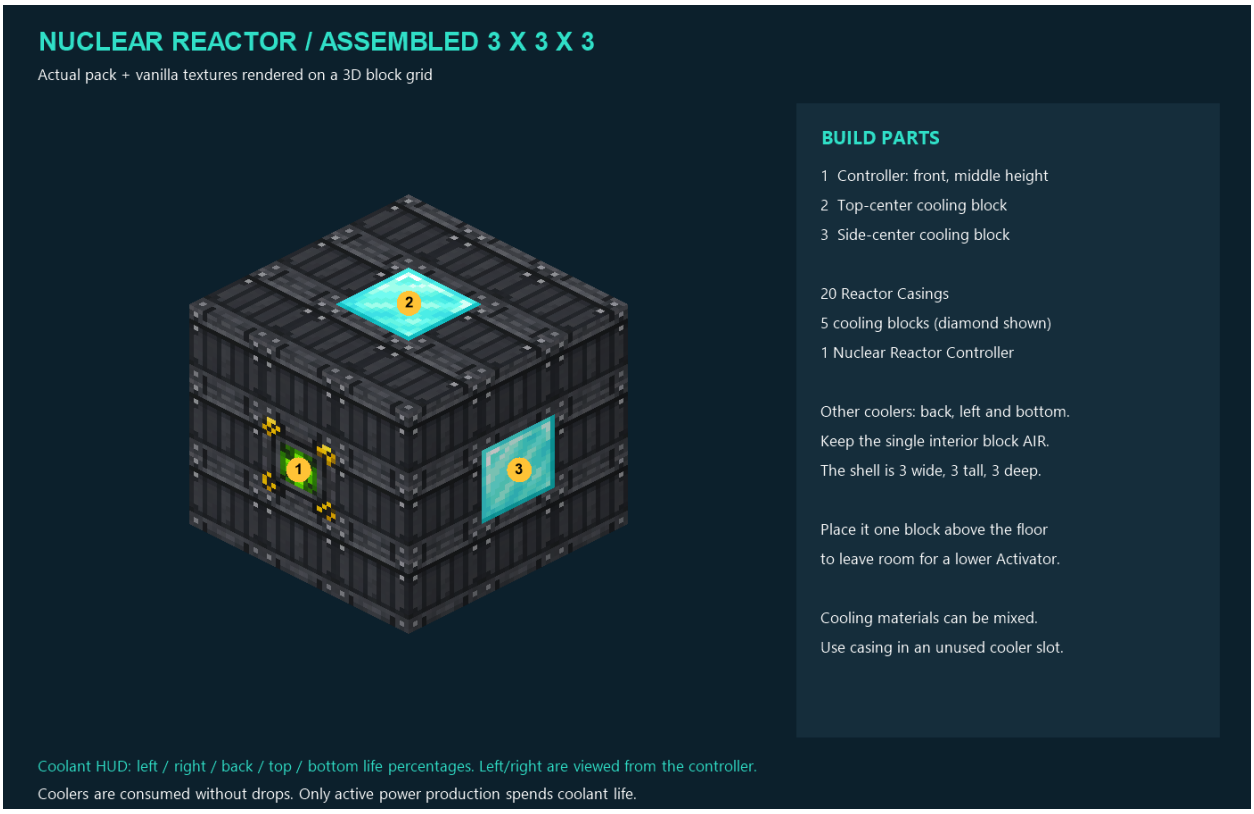
Raw Iron 35%; Raw Copper 25%; Raw Gold 12%; Crushed Tin 7%; Crushed Silver 6%; Crushed Lead 5%; Crushed Nickel 5%; Crushed Aluminum 4%; Crushed Platinum 1%. Process outputs using the existing furnace, pulverizer, smeltery and crafting routes; these are not finished ingots.

If it stops or the beam is missing

Check the Controller status first: complete the ring, clear the entire shaft, supply 400 W per working second, remove unsupported lens-slot items and extract full output. Waiting conditions preserve progress and resume when fixed. The beam appears only during powered production. The machine does not force chunks to stay loaded.

JUMP TO: *Build rendering* / *Crafting recipes* / *Power and transport builds* / *Contents*

Nuclear and wireless power



Build a hollow 3 x 3 x 3 shell with the controller in the center of a front wall. Keep the middle block air. Place coolers at the five other face centers. The casing accepts Watt Pipes and wireless transmitters.

One fuel rod supplies 1,200,000 W. Insert rods in the upper input row and extract spent rods from the lower output row. Copper, iron, gold, diamond and netherite coolers increase output in that order. Coolers wear during generation and are destroyed on overheating. Use inward-facing Activators to replace them.

To pair wireless power, place a transmitter next to the source or connect it with Watt Pipes. Note its assigned name. Place another relay at the machine, use Find, and select the transmitter. Keep both ends loaded in the same dimension.

BACK *Contents*

Wither Chamber and Star Forge

Farm Nether Stars, then spend them on Star Ingots, gear charging or ore replication.

1. Set up mob handling p. 70
Use the mob systems to support the factory and skull supply.

2. Build and feed a Wither Chamber p. 71
Route skulls and Soul Sand in and export Nether Stars.

3. Connect the Star Forge supply lines p. 73
Bring Nether Stars, Crystalline Alloy and circuits together.

4. Automate gear charging and ore replication p. 74
Use separate input and output routes to keep items moving.

Look up ingredients and recipes p. 75
Find the illustrated crafting grids and machine input amounts.

Mobs, Safari Nets, meat, slime and Bacon

 *Safari Net*	 *Mob Grinder* Crafting recipe	 *Slaughter Factory* Crafting recipe	 *Mob Duplicator* Crafting recipe
 *Bacon Dispenser*			

Hostile mobs need suitable darkness, full-block spawning space, headroom and distance from the player. Passive animals need suitable grass, light, space and their spawning conditions. A platform directly under where you stand is not an adequate spawning test. For a practical test, stand about 30 blocks from the platform while keeping it inside the loaded simulation area. Passive spawning can take several minutes and depends on biome and existing populations.

A Safari Net captures a supported mob for safe release or the applicable machine route. It is obtainable in Survival. Keep valuable named mobs out of initial machine tests and provide safe release space.

Face the Mob Grinder's blades toward the mobs. The HUD names the direction and range it checks. Base cost is 80 W per mob. Transfer Nodes can extract item drops, and separate tank groups can collect Liquid XP and Pink Essence. Full fluid buffers preserve their stored contents and skip new overflow. Full item output still stops processing.

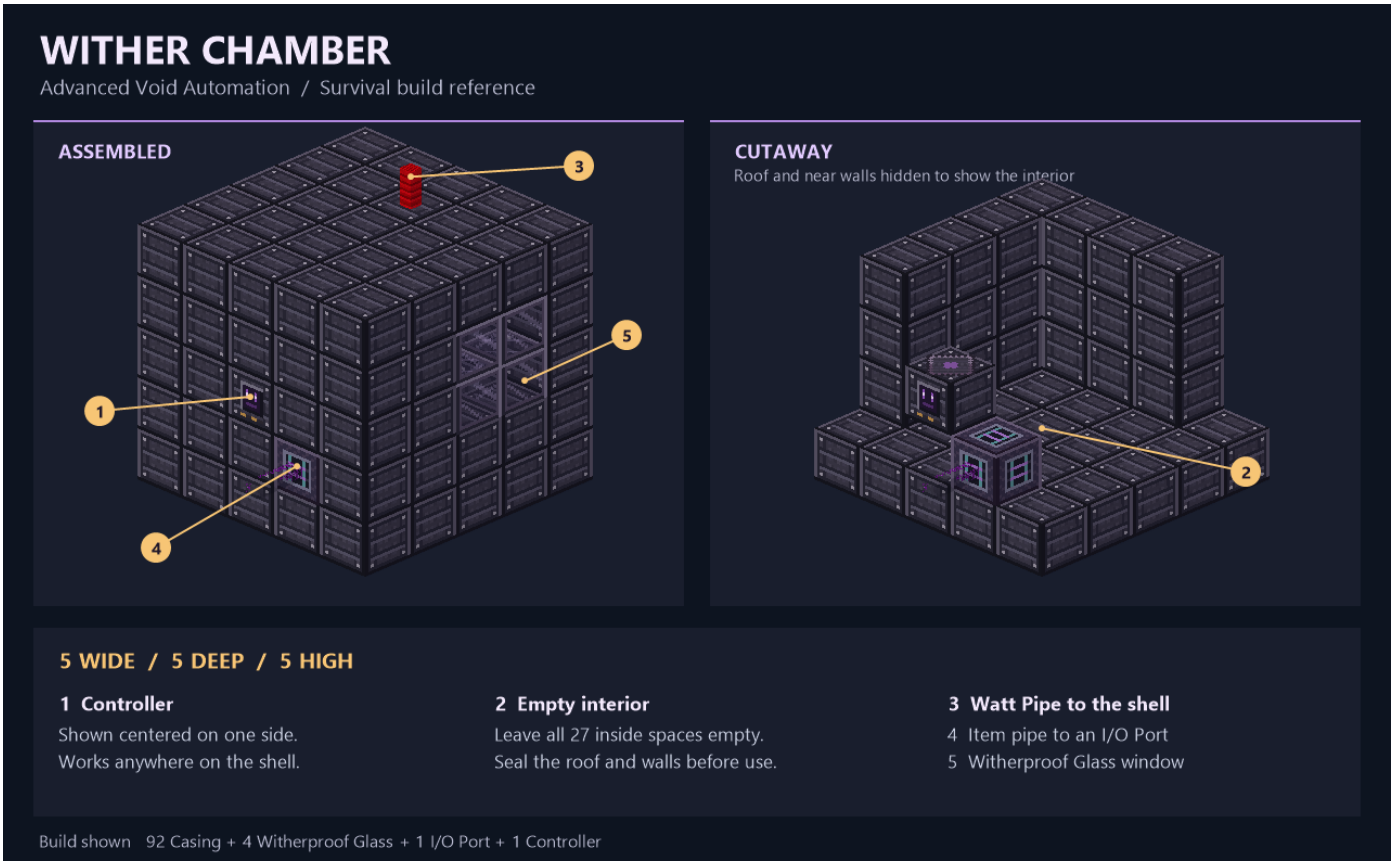
The Pink Slime Generator turns late mob production into power. Raw/cooked meat block packing and unpacking routes appear in the atlas.

The Bacon Dispenser stores more than one visual stack. Load while carrying Bacon using the supported use action and read its stored status. Its automatic feeding works in supported non-Creative modes when hungry. A native single-stack preview is not the authoritative stored total.

JUMP TO: *Machine directory* | *Block recipes* | *Drop tables* | *Essence equipment recipes* | *Tree to Supremium costs*

Wither Chamber

Build, power and automate a contained Wither fight



1. Build and seal the shell

Build 5 x 5 x 5 and leave the 3 x 3 x 3 interior empty. Place the controller anywhere on the shell. Face it toward an accessible side for the HUD. Witherproof Glass and I/O Ports can replace casing on any shell position, including corners, floor and roof. Leave the controller face open.

2. Connect power to the shell

Attach a Watt Pipe to a shell block away from the controller face. The chamber draws 500 W/s during charging and the first phase, then 800 W/s during the armored phase.

3. Supply materials and collect the Nether Star

Load 3 Wither Skeleton Skulls and 4 Soul Sand per fight. Send inputs from a chest with a Transfer Node and Transfer Pipes, or use filtered MV Pull Links into an I/O Port. Point an extraction Transfer Node at a port to export Nether Stars. Ports share the controller inventory.

4. Keep the chamber sealed

Each completed fight yields 1 Nether Star. Losing power or breaching the shell suspends progress. Vanilla TNT particles and sound mark the end of charging, half health, and defeat. The effects are harmless.

Wither Chamber recipes

Crafting Table or Universal Crafter



Find Wither Skeleton Skulls

In a void Nether, build a dark spawning platform out of Cobblestone, Stone Bricks or Nether Bricks. Wither Skeletons can spawn there. Kill them directly, or process them in the Mob Grinder for its 2.5% skull roll per Wither Skeleton.

Make Soul Sand

Sieve Sand or Red Sand for Ancient Spores, then use them on Dirt to make Mycelium. Put a water-filled Composter on the Mycelium, wait for Witch Water, then add Sand to make Soul Sand.

Star Forge automation



Star Infuser



Star Generator



Star Fusion Replicator

Set up the supply lines

1. Connect each machine and each MV Link to a power source using **Watt Pipes**. Use purple **Transfer Pipes** for item routes.

2. For an MV-fed machine, bind an **MV Link** to MV Storage, switch it to **Pull**, and select the ingredient you want to feed. You can also use a Transfer Node facing a supply chest and run Transfer Pipes to the machine.

3. Export finished items with a **Transfer Node facing the machine**. Send its pipe to a chest or an **MV Link in Transfer mode**. An adjacent Transfer-mode MV Link can also pull finished items directly. Keep the output route separate from the input route. On a mixed factory network, use filtered supply lines to give each machine the recipe you intend to run.

4. Use a connected water tank for continuous operation. For example, pump water from a Composter on a Moisture Collector into a regular Tank, then connect the Tank to the Pressure Reactor and Material Infuser by Transfer Pipes. Water Buckets can be fed into either processor as well. Empty buckets go into its output.

Produce the ingredients

Stage	Machine	Input per batch	Output
Polymer	Pressure Reactor	16 accepted crops or saplings, 1 charcoal, 1,000 mB water	4 Polymer
Crystalline Alloy	Material Infuser	2 diamonds, 4 Steel Ingots, 2 Nickel Ingots, 1 Polymer, 1,000 mB water	2 Crystalline Alloy
Polymer Circuit Board	Universal Crafter	2 Polymer, 1 Electrum Ingot, 1 redstone	1 Polymer Circuit Board
Advanced Circuit	Precision Assembler	2 Polymer Circuit Boards, 2 Electrum Ingots, 1 quartz, 2 Polymer	2 Advanced Circuits
Nether Star	Wither Chamber	3 Wither Skeleton Skulls, 4 Soul Sand	1 Nether Star
Star Ingot	Star Infuser	4 Nether Stars, 2 Netherite Ingots, 1 Crystalline Alloy, 2 Advanced Circuits	1 Star Ingot

The Pressure Reactor needs **400 W/s** at base settings. The Material Infuser needs **1,000 W/s** for Crystalline Alloy, and the Precision Assembler needs **500 W/s** for Advanced Circuits. Feed Wither Chamber ingredients through a **Wither Chamber I/O Port** and extract stars from a port. Keep the controller accessible. At base speed, Star Ingot production needs **2,000 W/s for 60 seconds**.

Recharge, generate and replicate

Automate gear recharging

1. Feed Nether Stars from your Wither Chamber or MV stock into the **Star Infuser**. Automated stars go into the top-middle input slot.

2. Place a tool, weapon, or armor piece that needs charging in a supply chest and point a Transfer Node at the chest. Route its output to the Infuser. Incoming gear uses the top-left input slot, so ingredients can keep arriving separately. You can also feed gear from MV by selecting the corresponding gear item on an MV Pull Link.

3. Each recharge uses **one Nether Star** and **1,600 W/s for 25 seconds**. An ingot already in progress finishes before newly delivered gear takes over. Charging retains the item's name, enchantments, and durability.

4. Send the output to a gear chest or back to MV. Extractors see finished output only. The Infuser rejects fully charged gear on automated input, which prevents an accidental recharge loop.

Equipped gear stays on the player until you put it into the supply line.

Fuel and replicate

- **Star Generator.** Pipe in Nether Stars. Each supplies 20 seconds of generation at 16,000 W/s. The generator has a 640,000 W buffer and pauses fuel use when the buffer is nearly full.

- **Star Fusion Replicator.** Feed an eligible ore and Nether Stars on filtered routes. Each batch uses one ore and two Nether Stars to output two of that ore, at 4,000 W/s for 45 seconds. Eligible inputs include Ancient Debris, selected vanilla ore blocks and raw ores, and the pack's crushed metal ores. Full output waits for space.

Reference and recipes

Troubleshooting and safe updates 76

Machine and utility directory 77

Sieve drop probabilities 82

Item index and crafting alternatives 88

Added block crafting grids 92

Processor recipes and inputs 101

Essence equipment and charm crafting grids 107

Reactor and ship crafting layouts 127

Void Ore Miner and Focus Lens grids 128

Star Forge and Star gear grids 130

Troubleshooting and safe pack updates

			
Essence Tinkering Table	*Infusion Crystal*	*MV System*	
Crafting recipe		Crafting recipe	

Essence Table idle: craft the crystal at a normal Crafting Table first. Then insert essence + crystal, previous gear + next-tier ingots, or Supremium gear + charm. Its input grid is not a general crafting bench. It needs no Watt Pipe.

If native flight persists after removing Supremium armor, verify that the world is using matching current Behavior and Resource Packs and reload. Record the Bedrock version when reporting a persistent issue.

No essence from a few dirt blocks: it is a chance drop, not guaranteed. The six-click flow is tested at all mesh tiers. Process more dirt and collect physical drops safely.

When mined, supported machines with stored items can drop a Packed Machine. Place it to restore its contents. Empty machines drop normally. MV Links, Composters and barrels are excluded from packing. Leave hidden storage hosts intact.

Wrong transfers: check intake direction, active ALLOW/BLOCK mode, saved filter items, priority, MV Link direction, power, loaded storage and free destination space.

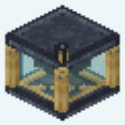
Two pack versions: an older embedded world pack can coexist with a new imported library version. Check BP and RP descriptions, then reload Minecraft. You do not have to delete the world.

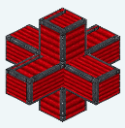
JUMP TO: *Machine directory* | *Block recipes* | *Drop tables* | *Essence equipment recipes* | *Tree to Supremium costs*

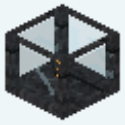
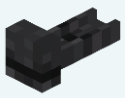
Machine and utility directory

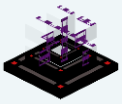
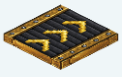
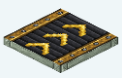
Identify the block, its job and the first setup steps. Use the numbered chapters for complete systems and the crafting appendix for the block recipe.

 **Sieve**	Turn dirt, gravel, sand and other supported inputs into resources. Install a mesh, load one input, and click six times. Drops eject into the world; use a hopper or collection system.
 **Moisture Collector**	Place beneath the water-producing composter arrangement to collect water over time. A Fluid Pump can extract the collected water.
 **Stone Barrel**	Holds lava. Add Redstone Dust for Netherrack, Glowstone Dust for End Stone, or a Water Bucket for Obsidian. Each conversion uses 100 mB of lava.
 **Kiln**	Melt one Cobblestone into 1,000 mB of lava. Take the lava with an empty bucket or move it with a Fluid Pump. No power required.
 **Magma Furnace**	Smelt supported raw materials into ingots and other products using power. Provide an output route before leaving a batch running.
 **Pulverizer**	Grind supported blocks, ores and materials. Processing yields depend on the input; powder is not a full ingot. Check the machine's recipe guide for supported inputs.
 **Dehydrator**	Powered drying recipes turn the supported wet/raw inputs into their listed outputs.
 **Glacial Chiller**	Powered cooling recipes use their listed inputs and water equivalents. Keep its water supply available.

 **Pressure Reactor**	Turn supported farm products into Polymer with charcoal, water and power. Feed every required ingredient, not just crops.
 **Precision Assembler**	Powered advanced-part assembly. Combine the listed ingredients for circuits and advanced factory components.
 **Material Infuser**	Powered infusion stage for advanced material upgrades. Supply the exact recipe inputs and extract the finished material.
 **Factory Core**	Late-game powered production endpoint. It processes its own recipes; it is not a replacement for every earlier machine.
 **Essence Tinkering Table**	No power required. Insert a real Infusion Crystal plus four lower-tier essence, previous-tier equipment plus matching ingots, or eligible gear plus a charm. The raw crystal recipe belongs at a normal crafting table.
 **Planter**	Supply supported crop seeds and power. It places crops on valid soil in range. Add a transfer route to replenish seeds.
 **Harvester**	Supply power and valid mature crops or trees in range. Collected drops stay in its output slots; attach a Transfer Node to export them.
 **Automated Sprinkler**	Place near crops with its required water supply. It supports local growth; it does not harvest or replant.
 **Activator**	Load tools or usable items, choose click mode and facing, and enable it. Tools wear one at a time in slot order. Speed upgrades shorten its action interval.

 **Mob Grinder**	Powered mob-killing and drop collection. Provide a safe mob chamber and output extraction.
 **Slaughter Factory**	Powered mob-processing route for meat and slime production. Feed mobs into its working area and extract its outputs.
 **Mob Duplicator**	Use a supported captured mob and its listed resources to duplicate mobs. Build a secure chamber before activation.
 **Combustion Generator**	Fuel-based power source. Active generation feeds the Watt network; it stops when its usable fuel is exhausted.
 **Pink Slime Generator**	Late-game power from the pack's pink-slime production chain. Maintain its fuel/input supply.
 **Solar Panel**	Solar power source. Connect it to Watt Pipes; generation depends on its operating conditions.
 **Advanced Solar Panel**	Higher-tier solar generation. Multiple active sources contribute to the connected power network and battery charging.
 **Battery Bank**	Charge from an active source through Watt Pipes. Adjacent batteries combine capacity. Full batteries stop charging; stored energy is reserved for when active primary sources are unavailable.
 **Watt Pipe**	Connect power sources, batteries and consumers. It carries network power; it is not a replacement generator.

 **Fluid Pump**	Move supported fluid from an adjacent source into a connected compatible tank route. A water composter on a Moisture Collector is a valid water source.
 **Tank**	Connect matching tanks to increase fluid capacity. Connected storage fills from the bottom upward. Check the displayed fluid before mixing supplies.
 **Seared Tank**	Smeltery-compatible fluid/fuel tank. Include at least one in the first wall ring and provide lava for smelting.
 **Smeltery Controller**	The only Controller in a valid seared smeltery. Open it to load meltable inputs, inspect metals and select the intended pour.
 **Smeltery Drain**	Outlet in the smeltery wall. Attach a Faucet and position a casting receiver below the pour.
 **Seared Faucet**	Transfer selected molten material from the Drain into the receiver below. An occupied or incompatible receiver stops the pour.
 **Casting Table**	Cast ingots, tool parts and reusable casts using the appropriate pattern/cast and molten material. Wait for cooling before extraction.
 **Casting Basin**	Receiver for bulk block casting. Place below a Faucet and supply enough of a compatible molten metal.
 **Tool Forge**	Assemble modular tools from compatible cast parts; apply supported modifiers through its listed recipes.

 **Transfer Node**	Extract items from the attached inventory or machine and route them through Transfer Pipes. Install a circuit board for persistent smart filters and destination priorities.
 **Transfer Pipe**	Item transport connections for Transfer Nodes. These are distinct from fluid and Watt networks.
 **Vacuum Hopper**	Collect nearby loose item drops into its inventory. Attach a transfer route to export the collected items.
 **MV System**	Shared item storage with search, withdrawal, deposit and an integrated 3 x 3 crafting pattern. Build the network before adding remote access.
 **MV Link**	Connect external item transfer to the MV network. Supports item insertion and extraction through compatible transfer routes.
 **Universal Crafter**	Choose a supported recipe and supply its ingredients. It supports the pack's registered crafting recipes; browsing an output alone does not provide ingredients.
 **Black Hole Unit**	Bulk storage for large quantities of one item type. Compatible transfer routes can insert and extract.
 **Conveyor Belt**	Move items/entities along the belt's direction. Build a continuous route and test its orientation before scaling.
 **Express Conveyor**	Faster conveyor tier. Use for high-volume surface transport; it is separate from Transfer Pipe routing.

Sieve drop tables: every block and mesh

Chance means the probability of getting at least one of that item from one completed sieve input. Repeated rolls for the same item are combined. Several different items can drop together. The figures include the weighted fallback used when every independent roll fails. Rounded to two decimals; very rare results are marked <0.01%.

Amounts show the possible quantity of that item. Dirt guarantees 4 regular pebbles before extra rolls; it is not a 15% pebble drop. These tables document current behavior, without changing it.

Crooks, infested leaves and hammers

Crooks add a 15% matching-sapling roll and an independent 3% Silkworm roll per natural leaf, in addition to normal leaf loot. Vein mining performs the same rolls separately for each leaf. Better crook materials do not raise these percentages. Infested leaves give 1-3 String when crook-mined through the special handling.

Hammer conversions are guaranteed for supported blocks: Cobblestone -> Gravel -> Sand -> Dust; Netherrack -> Crushed Netherrack. Normally one output per block. Tool tier affects the tool's stats, not a random ore-drop bonus. A supported Tool Forge extra-product modifier can produce an additional output.

Crushed Netherrack

Drop / Basic amount	Basic	Iron	Diamond	Netherite
 Quartz (1)	100%	100%	100%	100%
 Gold Nugget (0-1)	20%	25%	32%	40%
 Netherite Scrap (0-1)	1%	1.25%	1.60%	2%

Dirt

Drop / Basic amount	Basic	Iron	Diamond	Netherite
 Andesite Pebble (0-2)	9.84%	12.25%	15.59%	19.36%
 Basalt Pebble (0-2)	9.84%	12.25%	15.59%	19.36%
 Blackstone Pebble (0-2)	9.84%	12.25%	15.59%	19.36%
 Diorite Pebble (0-2)	55%	67.19%	83.20%	100%
 Granite Pebble (0-2)	9.84%	12.25%	15.59%	19.36%
 Pebble (4-6)	100%	100%	100%	100%
 Bamboo (0-1)	5%	6.25%	8%	10%
 Beetroot Seeds (0-1)	5%	6.25%	8%	10%
 Sweet Berries (0-1)	5%	6.25%	8%	10%
 Birch Sapling (0-1)	5%	6.25%	8%	10%
 Carrot (0-1)	5%	6.25%	8%	10%

Drop / Basic amount	Basic	Iron	Diamond	Netherite
Dark Oak Sapling (0-1)	5%	6.25%	8%	10%
Grass Seeds (0-1)	7%	8.75%	11.20%	14%
Melon Seeds (0-1)	5%	6.25%	8%	10%
Oak Sapling (0-1)	5%	6.25%	8%	10%
Potato (0-1)	5%	6.25%	8%	10%
Pumpkin Seeds (0-1)	5%	6.25%	8%	10%
Spruce Sapling (0-1)	5%	6.25%	8%	10%
Wheat Seeds (0-1)	7%	8.75%	11.20%	14%
Acacia Sapling (0-1)	4%	5%	6.40%	8%
Jungle Sapling (0-1)	4%	5%	6.40%	8%
Cocoa Beans (0-1)	4%	5%	6.40%	8%
Cactus (0-1)	4%	5%	6.40%	8%
Cocoa Seeds (0-1)	4%	5%	6.40%	8%
Sugar Cane Seeds (0-1)	4%	5%	6.40%	8%
Inferium Essence (0-1)	5%	6.25%	8%	10%

Regular pebbles: Basic 4-6 (average 5.25), Iron 4-6 (5.56), Diamond 5-6 (5.8), Netherite exactly 6. Inferium Essence: 5%, 6.25%, 8%, 10% respectively.

Gravel

Drop / Basic amount	Basic	Iron	Diamond	Netherite
Coal (0-1)	16.79%	18.81%	22.04%	26.17%
Diamond (0-1)	2.02%	2.26%	2.65%	3.14%
Emerald (0-1)	2.02%	2.26%	2.65%	3.14%
Flint (0-1)	33.58%	37.62%	44.08%	52.34%
Lapis Lazuli (0-1)	6.72%	7.52%	8.82%	10.47%
Aluminum Scrap (0-1)	6.72%	7.52%	8.82%	10.47%
Copper Fragment (0-1)	8.06%	9.03%	10.58%	12.56%
Gold Fragment (0-1)	4.03%	4.51%	5.29%	6.28%

Drop / Basic amount	Basic	Iron	Diamond	Netherite
Iron Fragment (0-1)	26.87%	30.09%	35.27%	41.88%
Lead Fragment (0-1)	4.03%	4.51%	5.29%	6.28%
Nickel Fragment (0-1)	4.03%	4.51%	5.29%	6.28%
Platinum Fragment (0-1)	6.72%	7.52%	8.82%	10.47%
Silver Fragment (0-1)	4.03%	4.51%	5.29%	6.28%
Tin Fragment (0-1)	8.06%	9.03%	10.58%	12.56%

Sand

Drop / Basic amount	Basic	Iron	Diamond	Netherite
Crushed Iron Ore (0-1)	23.14%	27.11%	33.19%	40.60%
Crushed Gold Ore (0-1)	4.63%	5.42%	6.64%	8.12%
Crushed Copper Ore (0-1)	11.57%	13.55%	16.59%	20.30%
Crushed Silver Ore (0-1)	5.78%	6.78%	8.30%	10.15%
Crushed Lead Ore (0-1)	5.78%	6.78%	8.30%	10.15%
Crushed Nickel Ore (0-1)	5.78%	6.78%	8.30%	10.15%
Crushed Aluminum Ore (0-1)	9.25%	10.84%	13.28%	16.24%
Crushed Tin Ore (0-1)	9.25%	10.84%	13.28%	16.24%
Crushed Platinum Ore (0-1)	2.31%	2.71%	3.32%	4.06%
Certus Quartz (0-1)	5.78%	6.78%	8.30%	10.15%
Bone Meal (0-1)	13.88%	16.27%	19.91%	24.36%
Redstone (0-1)	9.25%	10.84%	13.28%	16.24%
Glowstone Dust (0-1)	4.63%	5.42%	6.64%	8.12%
Blaze Powder (0-1)	2.31%	2.71%	3.32%	4.06%
Gunpowder (0-1)	4.63%	5.42%	6.64%	8.12%
Cactus (0-1)	5.78%	6.78%	8.30%	10.15%
Cocoa Seeds (0-1)	4.63%	5.42%	6.64%	8.12%
Sugar Cane Seeds (0-1)	5.78%	6.78%	8.30%	10.15%

Drop / Basic amount	Basic	Iron	Diamond	Netherite
Cocoa Beans (0-1)	3.47%	4.07%	4.98%	6.09%
Wheat Seeds (0-1)	3.47%	4.07%	4.98%	6.09%
Melon Seeds (0-1)	3.47%	4.07%	4.98%	6.09%
Ancient Spores (0-1)	5.78%	6.78%	8.30%	10.15%
Prosperity Shard (0-1)	9.25%	10.84%	13.28%	16.24%
Cactus Seeds (0-1)	5.78%	6.78%	8.30%	10.15%

Dust

Drop / Basic amount	Basic	Iron	Diamond	Netherite
Blaze Powder (0-1)	6.21%	7.11%	8.52%	10.28%
Bone Meal (0-1)	24.84%	28.44%	34.08%	41.13%
Glowstone Dust (0-1)	7.76%	8.89%	10.65%	12.85%
Gunpowder (0-1)	8.69%	9.95%	11.93%	14.40%
Redstone (0-1)	15.53%	17.77%	21.30%	25.71%
Iron Ore Powder (0-1)	24.84%	28.44%	34.08%	41.13%
Gold Ore Powder (0-1)	4.97%	5.69%	6.82%	8.23%
Copper Ore Powder (0-1)	12.42%	14.22%	17.04%	20.56%
Silver Ore Powder (0-1)	6.21%	7.11%	8.52%	10.28%
Lead Ore Powder (0-1)	6.21%	7.11%	8.52%	10.28%
Nickel Ore Powder (0-1)	6.21%	7.11%	8.52%	10.28%
Aluminum Ore Powder (0-1)	9.94%	11.38%	13.63%	16.45%
Tin Ore Powder (0-1)	9.94%	11.38%	13.63%	16.45%
Platinum Ore Powder (0-1)	2.48%	2.84%	3.41%	4.11%

Soul Sand

Drop / Basic amount	Basic	Iron	Diamond	Netherite
Quartz (1)	100%	100%	100%	100%
Nether Wart (0-1)	15%	18.75%	24%	30%

Drop / Basic amount	Basic	Iron	Diamond	Netherite
Ghast Tear (0-1)	2%	2.50%	3.20%	4%

End Stone

Drop / Basic amount	Basic	Iron	Diamond	Netherite
Ender Pearl (0-1)	23.65%	24.87%	27%	30%
Chorus Flower (0-1)	7.88%	8.29%	9%	10%
Quartz (0-1)	78.84%	82.90%	89.99%	100%

Red Sand

Drop / Basic amount	Basic	Iron	Diamond	Netherite
Crushed Iron Ore (0-1)	23.14%	27.11%	33.19%	40.60%
Crushed Gold Ore (0-1)	4.63%	5.42%	6.64%	8.12%
Crushed Copper Ore (0-1)	11.57%	13.55%	16.59%	20.30%
Crushed Silver Ore (0-1)	5.78%	6.78%	8.30%	10.15%
Crushed Lead Ore (0-1)	5.78%	6.78%	8.30%	10.15%
Crushed Nickel Ore (0-1)	5.78%	6.78%	8.30%	10.15%
Crushed Aluminum Ore (0-1)	9.25%	10.84%	13.28%	16.24%
Crushed Tin Ore (0-1)	9.25%	10.84%	13.28%	16.24%
Crushed Platinum Ore (0-1)	2.31%	2.71%	3.32%	4.06%
Certus Quartz (0-1)	5.78%	6.78%	8.30%	10.15%
Bone Meal (0-1)	13.88%	16.27%	19.91%	24.36%
Redstone (0-1)	9.25%	10.84%	13.28%	16.24%
Glowstone Dust (0-1)	4.63%	5.42%	6.64%	8.12%
Blaze Powder (0-1)	2.31%	2.71%	3.32%	4.06%
Gunpowder (0-1)	4.63%	5.42%	6.64%	8.12%
Cactus (0-1)	5.78%	6.78%	8.30%	10.15%
Cocoa Seeds (0-1)	4.63%	5.42%	6.64%	8.12%
Sugar Cane Seeds (0-1)	5.78%	6.78%	8.30%	10.15%

Drop / Basic amount	Basic	Iron	Diamond	Netherite
 Cocoa Beans (0-1)	3.47%	4.07%	4.98%	6.09%
 Wheat Seeds (0-1)	3.47%	4.07%	4.98%	6.09%
 Melon Seeds (0-1)	3.47%	4.07%	4.98%	6.09%
 Ancient Spores (0-1)	5.78%	6.78%	8.30%	10.15%
 Prosperity Shard (0-1)	9.25%	10.84%	13.28%	16.24%
 Cactus Seeds (0-1)	5.78%	6.78%	8.30%	10.15%

Ingredient alternatives

Recipe cards name A001-style groups when any of several ingredients can fill one ingredient role. The representative picture is only the first option. Every permitted alternative is listed here.

Alphabetical item and recipe index

Click a recipe reference to jump to its crafting grid. This index contains only the added blocks with crafting recipes shown in this guide.

Item / block	Obtain / recipe references
Activator	#122
Advanced Solar Panel	#95
Aluminum Block	#97
Automated Sprinkler	#121
Battery Bank	#126
Black Hole Unit	#127
Bronze Block	#130
Casting Basin	#143
Casting Table	#144
Combustion Generator	#152
Conveyor Belt	#153 - #154
Cooked Meat Block	#155 - #156
Dehydrator	#165
Electrum Block	#177
Essence Tinkering Table	#190
Express Conveyor	#191
Factory Core	#192
Fluid Pump	#194
Glacial Chiller	#196
Harvester	#206
Invar Block	#234
Kiln	#255
Lead Block	#261
Magma Furnace	#281
Material Infuser	#283
Mob Duplicator	#285
Mob Grinder	#286

Item / block	Obtain / recipe references
Moisture Collector	#287
MV Link	#288
MV System	#289
Nickel Block	#295
Pink Slime Generator	#324
Planter	#325
Platinum Block	#327
Precision Assembler	#346
Pressure Reactor	#347
Pulverizer	#352
Raw Meat Block	#358
Seared Bricks	#366
Seared Faucet	#367
Seared Glass	#368
Seared Tank	#369
Sieve	#370
Silver Block	#373
Slaughter Factory	#392
Smeltery Controller	#393
Smeltery Drain	#394
Solar Panel	#396
Steel Block	#400
Stone Barrel	#412
Tank	#421
Tin Block	#423
Tool Forge	#442
Transfer Node	#443
Transfer Pipe	#444
Universal Crafter	#445

Item / block	Obtain / recipe references
 Vacuum Hopper	#447
 Watt Pipe	#449

Added-block crafting recipes

Only added machine, storage, utility and building-block crafting recipes appear here. Vanilla, tool, armor and seed crafting grids are omitted. Machine uses are described in the system chapters. Ingredient pictures show the current pack artwork.

Crafting table / Universal Crafter - 1 - 8 of 60

#122 Activator x1
Shapeless: any arrangement



1 Piston; 1 Chest; 2 Iron Ingot; 2 Tin Ingot; 1 Redstone
BUILD & USE THIS BLOCK

#95 Advanced Solar Panel x1
Shapeless: any arrangement



4 Solar Panel; 1 Diamond; 1 Quartz; 1 Silver Ingot; 1 Nickel Ingot; 1 Redstone Block
MACHINE USES

#97 Aluminum Block x1
Shapeless: any arrangement



9 Aluminum Ingot
MACHINE USES

#121 Automated Sprinkler x1
Shapeless: any arrangement



1 Clay Bucket; 1 Iron Ingot; 2 Tin Ingot; 1 Transfer Pipe; 1 Redstone
BUILD & USE THIS BLOCK

#126 Battery Bank x1
Shaped crafting grid



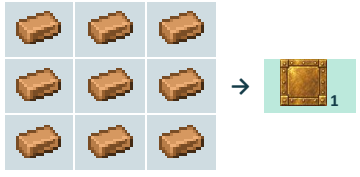
4 Iron Ingot; 2 Copper Ingot; 2 Redstone; 1 Quartz
BUILD & USE THIS BLOCK

#127 Black Hole Unit x1
Shaped crafting grid



4 Iron Ingot; 4 Redstone; 1 Chest
BUILD & USE THIS BLOCK

#130 Bronze Block x1
Shapeless: any arrangement



9 Bronze Ingot
MACHINE USES

#143 Casting Basin x1
Shaped crafting grid



7 Seared Brick
BUILD & USE THIS BLOCK

Crafting table / Universal Crafter - 9 - 16 of 60

#144 Casting Table x1
Shaped crafting grid



7 Seared Brick
BUILD & USE THIS BLOCK

#152 Combustion Generator x1
Shapeless: any arrangement



1 Furnace; 2 Copper Ingot; 1 Nickel Ingot; 1 Aluminum Ingot; 1 Redstone
MACHINE USES

#153 Conveyor Belt x8
Shapeless: any arrangement



2 Steel Ingot; 2 Bronze Ingot; 2 Aluminum Ingot; 1 Tin Ingot; 2 String
MACHINE USES

#154 Conveyor Belt x4
Shapeless: any arrangement



2 Bronze Ingot; 1 Tin Ingot; 3 String
MACHINE USES

#155 Cooked Meat Block x1
Shapeless: any arrangement



1 Raw Meat Block; 1 Charcoal
MACHINE USES

#156 Cooked Meat Block x1
Shapeless: any arrangement



1 Raw Meat Block; 1 Coal
MACHINE USES

#165 Dehydrator x1
Shaped crafting grid



4 Silver Ingot; 1 Iron Bars; 2 Iron Ingot; 1 Furnace; 1 Redstone
MACHINE USES

#177 Electrum Block x1
Shapeless: any arrangement



9 Electrum Ingot
MACHINE USES

Crafting table / Universal Crafter - 17 - 24 of 60

#190 Essence Tinkering Table x1
Shaped crafting grid



2 Prosperity Shard; 1 Inferium Essence; 5 Stone; 1 Crafting Table
BUILD & USE THIS BLOCK

#191 Express Conveyor x4
Shapeless: any arrangement



4 Conveyor Belt; 1 Steel Ingot; 2 Invar Ingot; 2 Electrum Ingot
MACHINE USES

#192 Factory Core x1
Shapeless: any arrangement



4 Reinforced Frame; 2 Conductive Core; 2 Advanced Circuit; 1 Crystalline Alloy
BUILD & USE THIS BLOCK

#194 Fluid Pump x1
Shapeless: any arrangement



1 Clay Bucket; 1 Transfer Pipe; 1 Iron Ingot; 2 Tin Ingot
BUILD & USE THIS BLOCK

#196 Glacial Chiller x1
Shaped crafting grid



4 Tin Ingot; 1 Glass; 2 Nickel Ingot; 1 Furnace; 1 Redstone
MACHINE USES

#206 Harvester x1
Shapeless: any arrangement



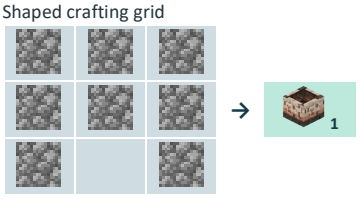
1 Transfer Node; 1 Iron Ingot; 1 Nickel Ingot; 1 Redstone; 1 Iron Hoe
BUILD & USE THIS BLOCK

#234 Invar Block x1
Shapeless: any arrangement



9 Invar Ingot
MACHINE USES

#255 Kiln x1
Shaped crafting grid



8 Cobblestone
MACHINE USES

Crafting table / Universal Crafter - 25 - 32 of 60

#261 Lead Block x1
Shapeless: any arrangement



9 Lead Ingot
MACHINE USES

#281 Magma Furnace x1
Shaped crafting grid



4 Iron Ingot; 1 Nickel Ingot; 2 Copper Ingot; 1 Furnace; 1 Tin Ingot
MACHINE USES

#283 Material Infuser x1
Shapeless: any arrangement



4 Reinforced Frame; 2 Advanced Circuit; 1 Fluid Pump; 2 Invar Ingot
BUILD & USE THIS BLOCK

#285 Mob Duplicator x1
Shapeless: any arrangement



1 Steel Ingot; 1 Diamond Block; 2 Ender Pearl; 2 Nickel Ingot; 1 Silver Ingot; 1 Safari Net; 1 Chest
MACHINE USES

#286 Mob Grinder x1
Shapeless: any arrangement



2 Steel Ingot; 2 Invar Ingot; 1 Electrum Ingot; 1 Nickel Ingot; 1 Platinum Ingot; 1 Diamond Sword; 1 Tank
MACHINE USES

#287 Moisture Collector x1
Shapeless: any arrangement



1 Kiln; 2 Aluminum Ingot; 1 Copper Ingot; 1 Glass; 1 Clay Bucket
BUILD & USE THIS BLOCK

#288 MV Link x1
Shapeless: any arrangement



1 Transfer Node; 1 Silver Ingot; 1 Redstone
BUILD & USE THIS BLOCK

#289 MV System x1
Shapeless: any arrangement



1 Netherite Ingot; 1 Diamond Block; 1 Chest; 1 Universal Crafter; 1 Steel Ingot; 1 Silver Ingot; 1 Lead Ingot; 1 Nickel Ingot; 1 Redstone Block
BUILD & USE THIS BLOCK

Crafting table / Universal Crafter - 33 - 40 of 60

#295 Nickel Block x1
Shapeless: any arrangement



9 Nickel Ingot
MACHINE USES

#324 Pink Slime Generator x1
Shapeless: any arrangement



2 Steel Ingot; 2 Invar Ingot; 2 Electrum Ingot; 1 Platinum Ingot; 1 Redstone Block; 1 Combustion Generator
MACHINE USES

#325 Planter x1
Shapeless: any arrangement



1 Transfer Node; 1 Iron Ingot; 1 Aluminum Ingot; 1 Redstone; 1 Flower Pot
BUILD & USE THIS BLOCK

#327 Platinum Block x1
Shapeless: any arrangement



9 Platinum Ingot
MACHINE USES

#346 Precision Assembler x1
Shapeless: any arrangement



4 Steel Ingot; 2 Polymer Circuit Board; 2 Piston; 1 Polymer
BUILD & USE THIS BLOCK

#347 Pressure Reactor x1
Shapeless: any arrangement



2 Steel Ingot; 2 Invar Ingot; 1 Bronze Ingot; 1 Piston; 1 Fluid Pump; 1 Furnace; 1 Redstone Block
BUILD & USE THIS BLOCK

#352 Pulverizer x1
Shaped crafting grid



4 Iron Ingot; 1 Piston; 2 Copper Ingot; 1 Furnace; 1 Redstone
MACHINE USES

#358 Raw Meat Block x1
Shaped crafting grid



9 Pink Essence Bucket
Returns nine empty Buckets.
MACHINE USES

Crafting table / Universal Crafter - 41 - 48 of 60

#366 Seared Bricks x1
Shaped crafting grid



4 Seared Brick
MACHINE USES

#367 Seared Faucet x1
Shaped crafting grid



3 Seared Brick
BUILD & USE THIS BLOCK

#368 Seared Glass x2
Shaped crafting grid



4 Seared Brick; 1 Glass
MACHINE USES

#369 Seared Tank x1
Shaped crafting grid



8 Seared Brick; 1 Glass
MACHINE USES

#370 Sieve x1
Shapeless: any arrangement



4 Oak Planks; 4 String
BUILD & USE THIS BLOCK

#373 Silver Block x1
Shapeless: any arrangement



9 Silver Ingot
MACHINE USES

#392 Slaughter Factory x1
Shapeless: any arrangement



1 Steel Ingot; 1 Diamond Sword; 1 Hopper; 1 Chest; 2 Nickel Ingot; 1 Lead Ingot; 1 Redstone; 1 Iron Block
MACHINE USES

#393 Smeltery Controller x1
Shaped crafting grid



8 Seared Brick; 1 Furnace
MACHINE USES

Crafting table / Universal Crafter - 49 - 56 of 60

#394 Smeltery Drain x1
Shaped crafting grid



6 Seared Brick
BUILD & USE THIS BLOCK

#396 Solar Panel x1
Shapeless: any arrangement



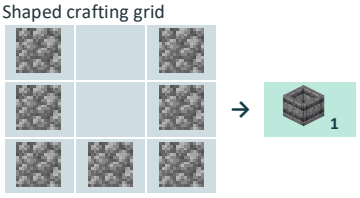
3 Glass; 2 Lapis Lazuli; 1 Quartz; 1 Silver Ingot; 1 Aluminum Ingot; 1 Redstone
BUILD & USE THIS BLOCK

#400 Steel Block x1
Shapeless: any arrangement



9 Steel Ingot
MACHINE USES

#412 Stone Barrel x1
Shaped crafting grid



7 Cobblestone
BUILD & USE THIS BLOCK

#421 Tank x1
Shapeless: any arrangement



2 Cobblestone; 2 Glass; 2 Lead Ingot
BUILD & USE THIS BLOCK

#423 Tin Block x1
Shapeless: any arrangement



9 Tin Ingot
MACHINE USES

#442 Tool Forge x1
Shaped crafting grid



3 Iron Block; 5 Seared Bricks; 1 Crafting Table
MACHINE USES

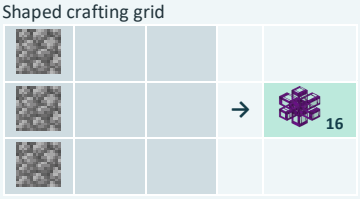
#443 Transfer Node x1
Shapeless: any arrangement



1 Transfer Pipe; 1 Chest; 1 Iron Ingot
BUILD & USE THIS BLOCK

Crafting table / Universal Crafter - 57 - 60 of 60

#444 Transfer Pipe x16
Shaped crafting grid



3 Cobblestone
BUILD & USE THIS BLOCK

#445 Universal Crafter x1
Shaped crafting grid



4 Aluminum Ingot; 1 Silver Ingot; 1 Nickel Ingot; 1 Crafting Table; 1 MV Link; 1 Redstone Block
MACHINE USES

#447 Vacuum Hopper x1
Shapeless: any arrangement



1 Hopper; 1 Chest; 1 Ender Pearl; 1 Aluminum Ingot; 1 Transfer Node
BUILD & USE THIS BLOCK

#449 Watt Pipe x16
Shapeless: any arrangement



2 Cobblestone; 1 Redstone
BUILD & USE THIS BLOCK

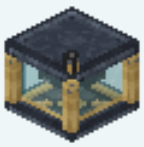
Recipes made inside production machines

These recipes run inside their named machine, not at a Crafting Table. Insert the ingredients into INPUT, connect Watt Pipes, supply water where listed, and leave output space. Recipe order in the input slots does not matter. The machine automatically processes the first complete supported recipe; its recipe-guide button only changes the displayed instructions.

Rates below are base values without upgrades. W/s is the working power draw; W/job is the total base energy for one successful batch. Machines wait when required ingredients, water, power or output space are missing.

MACHINE GUIDES: *pressure reactor* | *precision assembler* | *material infuser* | *factory core*

Pressure Reactor — machine recipes



16 supported biomass + Charcoal + Water → Polymer. Biomass may be mixed: crops, supported saplings and Mangrove Propagules. Supply water buckets or a connected water tank.

Output / batch	Inputs and operating cost
Polymer ×4 	16 Crops or saplings + 1 Charcoal Water: 1000 mB consumed per batch 8 seconds | 400 W/s | 3200 W/job

CONTINUE: *Production recipes* | *Machine directory* | *Contents*

Precision Assembler — machine recipes



Insert all listed ingredients in INPUT. Collect finished items from OUTPUT, or extract them through your item network. Use Next recipe guide to browse the machine's recipes.

Output / batch	Inputs and operating cost
Reinforced Frame ×1 	4 Steel Ingot + 2 Polymer + 1 Bronze Ingot Water: 0 mB consumed per batch 8 seconds | 250 W/s | 2000 W/job
Advanced Circuit ×2 	2 Polymer Circuit Board + 2 Electrum Ingot + 1 Quartz + 2 Polymer Water: 0 mB consumed per batch 10 seconds | 500 W/s | 5000 W/job
Conductive Core ×1 	2 Platinum Ingot + 2 Invar Ingot + 1 Redstone Block + 1 Advanced Circuit Water: 0 mB consumed per batch 15 seconds | 1000 W/s | 15000 W/job
Growth Essence ×4 	16 Wheat + 4 Bone Meal + 1 Polymer Water: 0 mB consumed per batch 6 seconds | 250 W/s | 1500 W/job

CONTINUE: *Production recipes* | *Machine directory* | *Contents*

Precision Assembler — machine recipes (continued)



Insert all listed ingredients in INPUT. Collect finished items from OUTPUT, or extract them through your item network. Use Next recipe guide to browse the machine’s recipes.

Output / batch	Inputs and operating cost
Reinforcement Plate ×1 	1 Reinforced Frame + 2 Crystalline Alloy Water: 0 mB consumed per batch 10 seconds | 750 W/s | 7500 W/job
Efficiency Module ×1 	2 Advanced Circuit + 1 Crystalline Alloy + 8 Redstone Water: 0 mB consumed per batch 10 seconds | 750 W/s | 7500 W/job
Fortune Module ×1 	2 Advanced Circuit + 2 Crystalline Alloy + 16 Lapis Lazuli + 2 Diamond Water: 0 mB consumed per batch 15 seconds | 1000 W/s | 15000 W/job

CONTINUE: *Production recipes* | *Machine directory* | *Contents*

Material Infuser — machine recipes



Insert all listed ingredients in INPUT. Collect finished items from OUTPUT, or extract them through your item network. Use Next recipe guide to browse the machine's recipes.

Output / batch	Inputs and operating cost
Crystalline Alloy ×2 	2 Diamond + 4 Steel Ingot + 2 Nickel Ingot + 1 Polymer Water: 1000 mB consumed per batch 15 seconds | 1000 W/s | 15000 W/job
Enriched Essence ×4 	8 Growth Essence + 4 Amethyst Shard + 1 Advanced Circuit Water: 1000 mB consumed per batch 12 seconds | 750 W/s | 9000 W/job

CONTINUE: *Production recipes* | *Machine directory* | *Contents*

Factory Core — machine recipes



Insert all listed ingredients in INPUT. Collect finished items from OUTPUT, or extract them through your item network. Use Next recipe guide to browse the machine's recipes.

Output / batch	Inputs and operating cost
Factory Trophy ×1 	8 Crystalline Alloy + 4 Conductive Core + 4 Advanced Circuit + 8 Polymer + 4 Platinum Ingot + 4 Pink Slime + 4 Enriched Essence + 1 Netherite Ingot Water: 8000 mB consumed per batch 60 seconds | 2000 W/s | 120000 W/job

CONTINUE: *Production recipes* | *Machine directory* | *Contents*

Essence armor and tools: every recipe

109 recipes from the pack: 14 equipment types at each of five tiers (70 equipment recipes), plus essence conversions, ingots, crystals, crop seeds, charms and watering cans. Crafting cards show exact grids where shaped. PROCESS cards show ingredient quantities only: insert them in any Essence Tinkering Table input slots, unpause, then wait for output. The table requires no power.

Inferium equipment

Inferium Axe | Inferium Boots | Inferium Chestplate | Inferium Crook | Inferium Hammer | Inferium Helmet | Inferium Hoe | Inferium Leggings | Inferium Paxel | Inferium Pickaxe | Inferium Scythe | Inferium Shovel | Inferium Sickle | Inferium Sword

Prudentium equipment

Prudentium Axe | Prudentium Boots | Prudentium Chestplate | Prudentium Crook | Prudentium Hammer | Prudentium Helmet | Prudentium Hoe | Prudentium Leggings | Prudentium Paxel | Prudentium Pickaxe | Prudentium Scythe | Prudentium Shovel | Prudentium Sickle | Prudentium Sword

Intermedium equipment

Intermedium Axe | Intermedium Boots | Intermedium Chestplate | Intermedium Crook | Intermedium Hammer | Intermedium Helmet | Intermedium Hoe | Intermedium Leggings | Intermedium Paxel | Intermedium Pickaxe | Intermedium Scythe | Intermedium Shovel | Intermedium Sickle | Intermedium Sword

Superium equipment

Superium Axe | Superium Boots | Superium Chestplate | Superium Crook | Superium Hammer | Superium Helmet | Superium Hoe | Superium Leggings | Superium Paxel | Superium Pickaxe | Superium Scythe | Superium Shovel | Superium Sickle | Superium Sword

Supremium equipment

Supremium Axe | Supremium Boots | Supremium Chestplate | Supremium Crook | Supremium Hammer | Supremium Helmet | Supremium Hoe | Supremium Leggings | Supremium Paxel | Supremium Pickaxe | Supremium Scythe | Supremium Shovel | Supremium Sickle | Supremium Sword

SUPPORT RECIPES: *Blank Essence Charm* / *Efficiency Essence Charm* / *Essence Crafting Seed* / *Fire Resistance Essence Charm* / *Fortune Essence Charm* / *Infusion Crystal* / *Master Infusion Crystal* / *Mining Aoe Essence Charm* / *Night Vision Essence Charm* / *No Fall Essence Charm* / *Prosperity Ingot* / *Resistance Essence Charm* / *Silk Touch Essence Charm* / *Speed Essence Charm* / *Strength Essence Charm* / *Water Breathing Essence Charm* / *Inferium Essence* / *Inferium Essence Seeds* / *Inferium Ingot* / *Inferium Tier2 Essence Seeds* / *Inferium Tier3 Essence Seeds* / *Inferium Tier4 Essence Seeds* / *Inferium Tier5 Essence Seeds* / *Inferium Watering Can* / *Prudentium Essence* / *Prudentium Essence* / *Prudentium Ingot* / *Prudentium Watering Can* / *Intermedium Essence* / *Intermedium Essence* / *Intermedium Ingot* / *Intermedium Watering Can* / *Superium Essence* / *Superium Essence* / *Superium Ingot* / *Superium Watering Can* / *Supremium Essence* / *Supremium Ingot* / *Supremium Watering Can*

Essence recipes 1 - 6

Blank Essence Charm x1
Crafting Table / Universal Crafter
Shapeless - any arrangement



1 Prosperity Ingot + 4 Supremium Essence
Essence recipe index

Efficiency Essence Charm x1
Crafting Table / Universal Crafter
Shapeless - any arrangement



1 Blank Essence Charm + 2 Supremium Essence + 2 Redstone Block
Essence recipe index

Essence Crafting Seed x1
Crafting Table / Universal Crafter
Shaped - copy the grid



4 Prosperity Shard + 1 Wheat Seeds
Essence recipe index

Fire Resistance Essence Charm x1
Crafting Table / Universal Crafter
Shapeless - any arrangement



1 Blank Essence Charm + 2 Supremium Essence + 2 Magma Cream
Essence recipe index

Fortune Essence Charm x1
Crafting Table / Universal Crafter
Shapeless - any arrangement



1 Blank Essence Charm + 2 Supremium Essence + 2 Diamond + 4 Lapis Lazuli
Essence recipe index

Infusion Crystal x1
Crafting Table / Universal Crafter
Shaped - copy the grid



4 Prosperity Shard + 4 Gold Ingot + 1 Diamond
Essence recipe index

Essence recipes 7 - 12

Master Infusion Crystal x1
Crafting Table / Universal Crafter
Shaped - copy the grid



4 Supremium Essence + 4 Prosperity Ingot + 1 Infusion Crystal
Essence recipe index

Mining Aoe Essence Charm x1
Crafting Table / Universal Crafter
Shapeless - any arrangement



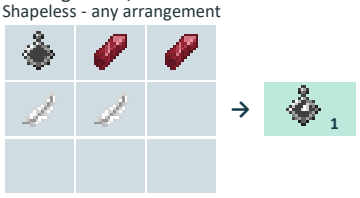
1 Blank Essence Charm + 2 Supremium Essence + 2 Piston
Essence recipe index

Night Vision Essence Charm x1
Crafting Table / Universal Crafter
Shapeless - any arrangement



1 Blank Essence Charm + 2 Supremium Essence + 2 Golden Carrot
Essence recipe index

No Fall Essence Charm x1
Crafting Table / Universal Crafter
Shapeless - any arrangement



1 Blank Essence Charm + 2 Supremium Essence + 2 Feather
Essence recipe index

Prosperity Ingot x1
Crafting Table / Universal Crafter
Shapeless - any arrangement



1 Iron Ingot + 4 Prosperity Shard
Essence recipe index

Resistance Essence Charm x1
Crafting Table / Universal Crafter
Shapeless - any arrangement



1 Blank Essence Charm + 2 Supremium Essence + 4 Obsidian
Essence recipe index

Essence recipes 13 - 18

Silk Touch Essence Charm x1
Crafting Table / Universal Crafter
Shapeless - any arrangement



1 Blank Essence Charm + 2 Supremium Essence + 4 String
Essence recipe index

Speed Essence Charm x1
Crafting Table / Universal Crafter
Shapeless - any arrangement



1 Blank Essence Charm + 2 Supremium Essence + 4 Sugar
Essence recipe index

Strength Essence Charm x1
Crafting Table / Universal Crafter
Shapeless - any arrangement



1 Blank Essence Charm + 2 Supremium Essence + 2 Blaze Powder
Essence recipe index

Water Breathing Essence Charm x1
Crafting Table / Universal Crafter
Shapeless - any arrangement



1 Blank Essence Charm + 2 Supremium Essence + 2 Pufferfish
Essence recipe index

Inferium Axe x1
Crafting Table / Universal Crafter
Shaped - copy the grid



3 Inferium Ingot + 2 Stick
Essence recipe index

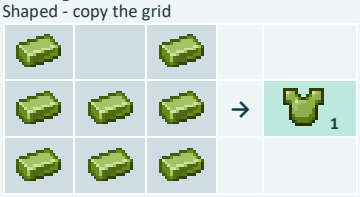
Inferium Boots x1
Crafting Table / Universal Crafter
Shaped - copy the grid



4 Inferium Ingot
Essence recipe index

Essence recipes 19 - 24

Inferium Chestplate x1
Crafting Table / Universal Crafter
Shaped - copy the grid



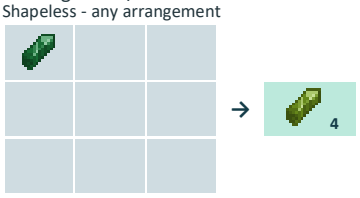
8 Inferium Ingot
Essence recipe index

Inferium Crook x1
Crafting Table / Universal Crafter
Shaped - copy the grid



3 Inferium Ingot + 2 Stick
Essence recipe index

Inferium Essence x4
Crafting Table / Universal Crafter
Shapeless - any arrangement



1 Prudentium Essence
Essence recipe index

Inferium Essence Seeds x1
Crafting Table / Universal Crafter
Shaped - copy the grid



4 Inferium Essence + 1 Essence Crafting Seed
Essence recipe index

Inferium Hammer x1
Crafting Table / Universal Crafter
Shaped - copy the grid



2 Inferium Ingot + 2 Stick
Essence recipe index

Inferium Helmet x1
Crafting Table / Universal Crafter
Shaped - copy the grid



5 Inferium Ingot
Essence recipe index

Essence recipes 25 - 30

Inferium Hoe x1
Crafting Table / Universal Crafter
Shaped - copy the grid



2 Inferium Ingot + 2 Stick
Essence recipe index

Inferium Ingot x1
Crafting Table / Universal Crafter
Shapeless - any arrangement



1 Prosperity Ingot + 4 Inferium Essence
Essence recipe index

Inferium Leggings x1
Crafting Table / Universal Crafter
Shaped - copy the grid



7 Inferium Ingot
Essence recipe index

Inferium Paxel x1
Crafting Table / Universal Crafter
Shapeless - any arrangement



1 Inferium Pickaxe + 1 Inferium Axe + 1 Inferium Shovel + 1 Prosperity Ingot
Essence recipe index

Inferium Pickaxe x1
Crafting Table / Universal Crafter
Shaped - copy the grid



3 Inferium Ingot + 2 Stick
Essence recipe index

Inferium Scythe x1
Crafting Table / Universal Crafter
Shaped - copy the grid



4 Inferium Ingot + 2 Stick
Essence recipe index

Essence recipes 31 - 36

Inferium Shovel x1
Crafting Table / Universal Crafter
Shaped - copy the grid



1 Inferium Ingot + 2 Stick
Essence recipe index

Inferium Sickle x1
Crafting Table / Universal Crafter
Shaped - copy the grid



3 Inferium Ingot + 2 Stick
Essence recipe index

Inferium Sword x1
Crafting Table / Universal Crafter
Shaped - copy the grid



2 Inferium Ingot + 1 Stick
Essence recipe index

Inferium Tier2 Essence Seeds x1
Crafting Table / Universal Crafter
Shaped - copy the grid



4 Prudentium Essence + 1 Inferium Essence Seeds
Essence recipe index

Inferium Tier3 Essence Seeds x1
Crafting Table / Universal Crafter
Shaped - copy the grid



4 Intermedium Essence + 1 Inferium Tier2 Essence Seeds
Essence recipe index

Inferium Tier4 Essence Seeds x1
Crafting Table / Universal Crafter
Shaped - copy the grid



4 Superium Essence + 1 Inferium Tier3 Essence Seeds
Essence recipe index

Essence recipes 37 - 42

Inferium Tier5 Essence Seeds x1
Crafting Table / Universal Crafter
Shaped - copy the grid



4 Supremium Essence + 1 Inferium Tier4 Essence Seeds
Essence recipe index

Inferium Watering Can x1
Crafting Table / Universal Crafter
Shaped - copy the grid



5 Inferium Ingot + 1 Bucket
Essence recipe index

Prudentium Axe x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Inferium Axe + 4 Prudentium Ingot

Output **1 Prudentium Axe**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Prudentium Boots x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Inferium Boots + 4 Prudentium Ingot

Output **1 Prudentium Boots**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Prudentium Chestplate x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Inferium Chestplate + 4 Prudentium Ingot

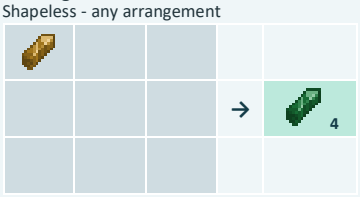
Output **1 Prudentium Chestplate**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Prudentium Crook x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Inferium Crook + 4 Prudentium Ingot

Output **1 Prudentium Crook**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Essence recipes 43 - 48

Prudentium Essence x4
Crafting Table / Universal Crafter
Shapeless - any arrangement



1 Intermedium Essence
Essence recipe index

Prudentium Essence x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
4 Inferium Essence + 1 Infusion Crystal

Output **1 Prudentium Essence**
4s per batch, up to 64 output. Crystal retained; one use per output, 1000 uses per normal crystal.
Essence recipe index

Prudentium Hammer x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Inferium Hammer + 4 Prudentium Ingot

Output **1 Prudentium Hammer**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Prudentium Helmet x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Inferium Helmet + 4 Prudentium Ingot

Output **1 Prudentium Helmet**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Prudentium Hoe x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Inferium Hoe + 4 Prudentium Ingot

Output **1 Prudentium Hoe**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Prudentium Ingot x1
Crafting Table / Universal Crafter
Shapeless - any arrangement



1 Inferium Ingot + 4 Prudentium Essence
Essence recipe index

Essence recipes 49 - 54

Prudentium Leggings x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Inferium Leggings + 4 Prudentium Ingot
Output  **1 Prudentium Leggings**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Prudentium Paxel x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Inferium Paxel + 4 Prudentium Ingot
Output  **1 Prudentium Paxel**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Prudentium Pickaxe x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Inferium Pickaxe + 4 Prudentium Ingot
Output  **1 Prudentium Pickaxe**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Prudentium Scythe x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Inferium Scythe + 4 Prudentium Ingot
Output  **1 Prudentium Scythe**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Prudentium Shovel x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Inferium Shovel + 4 Prudentium Ingot
Output  **1 Prudentium Shovel**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Prudentium Sickle x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Inferium Sickle + 4 Prudentium Ingot
Output  **1 Prudentium Sickle**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Essence recipes 55 - 60

Prudentium Sword x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Inferium Sword + 4 Prudentium Ingot

Output **1 Prudentium Sword**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Prudentium Watering Can x1
Crafting Table / Universal Crafter
Shaped - copy the grid



5 Prudentium Ingot + 1 Inferium Watering Can
Essence recipe index

Intermedium Axe x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Prudentium Axe + 4 Intermedium Ingot

Output **1 Intermedium Axe**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Intermedium Boots x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Prudentium Boots + 4 Intermedium Ingot

Output **1 Intermedium Boots**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Intermedium Chestplate x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Prudentium Chestplate + 4 Intermedium Ingot

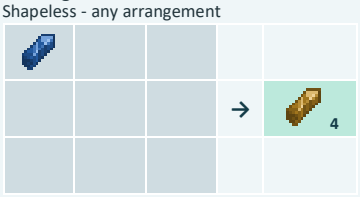
Output **1 Intermedium Chestplate**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Intermedium Crook x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Prudentium Crook + 4 Intermedium Ingot

Output **1 Intermedium Crook**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Essence recipes 61 - 66

Intermedium Essence x4
Crafting Table / Universal Crafter
Shapeless - any arrangement



1 Superium Essence
Essence recipe index

Intermedium Essence x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
4 Prudentium Essence + 1 Infusion Crystal

Output **1 Intermedium Essence**
4s per batch, up to 64 output. Crystal retained; one use per output, 1000 uses per normal crystal.
Essence recipe index

Intermedium Hammer x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Prudentium Hammer + 4 Intermedium Ingot

Output **1 Intermedium Hammer**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Intermedium Helmet x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Prudentium Helmet + 4 Intermedium Ingot

Output **1 Intermedium Helmet**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Intermedium Hoe x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Prudentium Hoe + 4 Intermedium Ingot

Output **1 Intermedium Hoe**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Intermedium Ingot x1
Crafting Table / Universal Crafter
Shapeless - any arrangement



1 Prudentium Ingot + 4 Intermedium Essence
Essence recipe index

Essence recipes 67 - 72

Intermedium Leggings x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Prudentium Leggings + 4 Intermedium Ingot
Output **1 Intermedium Leggings**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Intermedium Paxel x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Prudentium Paxel + 4 Intermedium Ingot
Output **1 Intermedium Paxel**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Intermedium Pickaxe x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Prudentium Pickaxe + 4 Intermedium Ingot
Output **1 Intermedium Pickaxe**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Intermedium Scythe x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Prudentium Scythe + 4 Intermedium Ingot
Output **1 Intermedium Scythe**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Intermedium Shovel x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Prudentium Shovel + 4 Intermedium Ingot
Output **1 Intermedium Shovel**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Intermedium Sickle x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Prudentium Sickle + 4 Intermedium Ingot
Output **1 Intermedium Sickle**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Essence recipes 73 - 78

Intermedium Sword x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Prudentium Sword + 4 Intermedium Ingot

Output **1 Intermedium Sword**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Intermedium Watering Can x1
Crafting Table / Universal Crafter
Shaped - copy the grid



5 Intermedium Ingot + 1 Prudentium Watering Can
Essence recipe index

Superium Axe x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Intermedium Axe + 4 Superium Ingot

Output **1 Superium Axe**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Superium Boots x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Intermedium Boots + 4 Superium Ingot

Output **1 Superium Boots**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Superium Chestplate x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Intermedium Chestplate + 4 Superium Ingot

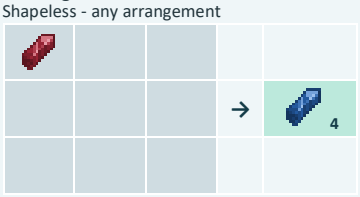
Output **1 Superium Chestplate**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Superium Crook x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Intermedium Crook + 4 Superium Ingot

Output **1 Superium Crook**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Essence recipes 79 - 84

Superium Essence x4
Crafting Table / Universal Crafter
Shapeless - any arrangement



1 Supremium Essence
Essence recipe index

Superium Essence x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
4 Intermedium Essence + 1 Infusion Crystal

Output **1 Superium Essence**
4s per batch, up to 64 output. Crystal retained; one use per output, 1000 uses per normal crystal.
Essence recipe index

Superium Hammer x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Intermedium Hammer + 4 Superium Ingot

Output **1 Superium Hammer**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Superium Helmet x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Intermedium Helmet + 4 Superium Ingot

Output **1 Superium Helmet**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Superium Hoe x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Intermedium Hoe + 4 Superium Ingot

Output **1 Superium Hoe**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Superium Ingot x1
Crafting Table / Universal Crafter
Shapeless - any arrangement



1 Intermedium Ingot + 4 Superium Essence
Essence recipe index

Essence recipes 85 - 90

Superium Leggings x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Intermedium Leggings + 4 Superium Ingot
Output  **1 Superium Leggings**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Superium Paxel x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Intermedium Paxel + 4 Superium Ingot
Output  **1 Superium Paxel**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Superium Pickaxe x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Intermedium Pickaxe + 4 Superium Ingot
Output  **1 Superium Pickaxe**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Superium Scythe x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Intermedium Scythe + 4 Superium Ingot
Output  **1 Superium Scythe**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Superium Shovel x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Intermedium Shovel + 4 Superium Ingot
Output  **1 Superium Shovel**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Superium Sickle x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Intermedium Sickle + 4 Superium Ingot
Output  **1 Superium Sickle**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Essence recipes 91 - 96

Superium Sword x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Intermedium Sword + 4 Superium Ingot
Output **1 Superium Sword**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Superium Watering Can x1
Crafting Table / Universal Crafter
Shaped - copy the grid



5 Superium Ingot + 1 Intermedium Watering Can
Essence recipe index

Supremium Axe x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Superium Axe + 4 Supremium Ingot
Output **1 Supremium Axe**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Supremium Boots x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Superium Boots + 4 Supremium Ingot
Output **1 Supremium Boots**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Supremium Chestplate x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Superium Chestplate + 4 Supremium Ingot
Output **1 Supremium Chestplate**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Supremium Crook x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Superium Crook + 4 Supremium Ingot
Output **1 Supremium Crook**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Essence recipes 97 - 102

Supremium Essence x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
4 Superium Essence + 1 Infusion Crystal

Output **1 Supremium Essence**
4s per batch, up to 64 output. Crystal retained; one use per output, 1000 uses per normal crystal.
Essence recipe index

Supremium Hammer x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Superium Hammer + 4 Supremium Ingot

Output **1 Supremium Hammer**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Supremium Helmet x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Superium Helmet + 4 Supremium Ingot

Output **1 Supremium Helmet**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Supremium Hoe x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Superium Hoe + 4 Supremium Ingot

Output **1 Supremium Hoe**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Supremium Ingot x1
Crafting Table / Universal Crafter
Shapeless - any arrangement



1 Superium Ingot + 4 Supremium Essence
Essence recipe index

Supremium Leggings x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Superium Leggings + 4 Supremium Ingot

Output **1 Supremium Leggings**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Essence recipes 103 - 108

Supremium Paxel x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Superium Paxel + 4 Supremium Ingot
Output  **1 Supremium Paxel**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Supremium Pickaxe x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Superium Pickaxe + 4 Supremium Ingot
Output  **1 Supremium Pickaxe**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Supremium Scythe x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Superium Scythe + 4 Supremium Ingot
Output  **1 Supremium Scythe**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Supremium Shovel x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Superium Shovel + 4 Supremium Ingot
Output  **1 Supremium Shovel**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Supremium Sickle x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Superium Sickle + 4 Supremium Ingot
Output  **1 Supremium Sickle**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Supremium Sword x1
Essence Tinkering Table
PROCESS - put ingredients in any input slots
1 Superium Sword + 4 Supremium Ingot
Output  **1 Supremium Sword**
Preserves name, enchantments, metadata and proportion of durability used.
Essence recipe index

Essence recipes 109 - 109

Supremium Watering Can x1
Crafting Table / Universal Crafter
Shaped - copy the grid



5 Supremium Ingot + 1 Superium Watering Can
Essence recipe index

Reactor and ship crafting layouts

Use these recipes in a normal Crafting Table. For the reactor, make the casing and the controller before gathering the cooling blocks and fuel.

Reactor Controller



Reactor Casing



Wireless Power Relay



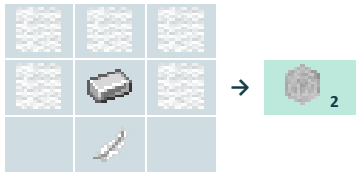
Uranium Chunk



Nuclear Fuel Rod



Ship Balloon



Ship Helm I



Ship Helm II



Ship Helm III



BUILDS *Nuclear power* | *Flying ships*

Void Ore Miner crafting recipes 1

The grid shows the ingredients. Each card labels the pictured output and its quantity.

Void Ore Miner

Crafting Table / Any arrangement

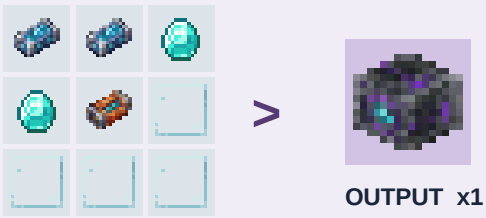


OUTPUT x1

4 Reinforced Frame + 2 Advanced Circuit + 1 Conductive Core + 1 Diamond + 1 Void Laser Head

Void Laser Head

Crafting Table / Any arrangement



OUTPUT x1

2 Crystalline Alloy + 2 Diamond + 1 Conductive Core + 4 Glass

Void Mining Frame

Crafting Table / Any arrangement



OUTPUT x4

2 Steel Ingot + 4 Seared Brick + 1 Reinforced Frame

Iron Focus Lens

Crafting Table / Any arrangement



OUTPUT x1

4 Glass + 4 Iron Ingot + 1 Light Gray Dye

Void Ore Miner crafting recipes 2

The grid shows the ingredients. Each card labels the pictured output and its quantity.

Copper Focus Lens

Crafting Table / Any arrangement



4 Glass + 4 Copper Ingot + 1 Orange Dye

Gold Focus Lens

Crafting Table / Any arrangement



4 Glass + 4 Gold Ingot + 1 Yellow Dye

Alloy Focus Lens

Crafting Table / Any arrangement



4 Glass + 4 Steel Ingot + 1 Purple Dye

Star Forge machines and materials

Machine recipe icons show batch amounts. A Water Bucket represents 1,000 mB of water supplied to the machine.

Star Infuser

Crafting Table / Match the layout



OUTPUT x1

2 Nether Star + 1 Conductive Core + 2 Reinforced Frame + 1 Material Infuser + 2 Advanced Circuit + 1 Precision Assembler

Star Generator

Crafting Table / Match the layout



OUTPUT x1

4 Star Ingot + 1 Nether Star + 2 Conductive Core + 1 Factory Core + 1 Advanced Solar Panel

Star Fusion Replicator

Crafting Table / Match the layout



OUTPUT x1

4 Star Ingot + 1 Nether Star + 2 Crystalline Alloy + 1 Factory Core + 1 Precision Assembler

Star Ingot

Star Infuser / 60 s at 2,000 W/s



OUTPUT x1

4 Nether Star + 2 Netherite Ingot + 1 Crystalline Alloy + 2 Advanced Circuit

Crystalline Alloy

Material Infuser / 1,000 W/s



OUTPUT x2

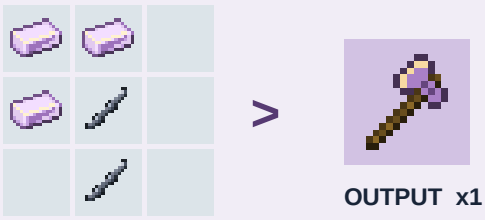
2 Diamond + 4 Steel Ingot + 2 Nickel Ingot + 1 Polymer + 1 Water Bucket

Star gear crafting recipes 1

Each pictured output shows what the Crafting Table makes and how many you get.

Star Axe

Crafting Table / Match the layout



3 Star Ingot + 2 Steel Tool Rod

Star Battleaxe

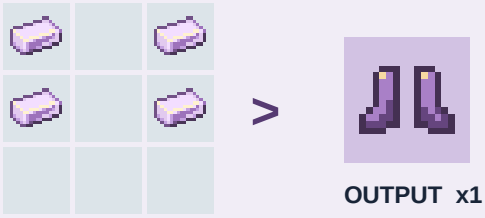
Crafting Table / Match the layout



5 Star Ingot + 2 Steel Tool Rod

Star Boots

Crafting Table / Match the layout



4 Star Ingot

Star Chestplate

Crafting Table / Match the layout



8 Star Ingot

Star Crook

Crafting Table / Match the layout



3 Star Ingot + 2 Steel Tool Rod

Star Hammer

Crafting Table / Match the layout



4 Star Ingot + 2 Steel Tool Rod

Star gear crafting recipes 2

Each pictured output shows what the Crafting Table makes and how many you get.

Star Helmet

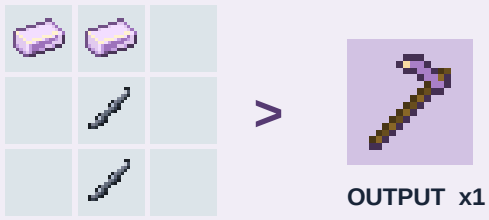
Crafting Table / Match the layout



5 Star Ingot

Star Hoe

Crafting Table / Match the layout



2 Star Ingot + 2 Steel Tool Rod

Star Leggings

Crafting Table / Match the layout



7 Star Ingot

Star Pickaxe

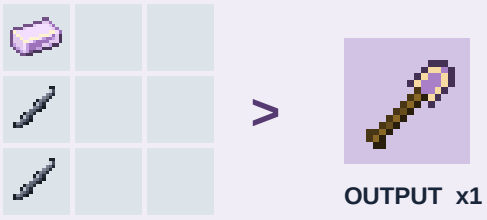
Crafting Table / Match the layout



3 Star Ingot + 2 Steel Tool Rod

Star Shovel

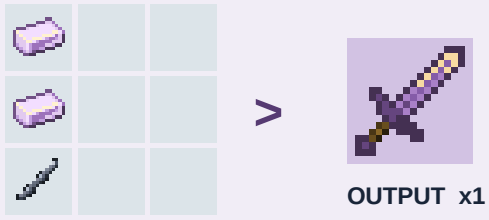
Crafting Table / Match the layout



1 Star Ingot + 2 Steel Tool Rod

Star Sword

Crafting Table / Match the layout



2 Star Ingot + 1 Steel Tool Rod

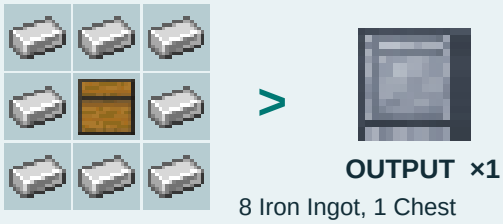
Upgraded chests

Start with a vanilla Chest. Upgrade to Iron, then Gold, Diamond and Netherite. In a Crafting Table, place the previous tier in the center and surround it with eight of the pictured material.

Place two matching chests side by side to join them. Each tier doubles its slots when paired. Use a Transfer Node on the source inventory and Transfer Pipes to route items. Try a small transfer into and out of a new chest before connecting the entire line.

Use the inventory to store tools and mixed batches. Keep a dedicated chest for the feed or output of a single machine when you need to trace where items went.

Iron Chest



OUTPUT ×1

8 Iron Ingot, 1 Chest

Gold Chest



OUTPUT ×1

8 Gold Ingot, 1 Iron Chest

Diamond Chest



OUTPUT ×1

8 Diamond, 1 Gold Chest

Netherite Chest



OUTPUT ×1

8 Netherite Ingot, 1 Diamond Chest

Slots per single / double chest

Iron 45 / 90 Gold 54 / 108 Diamond 63 / 126 Netherite 72 / 144

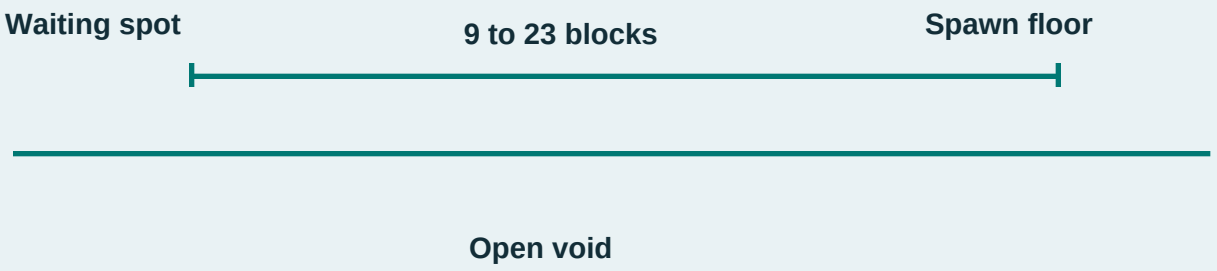
A spawning platform in the void Nether

Build a broad platform from solid, non-waterlogged blocks. Leave at least three blocks of open air above the floor and open void underneath the platform. Keep the spawning floor dark and put your waiting spot at about the same height, 9 to 23 blocks from the intended spawning area.

Enable mob spawning and use a difficulty above Peaceful. Stay in the Nether while the farm runs. The pack checks periodically, samples nearby floor positions, and limits nearby monsters. Give it time and clear existing mobs if the platform stops producing.

Possible spawns include Wither Skeletons, Blazes, Piglins, Zombified Piglins and Magma Cubes. Take the Wither Skeleton Skulls to the Wither Chamber, along with Soul Sand. A Mob Grinder facing the mobs can collect drops, including occasional skulls.

SIDE VIEW



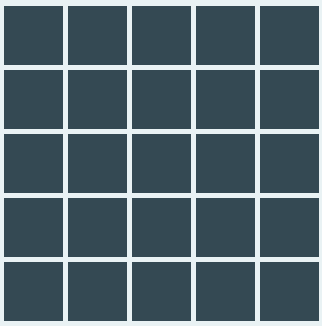
From skulls to Nether Stars

Build a sealed 5 × 5 × 5 shell around an empty 3 × 3 × 3 interior. Put one Wither Chamber Controller in an exterior wall. You can put it off-center. Use Witherproof Casing, Witherproof Glass or Chamber Ports for shell blocks, including edges and corners.

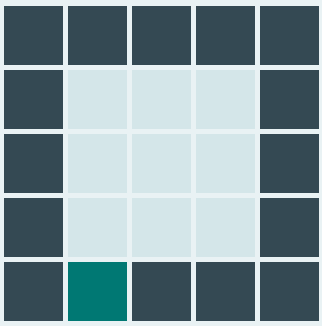
Run Watt Pipes into the shell so the Controller face stays clear. To automate, feed 3 Wither Skeleton Skulls and 4 Soul Sand through a Chamber Port. Route Nether Stars out with a Transfer Node attached to a Port. Use the Controller inventory to test the first cycle by hand.

The chamber uses a contained Wither entity and animates the assembly with Soul Sand and skulls. The explosion particles and sound are cosmetic. The chamber needs 500 W per work step at first and 800 W in the armored phase. Restore power or repair the shell if the cycle suspends. Each completed cycle makes one Nether Star.

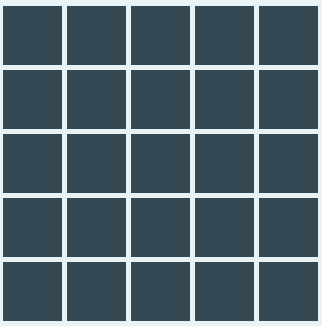
FLOOR



MIDDLE ×3



ROOF



Dark tiles show the shell. The green tile marks an example controller position. Use one controller in total. Leave the middle of all three wall layers empty.

A 5 × 5 × 5 shell uses 98 blocks in total, including the controller. Substitute glass or ports where needed.

Wither Chamber crafting recipes

Use a Crafting Table. Each card shows the recipe and its output quantity.

Wither Chamber Controller



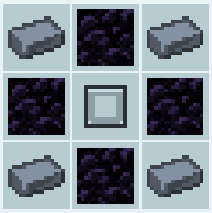
>



OUTPUT ×1

2 Netherite Ingot, 2 Conductive Core, 2 Reinforced Frame, 1 Mob Grinder, 2 Advanced Circuit

Witherproof Casing



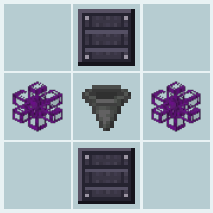
>



OUTPUT ×8

4 Steel Ingot, 4 Obsidian, 1 Reinforced Frame

Chamber Port



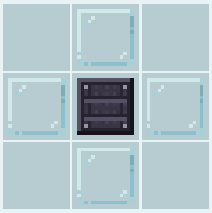
>



OUTPUT ×1

2 Witherproof Casing, 2 Transfer Pipe, 1 Hopper

Witherproof Glass



>



OUTPUT ×4

4 Glass, 1 Witherproof Casing

Star Forge and the new machine HUDs

Charge Star gear in the Star Infuser with one Nether Star and 1,600 W/s. The charge takes 10 seconds and lasts up to one Minecraft day of active use. The full charged armor set grants Resistance, Speed and Jump Boost I.

Use one eligible ore or ore product and two Nether Stars in the Star Fusion Replicator. Supply 4,000 W/s for 15 seconds to get two of the input. Diamond is supported. The Star Generator uses one Nether Star for 20 seconds at 16,000 W/s, with a 640,000 W buffer.

The red Watt Reserve gauge measures usable energy remaining on the connected supply. A green draw lamp means the machine recently drew power. A machine can keep working while the gauge looks empty if it uses power as fast as the network delivers it.

Read the work bar separately from the power gauge. Timed machines show job progress, the Harvester can show tree progress, and the Planter pulses on successful planting. Generators and machines with fuel, heat, Liquid XP or Pink Essence show those quantities on separate gauges. The Autonomous Activator is unpowered.



Star Infuser



Star Fusion Replicator



Mob Grinder



